

STABLE NONLOCAL CONES AND THE FRACTIONAL DE GIORGI CONJECTURE IN DIMENSION FOUR BELOW ONE HALF

ABSTRACT. We present a candidate proof of one-dimensional symmetry for every bounded entire solution of $(-\Delta)^s U = U - U^3$ in $\mathbb{R}^4$ with $U_4 > 0$, throughout $0 < s < 1/2$, without prescribed directional limits. The geometric statement is flatness of every smooth stationary stable nonlocal perimeter cone in $\mathbb{R}^3$ for every perimeter order $0 < p < 1$. A spherical Green identity controls the total normal flux; an angular stability inequality then forces that flux to be maximal. For small p , a truncated integral of the exact angular kernel replaces a singular power majorant. The estimates close with explicit rational bounds and allow disconnected spherical links. The PDE conclusion uses the stable-solution theory of Cabré–Cinti–Serra, ordered minimizing cone splitting, and improvement of flatness. The parameters satisfy $p = 2s$.

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1. INTRODUCTION AND MAIN RESULTS

We use the Fourier normalization $\widehat{(-\Delta)^s u}(\xi) = |\xi|^{2s} \widehat{u}(\xi)$. For $0 < s < 1/2$, the geometric limit of the fractional Allen–Cahn energy is a nonlocal perimeter with parameter $p = 2s \in (0, 1)$. For measurable $E \subset \mathbb{R}^n$ and bounded open D , put

$$\text{Per}_p(E; D) = \iint_{(E \cap D) \times E^c} \frac{dX dY}{|X - Y|^{n+p}} + \iint_{(E \setminus D) \times (E^c \cap D)} \frac{dX dY}{|X - Y|^{n+p}}.$$

A cone satisfies $tE = E$ for all $t > 0$. Stationarity means vanishing first variation, and stability means nonnegative second variation, for compact smooth ambient deformations

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supported away from the vertex. Sets are identified up to null sets and represented by their open phase when their boundary is smooth. A nontrivial cone has both phases of positive measure. Smoothness includes a locally one-sided boundary; a compact smooth link is an embedded compact one-dimensional submanifold, with finitely many components. A smooth open phase need not be connected: at each boundary point it occupies exactly one side of a local graph. All spherical phases below have this meaning; no connectedness of the phase or its boundary is implicit in the terminology.

Caffarelli–Roquejoffre–Savin [5] developed the basic theory of nonlocal minimal surfaces. Savin–Valdinoci [8] classified minimizing cones in dimension two. Cabré–Cinti–Serra [3] classified smooth stable cones in dimension three for perimeter orders sufficiently close to one. Their stable Allen–Cahn theory [4] transfers stable cone rigidity to stable entire solutions. Tsiamis [9] proves stable cone rigidity for orders sufficiently close to zero. We use the angular stability and translation identities from that work, which apply throughout $0 < p < 1$. Its qualitative normal-flux and isoperimetric pinching arguments [9, Theorem 4.2 and Corollary 4.3] also motivate the geometric organization below. Here we develop explicit estimates intended to close the entire interval.

Definition 1.1 (The normalized layer). For $0 < s < 1$, let q_s denote the increasing solution of

$$(-\partial_t^2)^s q_s = q_s - q_s^3 \quad \text{in } \mathbb{R}, \quad q_s(0) = 0, \quad \lim_{t \rightarrow \pm\infty} q_s(t) = \pm 1,$$

where the Fourier symbol is $|\xi|^{2s}$.

The existence and uniqueness implicit in this definition are justified in the final section.

Theorem 1.2 (Monotone rigidity in dimension four). *For every $s \in (0, 1/2)$, let U be a bounded entire classical solution of*

$$(-\Delta)^s U = U - U^3 \quad \text{in } \mathbb{R}^4, \quad \partial_{x_4} U > 0.$$

No assumption is imposed on the limits as $x_4 \rightarrow \pm\infty$. Then there are $e \in S^3$, with $e_4 > 0$, and $a \in \mathbb{R}$ such that

$$U(x) = q_s(e \cdot x - a).$$

Theorem 1.3 (Stable cone classification). *For every $0 < p < 1$, each nontrivial stationary p -perimeter cone in $\mathbb{R}^3$ with locally one-sided smooth boundary away from zero, compact smooth spherical link, and stability under compact variations away from zero is a halfspace. No convexity, symmetry, or link connectivity assumption is imposed.*

Corollary 1.4 (Stable rigidity in dimension three). *For every $0 < s < 1/2$, each bounded stable entire classical solution of $(-\Delta)^s u = u - u^3$ in $\mathbb{R}^3$ is either a constant well ± 1 or a layer $u(x) = q_s(e \cdot x - a)$ for some $e \in S^2$ and $a \in \mathbb{R}$, where q_s is defined in Definition 1.1.*

For the equation of order $p = 2s$, stability of a solution means

$$\int_{\mathbb{R}^n} |(-\Delta)^{p/4} \phi|^2 - \int_{\mathbb{R}^n} (1 - 3u^2) \phi^2 \geq 0 \quad (\phi \in C_c^\infty(\mathbb{R}^n)).$$

The integral/Fourier normalization and its weak-testing identity are proved for every $0 < p < 2$ in Lemma A.2. All constants and one-dimensional profiles are understood in the Fourier normalization fixed above.

1.1. The quantitative mechanism. Let $\Gamma = \partial E \cap S^2$, $L = |\Gamma|$, $V = \int_\Gamma \nu$, $v = |V|/L$, and $\delta = 1 - v$. The angular normal energy P and the normalization $b_p = 2/[(1+p)(2+p)]$ are defined below. A spherical Green identity and the spherical isoperimetric inequality imply

$$\frac{P}{b_p L} \geq F(v), \quad F(v) = \frac{\pi\sqrt{2}(1-v^2)}{\sqrt{3-v^2}}.$$

A row estimate anchored in the normal at its center gives the opposite bound. A translation Jacobi field then supplies a second stability test. We use two overlapping ranges:

- (1) For $1/3 \leq p < 1$, the row bound gives $v > 3/5$. A power estimate for the kernel shows that $0 < \delta < 2/5$ is impossible.
- (2) For $0 < p \leq 1/3$, the row bound gives $v > 38/125 + p$. We retain the exact truncated kernel integral $T_p(a) = \int_0^a W_p(1 - r^2/2) dr$. The resulting necessary pinching inequality is contradicted by an explicit decreasing majorant $\mathcal{B}(p)$ with $\mathcal{B}(0) = 812/815 < 1$.

Thus $\delta = 0$ in both ranges. The proof works for every smooth link, including disconnected links. The final section checks the hypotheses of the published PDE and minimizing-cone reductions.

2. ANGULAR KERNELS AND THE STABILITY FORM

All surface kernels in the geometric argument are unnormalized. A common positive factor has no effect on stationarity or stability. We write $d\mu$ for arclength on the spherical link and $\Gamma(z)$ for Euler's gamma function when its argument is a scalar. Unless an estimate is explicitly declared uniform, its constants may depend on the fixed order p and the fixed compact link. In particular, the fixed- p power majorant for W_p is not used at $p = 0$.

Lemma 2.1 (Kernel representations and comparison). *For $0 < p < 1$ and $-1 \leq z < 1$, define*

$$A_p(z) = \int_0^\infty \frac{\rho^{(1+p)/2}}{(1 + \rho^2 - 2\rho z)^{(3+p)/2}} d\rho,$$

$$B_p(z) = \int_0^\infty \frac{\rho}{(1 + \rho^2 - 2\rho z)^{(3+p)/2}} d\rho, \quad W_p = B_p - A_p.$$

Put $b_p = 2/[(1+p)(2+p)]$, $k = 1 - p$, and $m = (3+p)/2$. Their difference also has the exact representation

$$W_p(1 - r^2/2) = \int_0^\infty F_k(u)(u^2 + r^2)^{-m} du, \quad F_k(u) = \frac{2[\cosh(k \operatorname{arsinh}(u/2)) - 1]}{\sqrt{1 + u^2/4}},$$

where $0 < r \leq 2$, and

$$0 \leq F_k(u) \leq k^2 F_1(u), \quad F_1(u) = 2 - \frac{4}{\sqrt{u^2 + 4}}, \quad F_1(u) \leq \frac{2u^2}{u^2 + 8} \leq \frac{u^2}{4}.$$

For $0 \leq u \leq 1$, also $F_1(u) \geq u^2/5$. These kernels obey

$$B_p(z) \geq \frac{2}{(1+p)(2+p)} \frac{1}{1-z},$$

and

$$0 \leq W_p(z) \leq 2(1-p)^2 \int_0^\infty \frac{u^2 du}{(u^2 + 8)[u^2 + 2(1-z)]^{(3+p)/2}}.$$

W_p is increasing in z .

Proof. For fixed $z < 1$, the denominator is positive on $[0, \infty)$. At zero the integrands defining A_p and B_p are respectively $O(\rho^{(1+p)/2})$ and $O(\rho)$; at infinity they are $O(\rho^{-(5+p)/2})$ and $O(\rho^{-2-p})$. Thus both integrals converge. In subsequent link integrals the diagonal $z = 1$ is omitted; the cancellations below establish local integrability. Set $\tau = (1+z)/2 \in [0, 1)$. Expand the positive binomial series in

$$1 + \rho^2 - 2\rho z = (1 + \rho)^2 \left(1 - \frac{4\tau\rho}{(1 + \rho)^2}\right).$$

Tonelli's theorem, the beta integral, and the gamma duplication formula give

$$B_p(z) = \frac{1}{(1+p)(2+p)} \sum_{n=0}^{\infty} \frac{(2)_n (1+p)_n}{(2+p/2)_n n!} \tau^n,$$

where $(b)_n = b(b+1)\cdots(b+n-1)$ and $(b)_0 = 1$. For completeness, the coefficient before simplification is

$$\frac{4^n((3+p)/2)_n}{n!} \int_0^\infty \frac{\rho^{n+1}}{(1+\rho)^{3+p+2n}} d\rho = \frac{4^n((3+p)/2)_n \Gamma(n+2) \Gamma(n+1+p)}{n! \Gamma(2n+3+p)}.$$

Indeed, with $m = (3+p)/2$, duplication gives

$$\Gamma(2n+3+p) = \Gamma(3+p) 4^n (m)_n (2+p/2)_n,$$

which cancels the factor $4^n(m)_n$ in the numerator. The remaining coefficient is

$$\frac{(2)_n(1+p)_n}{(2+p/2)_n n!} = \prod_{j=0}^{n-1} \frac{(j+2)(j+1+p)}{(j+2+p/2)(j+1)} \geq 1,$$

with the empty product interpreted as one, since for $j \geq 0$

$$(j+2)(j+1+p) - (j+2+p/2)(j+1) = \frac{p(j+3)}{2} \geq 0.$$

Since $0 \leq 4\tau\rho/(1+\rho)^2 \leq \tau < 1$, the binomial expansion is valid throughout the radial integral; all its terms are nonnegative, so Tonelli applies also at $\tau = 0$. Summing the geometric series gives

$$B_p(z) \geq \frac{1}{(1+p)(2+p)(1-\tau)} = \frac{b_p}{1-z},$$

as asserted. The same calculation at $p = 0$ gives the exact identity $B_0(z) = (1-z)^{-1}$.

Split the integrals at $\rho = 1$ and replace ρ by $1/\rho$ in the second part. Writing $D_z(\rho) = 1 + \rho^2 - 2\rho z$ and $a = (1+p)/2$, the identity $D_z(1/\rho) = \rho^{-2} D_z(\rho)$ yields

$$B_p(z) = \int_0^1 \frac{\rho + \rho^p}{D_z(\rho)^m} d\rho, \quad A_p(z) = 2 \int_0^1 \frac{\rho^a}{D_z(\rho)^m} d\rho.$$

Their difference is therefore

$$W_p(z) = \int_0^1 \frac{(\rho^{1/2} - \rho^{p/2})^2}{(1 + \rho^2 - 2\rho z)^{(3+p)/2}} d\rho.$$

The numerator is nonnegative, while $D_z(\rho)$ decreases as z increases; this proves positivity and monotonicity by direct comparison. For $\rho = e^{-t}$, the numerator and denominator become

$$(\rho^{1/2} - \rho^{p/2})^2 = 2e^{-(1+p)t/2} [\cosh(kt/2) - 1],$$

$$D_z(e^{-t}) = e^{-t} [2(\cosh t - z)].$$

The exponential factors cancel with $d\rho = e^{-t} dt$ after reversing the limits, since $m-1-(1+p)/2 = 0$. Now put $u = 2 \sinh(t/2)$; then $2(\cosh t - z) = u^2 + 2(1-z)$ and $dt = du/\sqrt{1+u^2/4}$. Hence

$$W_p(z) = \int_0^\infty \frac{F_k(u)}{[u^2 + 2(1-z)]^m} du, \quad F_k(u) = \frac{2[\cosh(k \operatorname{arsinh}(u/2)) - 1]}{\sqrt{1+u^2/4}}.$$

For $0 \leq k \leq 1$ and $v \geq 0$, the nonnegative power series gives

$$0 \leq \cosh(kv) - 1 = \sum_{n \geq 1} \frac{k^{2n} v^{2n}}{(2n)!} \leq k^2 (\cosh v - 1).$$

At $k = 1$, rationalizing the numerator gives $F_1(u) = 2 - 4/\sqrt{u^2 + 4} = 2u^2/[\sqrt{u^2 + 4}(\sqrt{u^2 + 4} + 2)]$. The denominator is at least $u^2 + 8$, because $\sqrt{u^2 + 4} \geq 2$. Consequently

$$F_k(u) \leq \frac{2k^2 u^2}{\sqrt{u^2 + 4}(\sqrt{u^2 + 4} + 2)} \leq \frac{2k^2 u^2}{u^2 + 8}.$$

Also $2u^2/(u^2+8) \leq u^2/4$. Inserting the preceding bound in the positive representation of W_p proves its stated integral upper bound. The identity $F_1(u) = 2u^2/[\sqrt{u^2+4}(\sqrt{u^2+4}+2)]$ also gives $F_1(u) \geq u^2/5$ on $[0, 1]$, since the denominator is at most $5 + 2\sqrt{5} < 10$. $\square$

Lemma 2.2 (Localized perimeter and differentiation of deformations). *Let $0 < p < 1$ and let $E \subset \mathbb{R}^3$ be a cone with locally one-sided C^2 boundary away from zero. Its spherical link is a compact embedded C^2 one-dimensional manifold with finitely many components. The localized perimeter $\text{Per}_p(E; D)$ is finite for every bounded open set D . Let Ψ_t be a smooth family of ambient diffeomorphisms, $\Psi_0 = \text{Id}$, which are the identity outside one fixed compact subset of D . Then $t \mapsto \text{Per}_p(\Psi_t(E); D)$ is twice continuously differentiable for sufficiently small $|t|$. If E is stationary on the support of the family, this second derivative at zero depends only on its initial velocity, not on its acceleration. Thus testing autonomous flows suffices for testing arbitrary smooth families at a stationary set, with the same support restriction.*

Proof. The radial tangent to ∂E is transverse to S^2 . Its intersection with S^2 is closed, embedded, and C^2 , hence compact. Connected coordinate neighborhoods form a finite cover; every component contains one of these neighborhoods, so there are finitely many components. Denote their total length by L . Polar parametrization gives

$$\mathcal{H}^2(\partial E \cap \{a < |X| < b\}) = L \int_a^b r \, dr = \frac{L}{2}(b^2 - a^2).$$

The empty-link case has a constant phase off zero and is immediate. For $R > 0$ and $0 < \varepsilon < 2R$, the piecewise C^2 set $F_\varepsilon = E \cap (B_{2R} \setminus \overline{B}_\varepsilon)$ has ordinary perimeter at most

$$2LR^2 + 16\pi R^2 + 4\pi\varepsilon^2.$$

The two spherical cuts meet the cone transversely; their corners have zero surface measure. Apply the divergence theorem against each compact smooth vector field and pass to the limit as $\varepsilon \downarrow 0$. The definition of total variation shows that $F = E \cap B_{2R}$ has finite ordinary perimeter and finite volume. For such a set, mollification and the fundamental theorem of calculus give

$$\|\chi_F(\cdot + h) - \chi_F\|_1 \leq \min\{|h| \text{Per}(F), 2|F|\}.$$

Indeed the gradient integral of each mollification is at most $\text{Per}(F)$; integrate the gradient along segments and then use L^1 convergence. Tonelli's theorem therefore gives

$$\iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{|\chi_F(X) - \chi_F(Y)|}{|X - Y|^{3+p}} \, dX \, dY \leq C \left(\text{Per}(F) \int_0^1 r^{-p} \, dr + |F| \int_1^\infty r^{-1-p} \, dr \right) < \infty.$$

This controls the interactions in $\text{Per}_p(E; B_R)$ with both points in B_{2R} . The remaining interactions have one point in B_R and one outside B_{2R} and are bounded by $2|B_R| \int_{|h| \geq R} |h|^{-3-p} \, dh$. Every bounded D lies in some such ball, and its interaction region is a subset of that ball's interaction region, proving local finiteness.

Write $\mathcal{R}_D = \mathbb{R}^6 \setminus (D^c \times D^c)$ and $J_t(X) = \det D\Psi_t(X) > 0$ for small $|t|$. Since Ψ_t fixes D^c , bijectivity implies $\Psi_t(D) = D$. The change of variables formula gives

$$\text{Per}_p(\Psi_t(E); D) = \frac{1}{2} \iint_{\mathcal{R}_D} |\chi_E(X) - \chi_E(Y)| \frac{J_t(X)J_t(Y)}{|\Psi_t(X) - \Psi_t(Y)|^{3+p}} \, dX \, dY.$$

Uniform bi-Lipschitz bounds give $|\Psi_t(X) - \Psi_t(Y)| \geq c|X - Y|$. For $j = 1, 2$, smoothness and the common compact support give

$$|\partial_t^j \Psi_t(X) - \partial_t^j \Psi_t(Y)| \leq C|X - Y|.$$

The Jacobians and their first two time derivatives are uniformly bounded. Differentiating the transformed kernel thus gives, for $j = 0, 1, 2$,

$$\left| \partial_t^j \left(\frac{J_t(X)J_t(Y)}{|\Psi_t(X) - \Psi_t(Y)|^{3+p}} \right) \right| \leq C|X - Y|^{-3-p}.$$

The right-hand side multiplied by $|\chi_E(X) - \chi_E(Y)|\mathbf{1}_{\mathcal{R}_D}$ is integrable by local finiteness. Dominated differentiation proves both finiteness of the deformed perimeters and twice continuous differentiability, before any surface second-variation formula is invoked.

Finally let $X = \partial_t \Psi_t|_{t=0}$ and $Y = \partial_t^2 \Psi_t|_{t=0}$. Twice differentiating the same expression shows that its second derivative is the sum of a quadratic expression in X and the first variation applied to Y . The autonomous flow of X has acceleration $DX X$. The difference between the two second variations is therefore the first variation applied to $Y - DX X$. This is zero at a stationary set. All these fields have the original compact support; in particular a support away from the vertex remains away from the vertex. This proves the last assertion. $\square$

Proposition 2.3 (Angular stability and translation fields). *Let $0 < p < 1$ and let E be a stationary cone in $\mathbb{R}^3$ with locally one-sided smooth boundary away from zero and compact smooth link Γ . Let ν be the outward normal, $d\mu$ arclength, and use the kernels in Lemma 2.1. Put*

$$P_x = \int_{\Gamma} B_p(x \cdot y)(1 - \nu_x \cdot \nu_y) d\mu_y, \quad P = \int_{\Gamma} P_x d\mu_x,$$

and, for smooth real f on Γ , define

$$q_B[f] = \frac{1}{2} \iint_{\Gamma^2} B_p(x \cdot y)(f_x - f_y)^2 d\mu_x d\mu_y - \int_{\Gamma} P_x f_x^2 d\mu_x.$$

Then $q_B(\nu \cdot e, f) = 0$ for every constant vector e and every smooth f , where the parentheses denote polarization. If the cone is stable under compact variations away from zero, then

$$q_A[f] := q_B[f] + \iint_{\Gamma^2} W_p(x \cdot y) f_x f_y d\mu_x d\mu_y \geq 0.$$

In particular, $P \leq \iint W_p$ and $q_B[1 - \nu \cdot e] = -P$ for every constant unit vector e .

Proof. Use the kernels A_p, B_p, W_p defined in Lemma 2.1. Lemma 2.2 ensures that the localized perimeters and their first two variations are finite. The normal second variation, with an irrelevant common positive constant suppressed, is

$$Q[\zeta] = \frac{1}{2} \iint_{\partial E \times \partial E} \frac{(\zeta(X) - \zeta(Y))^2}{|X - Y|^{3+p}} d\sigma_X d\sigma_Y - \int_{\partial E} \zeta(X)^2 \int_{\partial E} \frac{1 - \nu_X \cdot \nu_Y}{|X - Y|^{3+p}} d\sigma_Y d\sigma_X.$$

The second-variation formula is recorded in Tsiamis [9, Section 2], and the translation identity in [9, Lemma 2.4 and equations (7), (8)]. They apply with dimension $N = 3$ and perimeter order $s = p$ to the stationary cone and its smooth link; convexity and connectivity are not required. Indeed, translating E by te preserves zero nonlocal mean curvature. Differentiation at $t = 0$, with the tangential component absorbed by reparametrization, makes the normal component $h(X) = \nu_X \cdot e$ a Jacobi field. Although this field has no compact support, its Jacobi integral has convergent tails: h and ν are bounded, and the radial bounds are $C\rho$ at zero and $C\rho^{-2-p}$ at infinity for $X \in \Gamma$. At $X = x \in \Gamma$, write $Y = \rho y$ and $d\sigma_Y = \rho d\rho d\mu_y$. As h and ν have degree zero, retain the physical cutoff by defining

$$B_{p,\varepsilon}(z) = \int_{\{\rho > 0: 1 + \rho^2 - 2\rho z > \varepsilon^2\}} \frac{\rho d\rho}{(1 + \rho^2 - 2\rho z)^{(3+p)/2}}, \quad P_{x,\varepsilon} = \int_{\Gamma} B_{p,\varepsilon}(x \cdot y)(1 - \nu_x \cdot \nu_y) d\mu_y.$$

This truncated kernel is symmetric in x, y . The physical Jacobi identity is the statement that

$$J_{\varepsilon} h(x) := \int_{\Gamma} B_{p,\varepsilon}(x \cdot y)(h_y - h_x) d\mu_y + P_{x,\varepsilon} h_x \longrightarrow 0.$$

The convergence is uniform for $x \in \Gamma$. To see this, use a finite collection of uniformly controlled tangent-plane charts in a compact annular neighborhood of the link. With $u \in \mathbb{R}^2$ centered at x , write $Y = x + u + g_x(u)\nu_x$, where $g_x(0) = Dg_x(0) = 0$. Uniformly in x ,

$$\begin{aligned} |Y - x| &= |u|(1 + O(|u|^2)), & d\sigma_Y &= (1 + O(|u|^2)) du, \\ h(Y) - h(x) &= L_x(u) + O(|u|^2), & 1 - \nu_x \cdot \nu_Y &= O(|u|^2), \end{aligned}$$

where L_x is linear. The leading kernel times $L_x(u)$ is odd and cancels on a centered tangent disk. The quadratic remainder and potential have omitted integral at most $C \int_0^{2\varepsilon} r^{-p} dr$. The physical-ball boundary has tangent radius $\varepsilon + O(\varepsilon^3)$, so its difference from that disk is a shell of area $O(\varepsilon^4)$; the first-order integrand is $O(\varepsilon^{-2-p})$ there, giving $O(\varepsilon^{2-p})$. The Jacobian and kernel corrections to the linear integrand have size $O(r^{-p})$ before integration in du , and omitted integral $O(\varepsilon^{2-p})$. The Jacobi identity therefore gives $\sup_{x \in \Gamma} |J_\varepsilon h(x)| \leq C(\varepsilon^{1-p} + \varepsilon^{2-p}) \rightarrow 0$. All constants here may depend on the fixed link and on p .

For each fixed $\varepsilon > 0$, Fubini and symmetry apply to the truncated integrals and give

$$\int_{\Gamma} f_x J_\varepsilon h(x) d\mu_x = -\frac{1}{2} \iint_{\Gamma^2} B_{p,\varepsilon}(x \cdot y)(h_x - h_y)(f_x - f_y) d\mu_x d\mu_y + \int_{\Gamma} P_{x,\varepsilon} h_x f_x d\mu_x.$$

Pass to the limit in these paired integrals. Here is the needed kernel estimate directly from its radial definition. Write $d = |x - y|$ and $m = (3 + p)/2$. In the defining radial integral,

$$1 + \rho^2 - 2\rho x \cdot y = (\rho - 1)^2 + \rho d^2.$$

For $0 < \rho < 1/2$ this denominator is at least $1/4$, and for $\rho > 2$ it is at least $\rho^2/4$. The corresponding integrals of $\rho[(\rho - 1)^2 + \rho d^2]^{-m}$ are therefore bounded by $4^m \int_0^{1/2} \rho d\rho$ and $4^m \int_2^\infty \rho^{-2-p} d\rho$, respectively, uniformly in x, y . On $1/2 \leq \rho \leq 2$, the same integral is at most

$$2 \int_{\mathbb{R}} (t^2 + d^2/2)^{-m} dt = C_p d^{-2-p},$$

by $t = dv$. The integral defining C_p is finite because $2m > 1$. Since $0 < d \leq 2$, the two bounded tail contributions can be absorbed by enlarging the constant. Thus $B_p(x \cdot y) \leq C_p^B d^{-2-p}$ for a finite constant C_p^B . Positivity of A_p and $W_p = B_p - A_p$ also gives $0 < A_p \leq B_p$. The positive representation and $F_{1-p}(u) \leq (1-p)^2 u^2/4$ also give, with $w_p(r) = W_p(1 - r^2/2)$,

$$w_p(r) \leq \frac{(1-p)^2}{4} \int_0^\infty \frac{u^2 du}{(u^2 + r^2)^{(3+p)/2}} = \frac{(1-p)^2}{4} r^{-p} \int_0^\infty \frac{t^2 dt}{(1+t^2)^{(3+p)/2}}.$$

The last integral is finite: its integrand is $O(t^2)$ at zero and $O(t^{-1-p})$ at infinity. Thus $w_p(r) \leq \mathcal{D}_p r^{-p}$ for a finite constant $\mathcal{D}_p$ depending only on p . On each local link chart, $|(h_x - h_y)(f_x - f_y)| + |1 - \nu_x \cdot \nu_y| \leq C|x - y|^2$. The resulting majorant is $C|x - y|^{-p}$ in one-dimensional arclength and is integrable. Compactness gives uniform chart bounds; distinct components and pairs outside the diagonal charts have positive separation. This proves that P_x, P , and all polarized q_B integrals are finite. Since $0 \leq B_{p,\varepsilon} \leq B_p$ and $B_{p,\varepsilon} \rightarrow B_p$ off the diagonal, dominated convergence in the preceding identity and uniform convergence of $J_\varepsilon h$ give

$$q_B(h, f) = \frac{1}{2} \iint_{\Gamma^2} B_p(x \cdot y)(h_x - h_y)(f_x - f_y) d\mu_x d\mu_y - \int_{\Gamma} P_x h_x f_x d\mu_x = 0.$$

The same estimates prove that the paired A_p form and the W_p term defining q_A are finite. In particular, no absolute integrability of the unpaired first difference $B_p(h_y - h_x)$ is asserted.

For angular stability choose $R \geq 1$ and a smooth cutoff $0 \leq \chi_R \leq 1$ that equals one on $[-R, R]$, vanishes outside $[-R-1, R+1]$, and satisfies $\|\chi_R'\|_2^2 \leq C$ independently of R . Such cutoffs are obtained from two translates of a fixed smooth transition function. Put

$$\beta = \frac{1-p}{2}, \quad \zeta_R(rx) = r^{-\beta} \chi_R(\log r) f(x),$$

$$\mathcal{K}_p(\tau, z) = [2(\cosh \tau - z)]^{-(3+p)/2}, \quad N_R = \int_{\mathbb{R}} \chi_R^2,$$

$$c_R(\tau) = \frac{1}{N_R} \int_{\mathbb{R}} \chi_R(t) \chi_R(t + \tau) dt.$$

In particular, $2R \leq N_R \leq 2R+2$ and c_R is even. The normal speed ζ_R has compact support in a smooth annulus away from the vertex. Extend $\zeta_R \nu$ by a tubular projection of a slightly larger compact surface piece, multiply by an ambient cutoff equal to one on its support,

and use the flow of the resulting smooth compact vector field. This realizes an admissible variation with normal speed ζ_R , so stability gives $Q[\zeta_R] \geq 0$. The displayed second variation is finite: locally its differences and normal defect vanish quadratically, and its far-field terms converge by compact support and the radial tail bounds above.

We derive the angular limit directly, in agreement with [9, Lemma 2.7 and Proposition 2.10]. In $X = e^t x$, $Y = e^{t+\tau} y$, write $z = x \cdot y$ and $n_{xy} = \nu_x \cdot \nu_y$. The product of the surface measures and the distance kernel is

$$\frac{d\sigma_X d\sigma_Y}{|X - Y|^{3+p}} = e^{2\beta t + \beta\tau} \mathcal{K}_p(\tau, z) dt d\tau d\mu_x d\mu_y.$$

First restrict to the symmetric region $\tau^2 + |x - y|^2 > \delta^2$. The squared X term has t -integral $e^{\beta\tau} N_R f_x^2$, the squared Y term has $e^{-\beta\tau} N_R f_y^2$, and the mixed product has $N_R c_R(\tau) f_x f_y$. The potential has $e^{\beta\tau} N_R (1 - n_{xy}) f_x^2$. Evenness in τ and symmetry in x, y turn the resulting integrand for $Q[\zeta_R]/N_R$ into $\mathcal{K}_p(\tau, z)$ times

$$\begin{aligned} & \frac{1}{2}(f_x - f_y)^2 + \frac{1}{2}[\cosh(\beta\tau) - 1](f_x^2 + f_y^2) \\ & + [1 - c_R(\tau)]f_x f_y - \frac{1}{2}\cosh(\beta\tau)(1 - n_{xy})(f_x^2 + f_y^2). \end{aligned}$$

Thus no divergent diagonal squared terms are integrated separately. Near $\tau = 0$, $x = y$, arclength coordinates s, s' give $2(\cosh \tau - x \cdot y) \asymp \tau^2 + |s - s'|^2$. The first, second, and fourth terms in the bracket have size $O(|s - s'|^2)$, $O(\tau^2)$, and $O(|s - s'|^2)$, respectively. Each has an integrable majorant whose radial integral in the two variables $(\tau, s - s')$ is $C \int_0^1 r^{-p} dr$. For large $|\tau|$, the factors involving $\cosh(\beta\tau)$ are bounded by $Ce^{-(1+p)|\tau|}$ after multiplication by $\mathcal{K}_p$. Off the local diagonal charts compactness gives positive separation for bounded τ .

For the remaining term, translation invariance of the L^2 norm and the fundamental theorem of calculus give

$$0 \leq 1 - c_R(\tau) = \frac{\|\chi_R - \chi_R(\cdot + \tau)\|_2^2}{2N_R} \leq \frac{\tau^2 \|\chi'_R\|_2^2}{2N_R} \leq \frac{C\tau^2}{R}.$$

Here the middle estimate follows by integrating $\chi_R(t + \tau) - \chi_R(t) = \int_0^\tau \chi'_R(t + v) dv$ and applying Cauchy–Schwarz and Fubini (with reversed limits when $\tau < 0$). Consequently

$$M_{p,\Gamma} := \int_{\mathbb{R}} \tau^2 \iint_{\Gamma^2} \mathcal{K}_p(\tau, x \cdot y) d\mu_x d\mu_y d\tau < \infty.$$

Indeed, the same two-variable estimate gives $C \int_0^1 r^{-p} dr$ near the diagonal, while the tail is bounded by $C \int_1^\infty \tau^2 e^{-(3+p)\tau/2} d\tau$. This proves absolute integrability of the entire combined bracket. Dominated convergence removes the symmetric cutoff; the original finite difference and potential integrals converge to $Q[\zeta_R]$ under that cutoff as well.

The change of variables $\rho = e^\tau$ in the radial definitions, and the evenness of $\mathcal{K}_p$, give, for $z < 1$,

$$\begin{aligned} & \int_{\mathbb{R}} \mathcal{K}_p(\tau, z) d\tau = A_p(z), \\ & \int_{\mathbb{R}} \cosh(\beta\tau) \mathcal{K}_p(\tau, z) d\tau = \int_{\mathbb{R}} e^{\beta\tau} \mathcal{K}_p(\tau, z) d\tau = B_p(z), \\ & \int_{\mathbb{R}} [\cosh(\beta\tau) - 1] \mathcal{K}_p(\tau, z) d\tau = W_p(z). \end{aligned}$$

Integrating the combined bracket therefore gives the exact identity

$$\begin{aligned} \frac{Q[\zeta_R]}{N_R} &= \frac{1}{2} \iint_{\Gamma^2} A_p(x \cdot y) (f_x - f_y)^2 d\mu_x d\mu_y + \int_{\Gamma} f_x^2 \int_{\Gamma} W_p(x \cdot y) d\mu_y d\mu_x \\ &\quad - \int_{\Gamma} P_x f_x^2 d\mu_x + \mathcal{E}_R[f], \\ \mathcal{E}_R[f] &= \int_{\mathbb{R}} \iint_{\Gamma^2} \mathcal{K}_p(\tau, x \cdot y) [1 - c_R(\tau)] f_x f_y d\mu_x d\mu_y d\tau. \end{aligned}$$

The error obeys $|\mathcal{E}_R[f]| \leq C \|f\|_{\infty}^2 M_{p,\Gamma}/R \rightarrow 0$ for fixed p and Γ . Using $A_p = B_p - W_p$ and symmetry, the three main terms equal $q_B[f] + \iint_{\Gamma^2} W_p(x \cdot y) f_x f_y d\mu_x d\mu_y = q_A[f]$; all these integrals are finite by the estimates already proved. Thus $Q[\zeta_R]/N_R \rightarrow q_A[f]$, and stability proves $q_A[f] \geq 0$.

Finally $q_B[1] = -P$, so $0 \leq q_A[1] = -P + \iint_{\Gamma^2} W_p$ proves the first consequence. With $h = \nu \cdot e$, the weak Jacobi identity with $f = 1$ and $f = h$ gives $q_B(h, 1) = q_B[h] = 0$. Polarization then yields $q_B[1 - h] = q_B[1] - 2q_B(1, h) + q_B[h] = -P$, as asserted. $\square$

3. TWO EXACT KERNEL COMPUTATIONS

Lemma 3.1 (A lower ratio for the angular kernels). *Use A_p, B_p from Lemma 2.1 and put $b_p = 2/[(1+p)(2+p)]$. For $0 < p < 1$,*

$$\frac{A_p(z)}{B_p(z)} \geq \kappa_p := \frac{\Gamma((3+p)/2)^2}{\Gamma(1+p)} \geq \frac{\pi}{4} \quad (-1 \leq z < 1).$$

Proof. The radial inversion used in Lemma 2.1 gives, with $D_z(\rho) = 1 + \rho^2 - 2\rho z$ and $M = (3+p)/2$,

$$A_p(z) = \int_0^1 \frac{2\rho^{(1+p)/2}}{D_z(\rho)^M} d\rho, \quad B_p(z) = \int_0^1 \frac{\rho + \rho^p}{D_z(\rho)^M} d\rho.$$

Define $w_z(\rho) = (\rho + \rho^p) D_z(\rho)^{-M}$ and $d\pi_z(\rho) = w_z(\rho) d\rho / B_p(z)$. The preceding lemma gives $0 < B_p(z) < \infty$, so π_z is a probability measure on $(0, 1)$. Thus A_p/B_p is the expectation of

$$F_p(\rho) = \frac{2\rho^{(1+p)/2}}{\rho + \rho^p} = \operatorname{sech} \left(\frac{1-p}{2} \log \rho \right).$$

This function is increasing in ρ : with $k = 1 - p > 0$,

$$F'_p(\rho) = -\frac{k}{2\rho} \operatorname{sech} \left(\frac{k}{2} \log \rho \right) \tanh \left(\frac{k}{2} \log \rho \right) > 0 \quad (0 < \rho < 1).$$

The logarithmic derivative of the unnormalized density is

$$S_z(\rho) := \partial_z \log w_z(\rho) = (3+p) \frac{\rho}{D_z(\rho)}, \quad \partial_\rho S_z(\rho) = (3+p) \frac{1 - \rho^2}{D_z(\rho)^2} > 0.$$

For z in a compact subinterval of $[-1, 1)$, $D_z(\rho)$ has a positive lower bound on $0 \leq \rho \leq 1$. Since $0 < F_p \leq 1$, the integrands $w_z, F_p w_z$ and their z derivatives are bounded by $C(\rho + \rho^p)$, which is integrable. Their integrals can therefore be differentiated under the integral sign, and are continuous also at $z = -1$. Quotient differentiation now gives

$$\begin{aligned} \frac{d}{dz} \frac{A_p(z)}{B_p(z)} &= \int F_p S_z d\pi_z - \left(\int F_p d\pi_z \right) \left(\int S_z d\pi_z \right) \\ &= \frac{1}{2} \iint [F_p(\rho) - F_p(\sigma)] [S_z(\rho) - S_z(\sigma)] d\pi_z(\rho) d\pi_z(\sigma) \geq 0. \end{aligned}$$

All terms are finite; the last sign follows from the common monotonicity of F_p and S_z . Hence A_p/B_p is increasing in z , and its minimum on $[-1, 1)$ is attained at $z = -1$.

At $z = -1$, the original radial denominators are $(1 + \rho)^{3+p}$. The beta identity

$$\int_0^\infty \frac{\rho^{a-1}}{(1 + \rho)^{a+b}} d\rho = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad (a, b > 0)$$

applies with $(a, b) = (M, M)$ for A_p and $(2, 1 + p)$ for B_p . It gives

$$A_p(-1) = \frac{\Gamma((3+p)/2)^2}{\Gamma(3+p)}, \quad B_p(-1) = \frac{\Gamma(1+p)}{\Gamma(3+p)},$$

and their ratio is exactly κ_p . Writing $\psi = \Gamma'/\Gamma$, the logarithmic derivative is

$$\frac{d}{dp} \log \kappa_p = \psi((3+p)/2) - \psi(1+p) \geq 0,$$

because $(3+p)/2 \geq 1+p$ for $0 \leq p \leq 1$ and ψ is increasing. The latter follows from $\psi'(x) = \sum_{j=0}^\infty (j+x)^{-2} > 0$ for $x > 0$; the series converges locally uniformly on that range. The gamma quotient extends continuously to $p = 0$, where $\kappa_0 = \Gamma(3/2)^2/\Gamma(1) = (\sqrt{\pi}/2)^2 = \pi/4$. Therefore $\kappa_p \geq \pi/4$. □

Lemma 3.2 (The exact equatorial row). *For the kernels in Lemma 2.1 let*

$$H_p = \int_0^{2\pi} W_p(\cos \theta) d\theta, \quad b_p = \frac{2}{(1+p)(2+p)}.$$

Then

$$H_p = \frac{2\pi\Gamma((3+p)/4)^2\Gamma((1-p)/2)}{(1+p)\Gamma((3+p)/2)\Gamma((1-p)/4)^2}.$$

Proof. We first compute the equatorial row. Put $\alpha = 1+p \in (1, 2)$, $\beta = (2-\alpha)/2 = (1-p)/2$, and $v_\gamma(x) = |x|^{-\gamma}$. For the unnormalized fractional operator on $\mathbb{R}^2$, use the physical principal value

$$\mathcal{L}_\alpha v(x) = \lim_{\varepsilon \downarrow 0} \int_{|x-y|>\varepsilon} \frac{v(x) - v(y)}{|x-y|^{2+\alpha}} dy.$$

For $v = v_\beta$ this exists at every $x \neq 0$: the linear Taylor term cancels on centered annuli, and the quadratic remainder has radial integral $C \int_0^1 r^{1-\alpha} dr < \infty$. The singularity of v_β at $y = 0$ is locally integrable because $\beta < 2$, and the tail is integrable because $\alpha > 0$.

At $x = e_1$, polar coordinates give the radial numerator $\rho(1 - \rho^{-\beta})$ over $D(\rho, \theta)^{(2+\alpha)/2}$, where $D = 1 + \rho^2 - 2\rho \cos \theta$. Split at $\rho = 1$ and invert the outer part. Its numerator becomes $\rho^{\alpha-1}(1 - \rho^\beta)$ on $(0, 1)$, since the radial measure and denominator contribute the factor $\rho^{\alpha-1}$ after inversion. The outer cutoff $|e_1 - y| > \varepsilon$ becomes the inner cutoff $|e_1 - y| > \varepsilon|y|$. Its difference from the original inner cutoff is confined to

$$\varepsilon\rho < r := |e_1 - y| \leq \varepsilon, \quad 0 < \rho = |y| < 1.$$

Here $\rho \geq 1 - r$, hence $\varepsilon/(1 + \varepsilon) < r \leq \varepsilon$. The shell therefore has thickness $O(\varepsilon^2)$ and area $O(\varepsilon^3)$. In it ρ is bounded away from zero, the transformed numerator is $O(1 - \rho) = O(r)$, and the denominator is $r^{2+\alpha}$. The cutoff discrepancy is consequently bounded by $C\varepsilon^{2-\alpha} \rightarrow 0$. After replacing both cutoffs by the common physical cutoff, the numerators combine to

$$\begin{aligned} \rho(1 - \rho^{-\beta}) + \rho^{\alpha-1}(1 - \rho^\beta) &= \rho + \rho^p - 2\rho^{(1+p)/2} \\ &= (\rho^{1/2} - \rho^{p/2})^2 \geq 0. \end{aligned}$$

Near $\rho = 1$, $\theta = 0$ modulo 2π , this numerator is $O((1 - \rho)^2)$ and $D \asymp (1 - \rho)^2 + \theta^2$. Its two-variable singular integral is bounded by $C \int_0^1 r^{-p} dr < \infty$. At $\rho = 0$ the numerator is $O(\rho^p)$; elsewhere the integrand is bounded. The paired integral is therefore absolutely convergent. The radial definitions in Lemma 2.1 and Tonelli's theorem now give

$$\mathcal{L}_\alpha v_\beta(e_1) = \int_0^{2\pi} \int_0^1 \frac{\rho + \rho^p - 2\rho^{(1+p)/2}}{(1 + \rho^2 - 2\rho \cos \theta)^{(3+p)/2}} d\rho d\theta = H_p.$$

This also proves that H_p is finite.

For completeness, we compute the power using the Fourier convention $\widehat{v}(\xi) = \int e^{-ix \cdot \xi} v(x) dx$, with inverse factor $(2\pi)^{-2}$ and multiplier $|\xi|^\alpha$ for $(-\Delta)^{\alpha/2}$. For $0 < \gamma < 2$, v_γ is a locally integrable tempered distribution, and the gamma integral gives

$$|x|^{-\gamma} = \frac{1}{\Gamma(\gamma/2)} \int_0^\infty t^{\gamma/2-1} e^{-t|x|^2} dt.$$

The Gaussian transform is $\widehat{e^{-t|x|^2}}(\xi) = \pi t^{-1} e^{-|\xi|^2/(4t)}$. When paired with a Schwartz test function, the Gaussian integrals are absolutely bounded by $C \min(1, t^{-1})$, on either the spatial side or the Fourier side with its prefactor t^{-1} included. Thus the t integrals converge at zero because $\gamma > 0$ and at infinity because $\gamma < 2$; Fubini is justified. Setting $u = |\xi|^2/(4t)$ gives

$$\begin{aligned} \widehat{v}_\gamma(\xi) &= \frac{\pi}{\Gamma(\gamma/2)} \int_0^\infty t^{\gamma/2-2} e^{-|\xi|^2/(4t)} dt \\ &= G(\gamma) |\xi|^{\gamma-2}, \quad G(\gamma) = 2^{2-\gamma} \pi \frac{\Gamma(1-\gamma/2)}{\Gamma(\gamma/2)}. \end{aligned}$$

The formula holds as tempered distributions: the displayed function is locally integrable at $\xi = 0$ because $\gamma > 0$, and grows at most polynomially at infinity. Both $\beta = (2 - \alpha)/2$ and $\beta + \alpha = (2 + \alpha)/2$ lie in $(0, 2)$. Here multiplication by $|\xi|^\alpha$ is an ordinary product of locally integrable functions, yielding $G(\beta) |\xi|^{\beta+\alpha-2}$, which is again locally integrable. Division by the transform constant $G(\beta + \alpha)$ therefore gives the Fourier-defined distribution

$$(-\Delta)^{\alpha/2} v_\beta = \frac{G(\beta)}{G(\beta + \alpha)} v_{\beta+\alpha} = 2^\alpha \frac{\Gamma((2 + \alpha)/4)^2}{\Gamma((2 - \alpha)/4)^2} |x|^{-\beta-\alpha}.$$

The principal-value normalization follows from its Fourier symbol. For a smooth compactly supported function ϕ , the symmetric second-difference formula is

$$\mathcal{L}_\alpha \phi(x) = \frac{1}{2} \int_{\mathbb{R}^2} \frac{2\phi(x) - \phi(x+h) - \phi(x-h)}{|h|^{2+\alpha}} dh.$$

The L^1 norm in x of the numerator is at most $C \min(|h|^2, 1)$, so Fubini gives

$$\widehat{\mathcal{L}_\alpha \phi}(\xi) = J_\alpha |\xi|^\alpha \widehat{\phi}(\xi).$$

For $a = \alpha/2 \in (0, 1)$, the Gaussian representation evaluates the symbol:

$$\begin{aligned} J_\alpha |\xi|^\alpha &:= \int_{\mathbb{R}^2} \frac{1 - \cos(\xi \cdot y)}{|y|^{2+\alpha}} dy = \frac{\pi}{\Gamma(1+a)} \int_0^\infty t^{a-1} \left(1 - e^{-|\xi|^2/(4t)}\right) dt \\ &= \frac{\pi \Gamma(1-a)}{a 4^a \Gamma(1+a)} |\xi|^\alpha. \end{aligned}$$

Tonelli applies because $1 - \cos(\xi \cdot y)$ is nonnegative. For $\xi \neq 0$, set $u = |\xi|^2/(4t)$ and integrate by parts:

$$\int_0^\infty (1 - e^{-u}) u^{-1-a} du = \frac{1}{a} \int_0^\infty e^{-u} u^{-a} du = \frac{\Gamma(1-a)}{a}.$$

The boundary terms vanish because $1 - e^{-u} = O(u)$ at zero and $a > 0$ at infinity; the case $\xi = 0$ is immediate. Consequently $c_{2,\alpha} = J_\alpha^{-1}$, namely

$$c_{2,\alpha} = \frac{\alpha 2^{\alpha-1} \Gamma(1 + \alpha/2)}{\pi \Gamma(1 - \alpha/2)}, \quad (-\Delta)^{\alpha/2} \phi = c_{2,\alpha} \mathcal{L}_\alpha \phi.$$

We finally justify the same identity for the singular power. Write $\mathcal{L}_{\alpha,\varepsilon}$ for the integral truncated at $|x - y| > \varepsilon$. Its symmetric formula, scaling, and rotation give

$$|\mathcal{L}_{\alpha,\varepsilon} v_\beta(x)| \leq C |x|^{-\beta-\alpha} \quad (x \neq 0),$$

uniformly in $\varepsilon > 0$. Indeed the full absolute symmetric integral at $x = e_1$ is finite: the numerator is $O(|h|^2)$ at $h = 0$, the singularities at $h = \pm e_1$ are integrable because $\beta < 2$, and

the tail is integrable because $\alpha > 0$. Since $\beta + \alpha < 2$, this bound is locally integrable in x . For $\phi \in C_c^\infty(\mathbb{R}^2)$, the symmetric formula bounds $\mathcal{L}_{\alpha,\varepsilon}\phi$ on bounded sets; outside a fixed large disk it is $-\int \phi(y)|x-y|^{-2-\alpha} dy$ for $\varepsilon < 1$. Hence $|\mathcal{L}_{\alpha,\varepsilon}\phi(x)| \leq C_\phi(1+|x|)^{-2-\alpha}$ uniformly for $\varepsilon < 1$. Symmetry and Fubini at a fixed cutoff, followed by dominated convergence on both sides, give

$$\int \phi \mathcal{L}_\alpha v_\beta = \lim_{\varepsilon \downarrow 0} \int \phi \mathcal{L}_{\alpha,\varepsilon} v_\beta = \lim_{\varepsilon \downarrow 0} \int v_\beta \mathcal{L}_{\alpha,\varepsilon} \phi = \int v_\beta \mathcal{L}_\alpha \phi.$$

To evaluate the last pairing, insert the Gaussian representation of v_β and Fourier inversion for $\mathcal{L}_\alpha \phi$. The spatial interchange is dominated by $\int v_\beta |\mathcal{L}_\alpha \phi| < \infty$; the Fourier interchange is dominated, after integrating the Gaussian parameter, by $\int |\xi|^{\beta+\alpha-2} |\widehat{\phi}(-\xi)| d\xi < \infty$. Here integrability at zero uses $\beta + \alpha > 0$, and the Fourier test function decays rapidly at infinity. It follows that

$$\begin{aligned} \int v_\beta \mathcal{L}_\alpha \phi &= \frac{J_\alpha G(\beta)}{(2\pi)^2} \int_{\mathbb{R}^2} |\xi|^{\beta+\alpha-2} \widehat{\phi}(-\xi) d\xi \\ &= J_\alpha \frac{G(\beta)}{G(\beta+\alpha)} \int v_{\beta+\alpha} \phi. \end{aligned}$$

Thus the physical principal value has exactly the Fourier-computed coefficient as a locally integrable distribution. Local Taylor bounds, together with integrability at $y = 0$ and infinity, also make the principal-value function continuous away from zero. The identity therefore holds pointwise there. Evaluating at e_1 and using $H_p = \mathcal{L}_\alpha v_\beta(e_1)$ gives

$$\begin{aligned} H_p &= \frac{2\pi\Gamma((2+\alpha)/4)^2\Gamma(1-\alpha/2)}{\alpha\Gamma(1+\alpha/2)\Gamma((2-\alpha)/4)^2} \\ &= \frac{2\pi\Gamma((3+p)/4)^2\Gamma((1-p)/2)}{(1+p)\Gamma((3+p)/2)\Gamma((1-p)/4)^2}, \end{aligned}$$

which is the asserted formula. □

4. SPHERICAL AREA AND AN ANGULAR ROW ESTIMATE

Lemma 4.1 (A spherical Green identity). *Let Ω be a nonempty proper smooth open set of S^2 , let $\Gamma = \partial\Omega$, and orient its spherical unit conormal ν outward from Ω . All boundary loops are included. Write $A = |\Omega|$ and $L = |\Gamma|$. Then, for every $x \in \Gamma$,*

$$\int_\Gamma \frac{-x \cdot \nu_y}{1 - x \cdot y} d\mu_y = 2\pi - A,$$

and

$$\iint_{\Gamma^2} \frac{1 - \nu_x \cdot \nu_y}{1 - x \cdot y} d\mu_x d\mu_y \geq (2\pi - A)^2.$$

For the kernel B_p of Lemma 2.1, with $b_p = 2/[(1+p)(2+p)]$, it follows that

$$P_p(\Gamma) := \iint_{\Gamma^2} B_p(x \cdot y)(1 - \nu_x \cdot \nu_y) d\mu_x d\mu_y \geq b_p(2\pi - A)^2.$$

The middle inequality is an equality for every spherical cap.

Proof. Fix $x \in \Gamma$ and set $f_x(y) = \log(1 - x \cdot y)$. If r is spherical distance from x , then $f'_x(r) = \cot(r/2)$ and

$$\Delta_{S^2} f_x = \frac{1}{\sin r} \frac{d}{dr} [\sin r \cot(r/2)] = -1 \quad (0 < r < \pi).$$

The function is smooth at the antipode, so this identity holds everywhere away from x . Furthermore

$$\partial_{\nu_y} f_x(y) = \frac{-x \cdot \nu_y}{1 - x \cdot y}.$$

To check the singularity, parametrize the boundary by unit-speed $y = \gamma(s)$ with $\gamma(0) = x$. Differentiating $\gamma(s) \cdot \nu_{\gamma(s)} = 0$ and using $\gamma'(0) \cdot \nu_x = 0$ gives $x \cdot (d\nu_{\gamma(s)}/ds)_{s=0} = 0$. Hence $x \cdot \nu_{\gamma(s)} = O(s^2) = O(r^2)$, while $1 - x \cdot \gamma(s)$ is comparable to s^2 . Thus this boundary integrand is bounded there and has an ordinary improper integral, with no principal value required. In geodesic normal coordinates at x , the boundary is a graph with height $O(|z|^2)$ over its tangent line, and Ω lies on one side. On the circle of radius ρ , the angular measure of the part in Ω is therefore $\pi + O(\rho)$. Apply Green's formula to $\Omega \setminus B_\rho^{S^2}(x)$ at sufficiently small regular radii. The added circular arc has length $\sin \rho[\pi + O(\rho)] = \pi\rho + O(\rho^2)$, and its outward conormal points toward x . Its flux is therefore $-\cot(\rho/2)[\pi\rho + O(\rho^2)] \rightarrow -2\pi$. The area integral tends to $-A$. Letting $\rho \downarrow 0$ proves the first formula. Each boundary loop retains its outward orientation; holes do not change this calculation.

Integrate this formula in x and symmetrize. Using $x \cdot \nu_x = y \cdot \nu_y = 0$ gives

$$L(2\pi - A) = \frac{1}{2} \iint_{\Gamma^2} \frac{(x - y) \cdot (\nu_x - \nu_y)}{1 - x \cdot y} d\mu_x d\mu_y.$$

Cauchy-Schwarz, $|x - y|^2 = 2(1 - x \cdot y)$, and $|\nu_x - \nu_y|^2 = 2(1 - \nu_x \cdot \nu_y)$ give

$$\begin{aligned} L^2(2\pi - A)^2 &\leq \left(\iint_{\Gamma^2} \frac{|x - y|^2}{2(1 - x \cdot y)} d\mu_x d\mu_y \right) \left(\iint_{\Gamma^2} \frac{|\nu_x - \nu_y|^2}{2(1 - x \cdot y)} d\mu_x d\mu_y \right) \\ &= L^2 \iint_{\Gamma^2} \frac{1 - \nu_x \cdot \nu_y}{1 - x \cdot y} d\mu_x d\mu_y. \end{aligned}$$

All integrals converge: the normal difference is first order along a smooth loop, and different compact loops have positive separation. The boundary is a nonempty compact smooth embedded one-dimensional manifold, so it has finitely many loops and $0 < L < \infty$. Divide by L^2 . The lower bound for B_p in Lemma 2.1 proves the last inequality; its integrand is nonnegative. For a cap $x \cdot e > h$, with $|e| = 1$ and $-1 < h < 1$, its boundary normal is $\nu_x = (hx - e)/\sqrt{1 - h^2}$. Hence the ratio in the middle inequality is $h^2/(1 - h^2)$, its length is $2\pi\sqrt{1 - h^2}$, and $2\pi - A = 2\pi h$. Substitution proves equality, also for $h = 0$. $\square$

Lemma 4.2 (An estimate for arbitrary smooth links). *Let $\Omega \subset S^2$ be a nonempty proper smooth open set, $\Gamma = \partial\Omega$, and ν its outward conormal. No convexity or connectivity is assumed. For $p \in (0, 1)$ use the kernels and b_p of Lemma 2.1, and put*

$$P_x = \int_{\Gamma} B_p(x \cdot y)(1 - \nu_x \cdot \nu_y) d\mu_y, \quad w_p(r) = W_p(1 - r^2/2).$$

There exist explicit constants

$$\eta_p = \frac{1}{b_p} \int_0^2 r w_p(r) dr < \frac{19}{50},$$

$$K_p = H_p + \int_{\sqrt{2}}^2 [-w'_p(r)][-2\sin(2\theta(r))] dr, \quad \theta(r) = 2\arcsin(r/2),$$

where H_p is the equatorial row from Lemma 3.2, such that

$$\int_{\Gamma} W_p(x \cdot y) d\mu_y \leq K_p + \eta_p P_x \quad (x \in \Gamma).$$

For $1/2 \leq p < 1$ one has $K_p/b_p < 1781/1792 < 1$. Consequently a stable stationary cone with this link satisfies

$$P := \int_{\Gamma} P_x d\mu_x < \frac{50}{31} b_p |\Gamma| \quad (1/2 \leq p < 1).$$

Proof. Fix $x \in \Gamma$, and let a cap centered at x have spherical radius $\theta \in (0, \pi)$ and chord radius $r = 2 \sin(\theta/2)$. Set $m_r = \mu(\Gamma \cap B_r(x))$ and $v_r = \int_{\Gamma \cap B_r(x)} \nu d\mu$. The direction $e = \nu_x$ is tangent to the sphere at x . Use the tangential vector field $e^T(y) = e - (e \cdot y)y$, for which $\operatorname{div}_{S^2} e^T = -2e \cdot y$ and $e^T \cdot \nu = e \cdot \nu$ on every spherical boundary arc. The spherical divergence theorem on the cap intersection gives

$$e \cdot v_r = -2 \int_{\Omega \cap B_r(x)} e \cdot y d\omega_y - \int_{\Omega \cap \partial B_r(x)} e \cdot \nu_{\text{cap}} d\mu.$$

The link is a compact smooth embedded one-dimensional manifold, so it consists of finitely many loops and has finite arclength. For a regular value $r^2 \in (0, 4)$ of the smooth function $y \mapsto |x - y|^2$ on Γ , the cap circle meets the link transversely at finitely many points. The boundary of $\Omega \cap B_r(x)$ is then a finite union of smooth arcs with corners, so the divergence theorem applies. Sard's theorem gives a dense set of such radii. All boundary components retain the outward conormal of Ω , including components surrounding holes. The bounds below extend to every open cap by choosing regular radii $r_j \uparrow r$: the cap indicators increase to that of $B_r(x)$, so $m_{r_j} \rightarrow m_r$ and $v_{r_j} \rightarrow v_r$, the latter by dominated convergence since $|\nu| = 1$ and $\mu(\Gamma) < \infty$. Regular radii decreasing to r give the corresponding closed-cap bounds if needed. Thus no zero-arclength assumption on an exceptional cap circle is required. Choose polar coordinates about x so that $e \cdot y = \sin t \cos \varphi$ and $e \cdot \nu_{\text{cap}} = \cos \theta \cos \varphi$. The largest possible contribution of the first signed integral over an arbitrary subset is its positive part on the entire cap:

$$2 \int_0^\theta \sin^2 t dt \int_0^{2\pi} (-\cos \varphi)_+ d\varphi = 4 \int_0^\theta \sin^2 t dt = 2\theta - \sin 2\theta.$$

Here $a_+ = \max\{a, 0\}$. The positive part of the second signed integral on the entire cap circle is $\sin \theta \int_0^{2\pi} (-\cos \theta \cos \varphi)_+ d\varphi = 2 \sin \theta |\cos \theta| = |\sin 2\theta|$. Thus

$$\nu_x \cdot v_r \leq G(\theta) := \begin{cases} 2\theta, & 0 \leq \theta \leq \pi/2, \\ 2\theta - 2 \sin 2\theta, & \pi/2 \leq \theta \leq \pi. \end{cases}$$

The use of the particular tangent direction ν_x is essential; this is not an estimate for $|v_r|$.

Since $B_p(x \cdot y) \geq 2b_p/r^2$ on the cap,

$$m_r - \nu_x \cdot v_r = \int_{\Gamma \cap B_r(x)} (1 - \nu_x \cdot \nu_y) d\mu_y \leq \frac{r^2}{2b_p} P_x.$$

Consequently $m_r \leq G(\theta(r)) + r^2 P_x / (2b_p)$. Letting $r \uparrow 2$ gives $|\Gamma| \leq 2\pi + 2P_x/b_p$: the only point omitted by the open chord cap of radius two is the antipode of x , which has zero arclength. Write $d = |x - y|$ and $m = (3 + p)/2$. In the defining radial integral,

$$1 + \rho^2 - 2\rho x \cdot y = (\rho - 1)^2 + \rho d^2.$$

For $0 < \rho < 1/2$ this denominator is at least $1/4$, and for $\rho > 2$ it is at least $\rho^2/4$. The corresponding integrals of $\rho[(\rho - 1)^2 + \rho d^2]^{-m}$ are therefore bounded by $4^m \int_0^{1/2} \rho d\rho$ and $4^m \int_2^\infty \rho^{-2-p} d\rho$, respectively, uniformly in x, y . On $1/2 \leq \rho \leq 2$, the same integral is at most

$$2 \int_{\mathbb{R}} (t^2 + d^2/2)^{-m} dt = \mathcal{C}'_p d^{-2-p},$$

by $t = dv$. The integral defining $\mathcal{C}'_p$ is finite because $2m > 1$. Since $0 < d \leq 2$, the two bounded tail contributions can be absorbed by enlarging this constant to a finite $\mathcal{C}_p$. Thus $B_p(x \cdot y) \leq \mathcal{C}_p d^{-2-p}$. The positive representation and $F_{1-p}(u) \leq (1-p)^2 u^2/4$ also give, with $w_p(r) = W_p(1 - r^2/2)$,

$$w_p(r) \leq \frac{(1-p)^2}{4} \int_0^\infty \frac{u^2 du}{(u^2 + r^2)^{(3+p)/2}} = \frac{(1-p)^2}{4} r^{-p} \int_0^\infty \frac{t^2 dt}{(1+t^2)^{(3+p)/2}}.$$

The last integral is finite: its integrand is $O(t^2)$ at zero and $O(t^{-1-p})$ at infinity. Thus $w_p(r) \leq \mathcal{D}_p r^{-p}$ for a finite constant $\mathcal{D}_p$ depending only on p . On every interval $\varepsilon \leq r \leq 2$, the bound $F_{1-p}(u) \leq \min\{u^2/4, 2\}$ supplies integrable majorants for the positive representation and its r derivative. Dominated differentiation therefore gives

$$-w'_p(r) = (3+p)r \int_0^\infty \frac{F_{1-p}(u)}{(u^2 + r^2)^{(5+p)/2}} du \geq 0.$$

Using $F_{1-p}(u) \leq (1-p)^2 u^2/4$ and substituting $u = rt$ also gives $-w'_p(r) \leq \mathcal{E}_p r^{-1-p}$; the remaining integral $\int_0^\infty t^2(1+t^2)^{-(5+p)/2} dt$ is finite. Smoothness gives $1 - \nu_x \cdot \nu_y = \frac{1}{2} |\nu_x - \nu_y|^2 \leq Cd^2$ in every sufficiently small link chart. Chord distance and arclength separation are comparable on such a chart. Consequently the diagonal integrand defining P_x is bounded by Cd^{-p} and is integrable because $p < 1$. The complement of the diagonal charts is compact and separated from the diagonal, including pairs from distinct link components; its contribution is finite. The chart constants and off-diagonal separation can be chosen uniformly for $x \in \Gamma$ by compactness. Hence $\sup_{x \in \Gamma} P_x < \infty$, and $P = \int_\Gamma P_x d\mu_x < \infty$. For $0 < d \leq 2$, write $w_p(d) = w_p(2) + \int_d^2 [-w'_p(r)] dr$. Integrating this identity over $y \in \Gamma$ and using Tonelli's theorem gives the layer formula below. It uses open caps: the endpoint $r = d$ has zero Lebesgue measure in the inner integral, even if a cap circle contains a boundary arc. Since $m_r = O_x(r)$ and $-w'_p(r) = O_p(r^{-1-p})$ at zero, the layer integral is finite for $p < 1$. Thus

$$\begin{aligned} \int_\Gamma W_p(x \cdot y) d\mu_y &= w_p(2)|\Gamma| + \int_0^2 [-w'_p(r)] m_r dr \\ &\leq 2\pi w_p(2) + \int_0^2 [-w'_p(r)] G(\theta(r)) dr + \eta_p P_x. \end{aligned}$$

The estimate $m_r = O_x(r)$ follows from $G(\theta(r)) = O(r)$ and the preceding cap bound. Equivalently, Stieltjes integration by parts has lower endpoint $w_p(r)m_r = O_{x,p}(r^{1-p}) \rightarrow 0$. The upper endpoint is exactly $w_p(2)|\Gamma|$. The coefficient of P_x in the estimate is

$$\frac{P_x}{2b_p} \left[4w_p(2) + \int_0^2 r^2 [-w'_p(r)] dr \right] = \frac{P_x}{b_p} \int_0^2 r w_p(r) dr = \eta_p P_x.$$

Here integration by parts cancels the terms at $r = 2$, while $r^2 w_p(r) \rightarrow 0$ at zero. This computation retains both endpoint contributions. For a great circle through x its cap length is $2\theta(r)$, so its identical layer formula identifies the part with G replaced by 2θ as H_p . This proves the asserted row estimate and formula for K_p .

Here are uniform bounds for its constants. Use the exact representation from Lemma 2.1,

$$w_p(r) = \int_0^\infty F_{1-p}(u)(u^2 + r^2)^{-(3+p)/2} du, \quad F_{1-p} \leq (1-p)^2 F_1, \quad F_1 = 2 - \frac{4}{\sqrt{u^2 + 4}}.$$

Define w_0 by this formula with $p = 0$, solely as a comparison kernel. Here A_0, B_0 mean the radial integrals in Lemma 2.1 with $p = 0$. For $0 \leq p \leq 1$, one has $F_{1-p}(u) \leq F_1(u) \leq \min\{u^2/4, 2\}$. On every interval $\varepsilon \leq r \leq 2$, the integrand defining w_p and the absolute value of its r derivative are bounded by $C_\varepsilon u^2$ for $0 < u < 1$, and respectively by $2u^{-3}$ and $16u^{-5}$ for $u \geq 1$, uniformly in p . These integrable bounds justify dominated differentiation, show local C^1 regularity also at $p = 0$, and permit passage to $p \downarrow 0$ in the integral and its derivative. For each fixed $z < 1$, the radial integrands defining $A_p(z)$ and $B_p(z)$ are uniformly integrable for $0 \leq p \leq 1$: near zero they are bounded by constant multiples of $\rho^{1/2}$ and ρ , respectively; at infinity by $\rho^{-5/2}$ and ρ^{-2} ; and on intervening compact radial intervals their denominators have a positive lower bound. Thus $A_p(z) \rightarrow A_0(z)$ and $B_p(z) \rightarrow B_0(z)$. Taking the limit in $w_p(r) = B_p(1 - r^2/2) - A_p(1 - r^2/2)$ proves

$$w_0(r) = B_0(1 - r^2/2) - A_0(1 - r^2/2), \quad 0 < r \leq 2.$$

Split the positive integral defining η_p into the quadrant disk $u^2 + r^2 < 1$ and its complement in $u > 0$, $0 < r < 2$. On the disk $u^2/5 \leq F_1(u) \leq u^2/4$. The extra factor r in η_p makes the

polar angular integral equal to $\int_0^{\pi/2} \cos^2 t \sin t dt = 1/3$. Thus the normalized positive-order contribution is at most

$$\frac{(1-p)^2}{12b_p(2-p)} \leq \frac{1}{24},$$

where the last inequality follows from

$$(1-p)^2(1+p)(2+p) - (2-p) = p^2(p^2 + p - 3) < 0.$$

Indeed, in coordinates $u = s \cos t$, $r = s \sin t$, the radial integral is $\int_0^1 s^{1-p} ds = (2-p)^{-1}$. The order-zero contribution on the disk is at least $(1/5)(1/3) \int_0^1 s ds = 1/30$. Off the disk, $(u^2 + r^2)^{-p/2} \leq 1$ and $(1-p)^2/b_p \leq 1$, because the preceding polynomial identity gives $(1-p)^2(1+p)(2+p) < 2-p < 2$. Thus the off-disk contribution to η_p is bounded by the order-zero off-disk contribution, whereas on the disk the excess is at most $1/24 - 1/30 = 1/120$. Hence

$$\eta_p \leq \int_0^2 r w_0(r) dr + \frac{1}{120} = \pi - 4 \log 2 + \frac{1}{120} < \frac{19}{50}.$$

For the equality, integration in r gives $\int_0^\infty F_1(u)[u^{-1} - (u^2 + 4)^{-1/2}] du$. Set $u = 2 \sinh t$, then $v = e^{-t}$; it becomes

$$\int_0^1 \frac{4v(1-v)}{(1+v)(1+v^2)} dv = \int_0^1 \left(\frac{4}{1+v^2} - \frac{4}{1+v} \right) dv.$$

The first three positive terms of $\log 2 = 2 \sum_{j \geq 0} 3^{-2j-1}/(2j+1)$ give $\log 2 > 842/1215$. Together with $\pi < 22/7$ this gives

$$\pi - 4 \log 2 + \frac{1}{120} < \frac{25799}{68040} = \frac{19}{50} - \frac{281}{340200} < \frac{19}{50}.$$

Suppose now $p \geq 1/2$. Duplication in the exact formula of Lemma 3.2 gives, for $a = (3+p)/4$ and $d = (1-p)/4$,

$$\frac{H_p}{b_p} = \pi(2+p)2^{-1-p} \frac{\Gamma(a)}{\Gamma(a+1/2)} \frac{\Gamma(d+1/2)}{\Gamma(d)}.$$

All gamma arguments are positive. Log convexity of Γ , which follows from Hölder's inequality in its Euler integral, gives

$$\Gamma(a+1)^2 \leq \Gamma(a+1/2)\Gamma(a+3/2) = (a+1/2)\Gamma(a+1/2)^2,$$

and therefore $\Gamma(a)/\Gamma(a+1/2) \leq \sqrt{a+1/2}/a$. Write $\psi = \Gamma'/\Gamma$. Its derivative satisfies $\psi'(t) = \sum_{j=0}^\infty (j+t)^{-2} > 0$ for $t > 0$, with locally uniform convergence. Thus the logarithmic derivative of $\Gamma(d+1/2)/\Gamma(d+1)$ is $\psi(d+1/2) - \psi(d+1) < 0$. This quotient extends continuously to $d = 0$ with value $\sqrt{\pi}$, so $\Gamma(d+1/2)/\Gamma(d) \leq d\sqrt{\pi}$. We obtain

$$\frac{H_p}{b_p} \leq \pi^{3/2}(2+p)2^{-1-p}d \frac{\sqrt{a+1/2}}{a}.$$

The right-hand side decreases for $p \geq 1/2$, since its logarithmic derivative is

$$\frac{1}{2+p} - \log 2 - \frac{1}{1-p} + \frac{1}{8(a+1/2)} - \frac{1}{4a} < 0:$$

here $1/(2+p) \leq 2/5$, $1/(1-p) \geq 2$, and the difference of the last two terms is negative. At $p = 1/2$ this yields

$$\frac{H_p}{b_p} \leq \frac{5\pi\sqrt{11\pi}}{112} < \frac{6}{7}.$$

For the strict numerical inequality, $\pi < 22/7$ implies

$$\left(\frac{5\pi\sqrt{11\pi}}{112} \right)^2 < \frac{366025}{537824} < \frac{395136}{537824} = \frac{36}{49}.$$

For $r \geq \sqrt{2}$, differentiating the positive integral formula gives

$$\frac{-w'_p(r)}{b_p} \leq D_p[-w'_0(r)], \quad D_p = \frac{(1-p)^2}{b_p} \frac{3+p}{3} \leq \frac{35}{64}.$$

Here the extra factor $(u^2 + r^2)^{-p/2}$ is at most one. The last bound follows by logarithmic differentiation of $(1-p)^2(1+p)(2+p)(3+p)/6$:

$$\frac{d}{dp} \log D_p = -\frac{2}{1-p} + \frac{1}{1+p} + \frac{1}{2+p} + \frac{1}{3+p} \leq -4 + \frac{2}{3} + \frac{2}{5} + \frac{2}{7} < 0 \quad (1/2 \leq p < 1).$$

Evaluation at $p = 1/2$ gives $D_{1/2} = 35/64$. Also $0 \leq -2 \sin(2\theta) \leq 2$ on this interval, whence

$$\frac{K_p - H_p}{b_p} \leq 2D_p[w_0(\sqrt{2}) - w_0(2)] \leq D_p(1 - \pi/4) < \frac{35}{256}.$$

For the second inequality, the order-zero radial integrals give

$$B_0(0) = \int_0^\infty \frac{\rho d\rho}{(1+\rho^2)^{3/2}} = 1, \quad B_0(-1) = \int_0^\infty \frac{\rho d\rho}{(1+\rho)^3} = \frac{1}{2},$$

and the beta integral gives $A_0(-1) = \Gamma(3/2)^2/\Gamma(3) = \pi/8$. Moreover $A_0(0) \geq \pi/4$, by taking $p \downarrow 0$ in Lemma 3.1 at $z = 0$. The radial majorants established above justify this limit. Consequently $w_0(\sqrt{2}) - w_0(2) = 1 - A_0(0) - 1/2 + \pi/8 \leq (1 - \pi/4)/2$. Finally $\pi > 3$ gives the strict last bound. Thus $K_p/b_p < 6/7 + 35/256 = 1781/1792 < 1$.

For a stable stationary cone the angular constant test in Proposition 2.3 gives $P \leq \iint_{\Gamma^2} W_p$. Integrating the row estimate gives $(1 - \eta_p)P \leq K_p|\Gamma|$. Since $1 - \eta_p > 31/50$ and $K_p < b_p$, this yields $P \leq K_p|\Gamma|/(1 - \eta_p) < (50/31)b_p|\Gamma|$, as claimed. $\square$

5. PERIMETER ORDERS FROM ONE THIRD TO ONE

Proposition 5.1 (Power-law normal-flux pinching). *Let $E \subset \mathbb{R}^3$ be a nontrivial stationary stable cone of perimeter order $p \in (0, 1)$ with locally one-sided smooth boundary away from zero and smooth compact link $\Gamma = \partial E \cap S^2$. Let ν be its outward unit normal and put*

$$L = |\Gamma|, \quad V = \int_\Gamma \nu d\mu, \quad \delta = 1 - \frac{|V|}{L}.$$

Define

$$b_p = \frac{2}{(1+p)(2+p)}, \quad a_p = \frac{(1-p)^2 \sqrt{\pi} \Gamma(p/2)}{16\Gamma((3+p)/2)}, \quad K_p^{\text{pin}} = \frac{2^{p+1} a_p}{(1-p)b_p}.$$

If $\delta = 0$, the cone is a half-space. If $0 < \delta < 1$, then necessarily

$$2 - \delta \leq K_p^{\text{pin}} \delta^{1-p} \left[\frac{4}{(1-\delta)^2} + \delta(2-\delta) \right]^{p/2}.$$

In particular, for $p = 1/2$, the condition $|V| \geq 3L/5$ forces the cone to be a half-space.

Proof. Use the kernels of Lemma 2.1 and the angular form of Proposition 2.3. Write

$$P = \iint_{\Gamma^2} B_p(x \cdot y)(1 - \nu_x \cdot \nu_y) d\mu_x d\mu_y.$$

Use the radial definition of B_p from Lemma 2.1. The energies used below are finite. To see this directly, put $d = |x - y|$ and $m = (3+p)/2$. In the radial integral defining B_p , the portions $\rho \leq 1/2$ and $\rho \geq 2$ are bounded uniformly in x, y , while on $[1/2, 2]$

$$1 + \rho^2 - 2\rho x \cdot y = (\rho - 1)^2 + \rho d^2 \geq (\rho - 1)^2 + d^2/2.$$

Scaling $\rho - 1$ therefore gives $B_p(x \cdot y) \leq C_p d^{-2-p}$ for $0 < d \leq 2$. On a compact smooth embedded link, $|\nu_x - \nu_y| \leq Cd$, and arclength and chord distance are comparable in uniform local charts. Thus

$$B_p(x \cdot y)(1 - \nu_x \cdot \nu_y) \leq C_p d^{-p},$$

which is integrable on the link because $p < 1$. The same argument applies to smooth differences $f_x - f_y$. Distinct components have positive separation; their cross terms are bounded. This proves $P < \infty$ and the finiteness of the difference terms in q_B . The bound on W_p established next also makes its row integrals finite. These observations apply to every smooth compact link considered below.

The stated global upper bound for W_p and $u^2 + 8 \geq 8$ imply, with $d = |x - y|$ and $m = (3 + p)/2$,

$$W_p(x \cdot y) \leq \frac{(1 - p)^2}{4} \int_0^\infty \frac{u^2}{(u^2 + d^2)^m} du = a_p d^{-p}.$$

The beta integral used here is $\int_0^\infty t^2(1 + t^2)^{-m} dt = \frac{1}{2}B(3/2, p/2) = \sqrt{\pi}\Gamma(p/2)/[4\Gamma(m)]$, obtained by $u = dt$ and then $s = t^2$. Its beta parameters are positive because $p > 0$.

We record explicitly the translation identity needed for the choice of test. Define

$$q_B[f] = \frac{1}{2} \iint_{\Gamma^2} B_p(x \cdot y)(f_x - f_y)^2 d\mu_x d\mu_y - \int_\Gamma f_x^2 \int_\Gamma B_p(x \cdot y)(1 - \nu_x \cdot \nu_y) d\mu_y d\mu_x.$$

Write $q_B(f_1, f_2)$ for its symmetric bilinear polarization. For a constant vector e , put $h = \nu \cdot e$. The translation-field conclusion of Proposition 2.3 gives exactly $q_B(h, f) = 0$ for every smooth f on Γ . Its hypotheses hold here: the cone is stationary, its boundary is locally one-sided and smooth away from zero, and its link is smooth and compact. Taking $f = h$ and $f = 1$ gives $q_B[h] = q_B(h, 1) = 0$. Since $q_B[1] = -P$, polarization yields $q_B[1 - h] = -P$. Direct expansion of the angular form gives

$$q_A[f] = q_B[f] + \iint_{\Gamma^2} W_p(x \cdot y) f_x f_y d\mu_x d\mu_y.$$

Indeed its difference from $q_B[f]$ is $\iint W_p f_x^2 - \frac{1}{2} \iint W_p (f_x - f_y)^2$; symmetry in x, y reduces this to $\iint W_p f_x f_y$. The finite row bounds established above justify this expansion. When $V \neq 0$, choose $e = V/|V|$ and $g = 1 - h$. Then $0 \leq g \leq 2$, $\int g = L\delta$, and $q_B[g] = -P$. Stability therefore yields

$$P \leq \iint_{\Gamma^2} W_p(x \cdot y) g_x g_y d\mu_x d\mu_y.$$

Here is a local length bound with its dependence on P made explicit. Set $C^2 = 16\pi^2 + 2P/b_p$. For every $x \in S^2$ and $r > 0$,

$$\mu(\Gamma \cap B_r(x)) \leq Cr,$$

where the ball is Euclidean and centered on the sphere. For $r \leq 1$, set $m_r = \mu(\Gamma \cap B_r(x))$ and $v_r = \int_{\Gamma \cap B_r(x)} \nu$. Put $\Omega = E \cap S^2$. At regular values r^2 of the smooth squared-distance function on Γ , spherical divergence of the tangential projections of constant vectors gives the vector identity

$$v_r = -2 \int_{\Omega \cap B_r(x)} y d\omega_y - \int_{\Omega \cap \partial B_r(x)} \nu_{\text{cap}} d\mu.$$

Here all link arcs have the outward conormal of Ω . The cap has area πr^2 and its circle has length $2\pi r \sqrt{1 - r^2/4} \leq 2\pi r$. Taking norms therefore gives $|v_r| \leq 2\pi r^2 + 2\pi r \leq 4\pi r$. Regular radii increasing to any prescribed r give the same bound for its open cap by monotone convergence of the arclength and dominated convergence of the conormal integral. On two points of this cap, $|y - z| \leq 2r$, so $B_p \geq b_p/(2r^2)$. Thus

$$m_r^2 - |v_r|^2 \leq \frac{2r^2 P}{b_p},$$

which proves the bound for $r \leq 1$. Here $m_r^2 - |v_r|^2$ is exactly $\iint_{(\Gamma \cap B_r(x))^2} (1 - \nu_y \cdot \nu_z) d\mu_y d\mu_z$; the full energy P bounds the energy of this subset because its integrand is nonnegative. For large radii, spherical divergence on the whole phase $\Omega = E \cap S^2$ gives $V = -2 \int_{\Omega} x d\omega$, so $|V| \leq 2\pi$: projection onto any unit direction is bounded by the integral of its positive part over S^2 , which equals π . Since $B_p \geq b_p/2$,

$$P \geq \frac{b_p}{2}(L^2 - |V|^2), \quad L \leq C.$$

More precisely $L^2 \leq 4\pi^2 + 2P/b_p \leq C^2$. This proves the local length bound also for $r \geq 1$.

The measure $d\lambda = g d\mu$ has mass $M = L\delta$ and satisfies $\lambda(B_r(x)) \leq 2Cr$. It has no point atoms, since g is bounded and arclength has none. Assume $0 < \delta < 1$, so $M > 0$, and set $r_* = M/(2C)$. Tonelli's theorem and $d^{-p} = p \int_d^\infty r^{-p-1} dr$ give the layer formula. Using both $\lambda(B_r(x)) \leq 2Cr$ and $\lambda(B_r(x)) \leq M$, we obtain

$$\begin{aligned} \int |x - y|^{-p} d\lambda(y) &= p \int_0^\infty r^{-p-1} \lambda(B_r(x)) dr \\ &\leq 2Cp \int_0^{r_*} r^{-p} dr + Mp \int_{r_*}^\infty r^{-p-1} dr \\ &= \frac{(2C)^p M^{1-p}}{1-p}. \end{aligned}$$

Consequently

$$P \leq \frac{a_p(2C)^p}{1-p} (L\delta)^{2-p}.$$

This follows by applying the preceding pointwise potential bound to the W_p estimate, then integrating once more against $d\lambda(x)$, whose total mass is $M = L\delta$.

One further elementary estimate removes L from the result. For each arclength parametrized boundary loop γ of length ℓ , choose its orientation so that $\nu = \gamma \times \gamma'$, and put $\bar{\gamma} = \ell^{-1} \int_0^\ell \gamma(s) ds$. Since $\int_0^\ell \gamma' = 0$, Cauchy-Schwarz and the componentwise periodic Wirtinger inequality give

$$\begin{aligned} \left| \int_0^\ell \nu ds \right| &= \left| \int_0^\ell (\gamma - \bar{\gamma}) \times \gamma' ds \right| \\ &\leq \left(\int_0^\ell |\gamma - \bar{\gamma}|^2 ds \right)^{1/2} \left(\int_0^\ell |\gamma'|^2 ds \right)^{1/2} \\ &\leq \frac{\ell}{2\pi} \int_0^\ell |\gamma'|^2 ds = \frac{\ell^2}{2\pi}. \end{aligned}$$

For component lengths ℓ_j , the triangle inequality and $\sum_j \ell_j^2 \leq (\sum_j \ell_j)^2 = L^2$ now give $|V| \leq L^2/(2\pi)$, and therefore $L \geq 2\pi(1 - \delta)$. Set $R = 2P/(b_p L^2)$. The preceding estimates give

$$R \geq \delta(2 - \delta), \quad R \leq K_p^{\text{pin}} \left(\frac{16\pi^2}{L^2} + R \right)^{p/2} \delta^{2-p} \leq K_p^{\text{pin}} \left(\frac{4}{(1 - \delta)^2} + R \right)^{p/2} \delta^{2-p}.$$

Here the identity

$$\frac{C}{L} = \left(\frac{16\pi^2}{L^2} + R \right)^{1/2}$$

was used in the upper bound; its coefficient is exactly $2^{p+1}a_p/[(1-p)b_p] = K_p^{\text{pin}}$. For fixed $A > 0$, the derivative

$$\frac{d}{dR} \frac{R}{(A+R)^{p/2}} = \frac{A + (1-p/2)R}{(A+R)^{1+p/2}} > 0$$

shows that this function increases on $[0, \infty)$. Apply this with $A = 4/(1 - \delta)^2$ to the last upper bound and with $R_0 = \delta(2 - \delta) \leq R$; then divide by $\delta > 0$ to obtain the asserted inequality.

If $\delta = 0$, then $V \neq 0$ and, with $e = V/|V|$,

$$\int_{\Gamma} |\nu - e|^2 d\mu = 2(L - |V|) = 0.$$

Continuity gives $\nu = e$ at every point of every component. As $x \cdot e = 0$, the link lies in the great circle $G = S^2 \cap e^\perp$. Every nonempty component of Γ is a one-dimensional embedded submanifold of G , hence is open in G , and compactness makes it closed. Connectedness of G therefore gives $\Gamma = G$. On each of the two hemispheres $S^2 \setminus G$, membership in the smooth phase $\Omega = E \cap S^2$ is constant, because there is no boundary there. The locally one-sided boundary condition forces exactly one of these hemispheres to belong to Ω . Thus the cone is a half-space, with the usual identification of sets up to null sets.

For the numerical consequence, the beta formula gives

$$K_{1/2}^{\text{pin}} = \frac{5\sqrt{2}}{8}I, \quad I = \int_0^{\pi/2} (1 + \sin^2 t)^{-1/2} dt \leq \frac{\pi}{4}(1 + 1/\sqrt{2}).$$

For the integral identity, substitute $x = \sin t$ and then set $s = x^4$:

$$I = \int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{1}{4}B(1/4, 1/2) = \frac{\Gamma(1/4)\sqrt{\pi}}{4\Gamma(3/4)}.$$

The upper bound follows from the chord bound for the convex function $(1+x)^{-1/2}$ on $[0, 1]$: it is at most $1 + (1/\sqrt{2} - 1)x$, and $\int_0^{\pi/2} \sin^2 t dt = \pi/4$. Thus $K_{1/2}^{\text{pin}} \leq 5\pi(\sqrt{2} + 1)/32 < 935/784 < 6/5$, using $\pi < 22/7$ and $\sqrt{2} < 10/7$. If $0 < \delta \leq 2/5$, then $4/(1-\delta)^2 + \delta(2-\delta) \leq 100/9 + 16/25 = 2644/225 < 12 < 16$ and $\sqrt{\delta} < 2/3$. The right side of the necessary inequality is strictly below $(6/5)(2/3)2 = 8/5$, whereas its left side is at least $8/5$. This contradiction proves the consequence. $\square$

Lemma 5.2 (Explicit row constants). *Let $1/3 \leq p < 1$ and let $\Omega \subset S^2$ be a nonempty proper smooth open set, with $\Gamma = \partial\Omega$ and outward conormal ν . Use the kernels of Lemma 2.1 and put $b_p = 2/[(1+p)(2+p)]$. Let H_p be the equatorial row of Lemma 3.2, and use P_x, w_p, η_p, K_p from Lemma 4.2. Then*

$$\eta_p < \frac{27}{100}, \quad \frac{H_p}{b_p} < \frac{597}{500}, \quad \frac{K_p - H_p}{b_p} < \frac{296}{3645}.$$

In particular the general row estimate holds with these constants:

$$\int_{\Gamma} W_p(x \cdot y) d\mu_y \leq K_p + \eta_p P_x.$$

If Γ is the link of a stationary cone stable under compact variations away from zero, then, writing $P = \int_{\Gamma} P_x d\mu_x$ and $L = |\Gamma|$,

$$\frac{P}{b_p L} < \frac{464813}{266085} < \frac{5024}{2875} < \frac{16\pi}{5\sqrt{33}}.$$

No convexity, symmetry, or connectivity hypothesis is imposed.

Proof. The geometric row estimate and the integral formulas for η_p and K_p are conclusions of Lemma 4.2. We estimate those integral formulas directly, with the exact representation in Lemma 2.1:

$$w_p(r) = \int_0^\infty F_{1-p}(u)(u^2 + r^2)^{-(3+p)/2} du, \quad F_{1-p} \leq (1-p)^2 F_1, \quad F_1 = 2 - \frac{4}{\sqrt{u^2 + 4}}.$$

Define w_0 by the same integral at $p = 0$, for comparison only. We will use its radial identification below, so we establish it here. We first verify the order-zero identities directly. For

$-1 \leq z < 1$ write $D_z(\rho) = 1 + \rho^2 - 2\rho z$ and

$$\tilde{B}_0(z) = \int_0^\infty \rho D_z(\rho)^{-3/2} d\rho, \quad \tilde{A}_0(z) = \int_0^\infty \rho^{1/2} D_z(\rho)^{-3/2} d\rho.$$

Both integrals converge: at zero their integrands are respectively $O(\rho)$ and $O(\rho^{1/2})$, and at infinity they are $O(\rho^{-2})$ and $O(\rho^{-5/2})$. There is no interior singularity because $z < 1$. Inversion $\rho \mapsto 1/\rho$ gives $\tilde{B}_0 = \int_0^\infty D_z(\rho)^{-3/2} d\rho$. The identity

$$\frac{d}{d\rho} \frac{\rho - 1}{\sqrt{D_z(\rho)}} = (1 - z)(\rho + 1)D_z(\rho)^{-3/2}$$

therefore gives

$$\tilde{B}_0(z) = \frac{1}{2} \int_0^\infty (\rho + 1)D_z(\rho)^{-3/2} d\rho = \frac{1}{1 - z}.$$

Here the primitive $(\rho - 1)/\sqrt{D_z(\rho)}$ has endpoint limits -1 and 1 , so the calculation includes $z = -1$. Splitting both radial integrals at one and inverting their outer pieces also gives

$$\tilde{B}_0(z) - \tilde{A}_0(z) = \int_0^1 \frac{(\sqrt{\rho} - 1)^2}{D_z(\rho)^{3/2}} d\rho.$$

Set $z = 1 - r^2/2$ and $\rho = e^{-2t}$ in this folded integral. Since $D_z(e^{-2t}) = e^{-2t}(4 \sinh^2 t + r^2)$, it becomes

$$4 \int_0^\infty (\cosh t - 1)(4 \sinh^2 t + r^2)^{-3/2} dt.$$

Finally $u = 2 \sinh t$, $du = 2 \cosh t dt$, transforms its numerator and differential into

$$\frac{2(\cosh t - 1)}{\cosh t} du = \left(2 - \frac{4}{\sqrt{u^2 + 4}} \right) du = F_1(u) du.$$

Thus the positive F_1 integral is exactly $\tilde{B}_0(1 - r^2/2) - \tilde{A}_0(1 - r^2/2)$. Write $A_0 = \tilde{A}_0$ and $B_0 = \tilde{B}_0 = 1/(1 - z)$. The calculation proves locally that $w_0(r) = B_0(1 - r^2/2) - A_0(1 - r^2/2)$; no order-zero cone or order-zero stability assertion is being used. All integrands used for the following changes of integration order are nonnegative. Put

$$c_p = \frac{(1 - p)^2}{b_p} = \frac{(1 - p)^2(1 + p)(2 + p)}{2}, \quad I_0 = \int_0^2 r w_0(r) dr.$$

Split the quadrant strip $u > 0, 0 < r < 2$ into the disk $u^2 + r^2 < 1$ and its complement. On the disk the bounds $u^2/5 \leq F_1(u) \leq u^2/4$ are part of the cited kernel lemma. Polar coordinates $u = \rho \cos t$, $r = \rho \sin t$ give the angular factor $\int_0^{\pi/2} \cos^2 t \sin t dt = 1/3$. Consequently the normalized positive-order disk contribution is at most $c_p/[12(2 - p)]$, whereas its order-zero counterpart is at least $1/30$. On the complement the extra factor $(u^2 + r^2)^{-p/2}$ is at most one. Keeping the factor c_p on both pieces therefore gives

$$\eta_p \leq c_p E_p, \quad E_p = I_0 + \frac{1}{12(2 - p)} - \frac{1}{30}.$$

The comparison moment can be evaluated without an estimate: integration in r , followed by $u = 2 \sinh t$ and $v = e^{-t}$, gives

$$\begin{aligned} I_0 &= \int_0^\infty F_1(u) \left(\frac{1}{u} - \frac{1}{\sqrt{u^2 + 4}} \right) du \\ &= \int_0^1 \frac{4v(1 - v)}{(1 + v)(1 + v^2)} dv \\ &= \int_0^1 \left(\frac{4}{1 + v^2} - \frac{4}{1 + v} \right) dv = \pi - 4 \log 2. \end{aligned}$$

The function $c_p E_p$ decreases on $[1/3, 1)$. Indeed

$$\frac{c'_p}{c_p} = -\frac{2}{1-p} + \frac{1}{1+p} + \frac{1}{2+p} \leq -\frac{51}{28}.$$

The derivative of this logarithmic derivative is $-2/(1-p)^2 - 1/(1+p)^2 - 1/(2+p)^2 < 0$. The logarithmic derivative itself equals $-51/28$ at $p = 1/3$. Also $E_p > 1/3$ and $E'_p = 1/[12(2-p)^2] \leq 1/12$, so $E'_p/E_p < 1/4$. Therefore $(c_p E_p)'/(c_p E_p) < -51/28 + 1/4 = -11/7 < 0$. For the lower bound on E_p it suffices to use $I_0 > 157/50 - 4(7/10) = 17/50 > 1/3$ and $1/[12(2-p)] - 1/30 > 0$. Here and below we use the classical rational bounds $157/50 < \pi < 22/7$. The bound $\log 2 < 7/10$ follows from $e^{7/10} > \sum_{j=0}^4 (7/10)^j/j! = 482921/240000 > 2$. The first three terms of

$$\log 2 = 2 \sum_{j=0}^{\infty} \frac{3^{-2j-1}}{2j+1}$$

give $\log 2 > 842/1215$. Thus

$$\eta_p < \frac{56}{81} \left(\frac{22}{7} - \frac{3368}{1215} + \frac{1}{60} \right) = \frac{26366}{98415} < \frac{27}{100}.$$

The last difference is $4121/1968300 > 0$.

Next use the exact gamma formula for H_p in the statement of Lemma 3.2. Duplication, with $a = (3+p)/4$ and $d = (1-p)/4$, gives

$$\frac{H_p}{b_p} = \pi(2+p)2^{-1-p} \frac{\Gamma(a)}{\Gamma(a+1/2)} \frac{\Gamma(d+1/2)}{\Gamma(d)}.$$

Log convexity at the midpoint $a+1$ implies $\Gamma(a+1)^2 \leq \Gamma(a+1/2)\Gamma(a+3/2)$, and hence

$$\frac{\Gamma(a)}{\Gamma(a+1/2)} \leq \frac{\sqrt{a+1/2}}{a}.$$

The increasing digamma function shows that $\Gamma(d+1/2)/\Gamma(d+1)$ decreases in $d > 0$. Its limit at zero is $\sqrt{\pi}$, so $\Gamma(d+1/2)/\Gamma(d) \leq d\sqrt{\pi}$. It follows that

$$\frac{H_p}{b_p} \leq U(p) := \pi^{3/2}(2+p)2^{-1-p}d \frac{\sqrt{a+1/2}}{a}.$$

This upper bound decreases for $p \geq 1/3$, since

$$\frac{U'(p)}{U(p)} = \frac{1}{2+p} - \log 2 - \frac{1}{1-p} + \frac{1}{8(a+1/2)} - \frac{1}{4a} < 0.$$

Indeed $1/(2+p) \leq 3/7$, $1/(1-p) \geq 3/2$, and the last two terms have negative sum. At the left endpoint,

$$U(1/3) = \frac{7\pi^{3/2}}{15\sqrt{3}2^{1/3}}.$$

The inequality $(19/12)^3 < 4$ gives $2^{2/3} > 19/12$. Together with $\pi < 22/7$, it yields

$$U(1/3)^2 < \frac{49}{675} \left(\frac{22}{7} \right)^3 \frac{12}{19} = \frac{42592}{29925} < \left(\frac{597}{500} \right)^2.$$

The last difference is $701573/299250000 > 0$, proving the asserted estimate for H_p/b_p .

For the remaining term in K_p , we justify differentiation before comparing the resulting integrals. Here and below differentiation of the positive integral has a uniform integrable majorant on the interval used for the row tail. Indeed, for $0 \leq p < 1$, $r \in [\sqrt{2}, 2]$, and $k = 1-p$, the bound $0 \leq F_k(u) \leq F_1(u) \leq \min\{u^2/4, 2\}$ gives

$$\left| \frac{\partial}{\partial r} [F_k(u)(u^2 + r^2)^{-(3+p)/2}] \right| = (3+p)rF_k(u)(u^2 + r^2)^{-(5+p)/2} \leq \begin{cases} 2u^2, & 0 < u \leq 1, \\ 16u^{-5}, & u \geq 1. \end{cases}$$

Both pieces are integrable in u . The undifferentiated integrand is similarly bounded by $u^2/4$ on $(0, 1]$ and $2u^{-3}$ on $[1, \infty)$. On an arbitrary compact interval $r \in [a, b] \subset (0, \infty)$ the same argument gives $C_{a,b}u^2$ for $u \leq 1$ and $8bu^{-5}$ for $u \geq 1$. Dominated differentiation is therefore valid, including at $p = 0$, and gives

$$-w'_p(r) = (3+p)r \int_0^\infty F_{1-p}(u)(u^2 + r^2)^{-(5+p)/2} du.$$

For $r \in [\sqrt{2}, 2]$ it follows that

$$\frac{-w'_p(r)}{b_p} \leq D_p[-w'_0(r)], \quad D_p = c_p \frac{3+p}{3}.$$

Indeed differentiation contributes $(3+p)r$, and the remaining factor $(u^2 + r^2)^{-p/2}$ is at most one. The function D_p decreases on $[1/3, 1]$: its logarithmic derivative is

$$-\frac{2}{1-p} + \frac{1}{1+p} + \frac{1}{2+p} + \frac{1}{3+p}.$$

This logarithmic derivative is strictly decreasing and equals $-213/140$ at $p = 1/3$. Hence $D_p \leq 560/729$. From the formula for K_p in Lemma 4.2 and $0 \leq -2\sin(2\theta) \leq 2$ for $\pi/2 \leq \theta \leq \pi$,

$$\frac{K_p - H_p}{b_p} \leq 2D_p[w_0(\sqrt{2}) - w_0(2)].$$

The defining radial integrals at order zero give

$$B_0(0) = 1, \quad B_0(-1) = \frac{1}{2}, \quad A_0(-1) = \frac{\pi}{8}, \quad A_0(0) = t := \frac{\Gamma(3/4)^2}{\sqrt{\pi}}.$$

For example the last equality is $\frac{1}{2}B(3/4, 3/4) = \Gamma(3/4)^2/\sqrt{\pi}$; the other three follow by elementary integration or the beta integral. Thus $2[w_0(\sqrt{2}) - w_0(2)] = 1 - 2t + \pi/4$.

Here is a rational lower bound for t . Define

$$I = \int_0^{\pi/2} (1 + \sin^2 u)^{-1/2} du.$$

The substitutions $x = \sin u$ and then $s = x^4$ give

$$I = \int_0^1 (1 - x^4)^{-1/2} dx = \frac{1}{4}B(1/4, 1/2) = \frac{\sqrt{\pi}\Gamma(1/4)}{4\Gamma(3/4)}.$$

Euler reflection therefore implies $tI = \pi/(2\sqrt{2})$. For the convex function $(1+x)^{-1/2}$, use its secants on $[0, 1/2]$ and $[1/2, 1]$, and split the integral at $u = \pi/4$. More explicitly, with $\alpha = \sqrt{2/3}$ and $\beta = 1/\sqrt{2}$, convexity gives

$$(1+x)^{-1/2} \leq \begin{cases} 1 + 2(\alpha - 1)x, & 0 \leq x \leq 1/2, \\ 2\alpha - \beta + 2(\beta - \alpha)x, & 1/2 \leq x \leq 1. \end{cases}$$

Since

$$\int_0^{\pi/4} \sin^2 u du = \frac{\pi}{8} - \frac{1}{4}, \quad \int_{\pi/4}^{\pi/2} \sin^2 u du = \frac{\pi}{8} + \frac{1}{4},$$

the resulting bound is

$$I \leq \sqrt{\frac{2}{3}} \left(\frac{\pi}{2} - 1 \right) + \frac{1 + 1/\sqrt{2}}{2} < \frac{49}{60} \frac{4}{7} + \frac{1 + 99/140}{2} = \frac{1109}{840} < \frac{37}{28}.$$

The radical estimates are certified by $(49/60)^2 - 2/3 = 1/3600$ and $(99/140)^2 - 1/2 = 1/19600$. Since $(99/70)^2 - 2 = 1/4900$,

$$t > \frac{14\pi}{37\sqrt{2}} > \frac{14(157/50)}{37(99/70)} = \frac{15386}{18315} > \frac{21}{25};$$

the last difference is 7/91575. Consequently

$$\frac{K_p - H_p}{b_p} < \frac{560}{729} \left(1 - \frac{42}{25} + \frac{11}{14} \right) = \frac{296}{3645}.$$

For a stable stationary cone, the constant-test conclusion of Proposition 2.3 is $P \leq \iint_{\Gamma^2} W_p$. For completeness, put $d = |x - y|$ and use the representations in Lemma 2.1. The radial denominator for B_p is $(\rho - 1)^2 + \rho d^2$. Split its integral into $(0, 1/2)$, $(1/2, 2)$, and $(2, \infty)$; the outer pieces are bounded, and the substitution $\rho - 1 = dv$ in the middle gives $B_p(x \cdot y) \leq C_p d^{-2-p}$. The bound $F_{1-p}(u) \leq (1-p)^2 u^2/4$ likewise gives

$$W_p(x \cdot y) \leq C_p \int_0^\infty \frac{u^2}{(u^2 + d^2)^{(3+p)/2}} du \leq C_p d^{-p},$$

where the scaled integral is finite for $p > 0$. On every smooth link chart, $|\nu_x - \nu_y|^2 \leq C_\Gamma d^2$, and arclength and chord distance are locally comparable. Thus the normal-defect integrand defining P is bounded by $C_{p,\Gamma} d^{-p}$, which is locally integrable on a curve because $p < 1$. Distinct components have positive separation. Compactness now gives $P < \infty$ and finite W_p rows. Integrating the row estimate therefore gives the finite inequality $P \leq K_p L + \eta_p P$. Since $\eta_p < 27/100 < 1$, we may subtract $\eta_p P$ and divide by $(1 - \eta_p)b_p L > 0$. Thus

$$\frac{P}{b_p L} \leq \frac{K_p/b_p}{1 - \eta_p} < \frac{597/500 + 296/3645}{73/100} = \frac{464813}{266085}.$$

Finally

$$\frac{5024}{2875} - \frac{464813}{266085} = \frac{94733}{152998875} > 0,$$

and $\pi > 157/50$, $\sqrt{33} < 23/4$ give $16\pi/(5\sqrt{33}) > 5024/2875$. This proves every assertion, including the endpoint $p = 1/3$. $\square$

Theorem 5.3 (Flatness for $1/3 \leq p < 1$). *Let $1/3 \leq p < 1$. Every nontrivial stationary p -perimeter cone in $\mathbb{R}^3$ with locally one-sided smooth boundary away from zero, compact smooth spherical link, and stability under compact variations away from zero is a half-space. No convexity, symmetry, or link connectivity is assumed.*

Proof. Put $\Omega = E \cap S^2$, $\Gamma = \partial\Omega$, $A = |\Omega|$, $L = |\Gamma|$, $V = \int_\Gamma \nu d\mu$, and $v = |V|/L \in [0, 1]$. Since the cone is nontrivial and locally one-sided, its smooth spherical phase satisfies $0 < A < 4\pi$ and its boundary has $0 < L < \infty$. The triangle inequality gives $|V| \leq L$. Use the kernels of Lemma 2.1 and set

$$P = \iint_{\Gamma^2} B_p(x \cdot y)(1 - \nu_x \cdot \nu_y) d\mu_x d\mu_y, \quad b_p = \frac{2}{(1+p)(2+p)}.$$

By the definition of P_x in Lemma 4.2, Tonelli's theorem gives $P = \int_\Gamma P_x d\mu_x$, initially as an identity in $[0, \infty]$. The upper estimate in Lemma 5.2 gives $P < \infty$ throughout the present range $1/3 \leq p < 1$. Thus the same energy is bounded above by that estimate and below by Lemma 4.1. The kernel lower bound in Lemma 2.1, and $1 - x \cdot y \leq 2$, give

$$\frac{P}{b_p} \geq \frac{L^2 - |V|^2}{2}.$$

The area estimate Lemma 4.1 and the classical spherical isoperimetric inequality give

$$\frac{P}{b_p} \geq (2\pi - A)^2 \geq 4\pi^2 - L^2.$$

To check the isoperimetric application, use its form for arbitrary sets of finite perimeter on the unit sphere. The smooth phase Ω has spherical perimeter L . Choose a geodesic cap of radius $\theta \in (0, \pi)$ with area $2\pi(1 - \cos \theta) = A$. The inequality compares the perimeter of the entire phase to the cap perimeter, giving $L \geq 2\pi \sin \theta$, and therefore $L^2 \geq A(4\pi - A)$.

This form imposes no restriction on components or holes, and includes $A > 2\pi$. For fixed $v < 1$, the first lower bound divided by L increases with L , and the second decreases. Their crossing is $L^2 = 8\pi^2/(3 - v^2)$, so

$$\frac{P}{b_p L} \geq \min_{L>0} \max \left\{ \frac{(1 - v^2)L}{2}, \frac{4\pi^2}{L} - L \right\} = F(v) := \frac{\pi\sqrt{2}(1 - v^2)}{\sqrt{3 - v^2}}.$$

Indeed, if L is below the crossing, the decreasing second term is at least its crossing value; above the crossing, the increasing first term is at least that same value. At the crossing they agree. Thus the displayed minimum is attained there. Differentiation gives

$$F'(v) = \frac{\pi\sqrt{2}v(v^2 - 5)}{(3 - v^2)^{3/2}},$$

which is negative for $0 < v < 1$. Thus F decreases on $[0, 1)$. If $v \leq 3/5$, then $F(v) \geq 16\pi/(5\sqrt{33})$, contradicting Lemma 5.2. Thus $v > 3/5$.

Set $\delta = 1 - v < 2/5$. If $\delta = 0$, the constant-normal conclusion of Proposition 5.1 already gives a half-space. Suppose $0 < \delta < 2/5$. That proposition asserts the necessary inequality

$$2 - \delta \leq K_p^{\text{pin}} \delta^{1-p} \left[\frac{4}{(1 - \delta)^2} + \delta(2 - \delta) \right]^{p/2}.$$

The gamma recurrence $\Gamma((3 + p)/2) = ((1 + p)/2)\Gamma((1 + p)/2)$ gives

$$K_p^{\text{pin}} = 2^{p-3}(1 - p)(2 + p)B(p/2, 1/2).$$

For every $x > 0$, subtracting $1/x$ from the beta integral gives

$$\begin{aligned} B(x, 1/2) - \frac{1}{x} &= \int_0^1 t^{x-1}((1 - t)^{-1/2} - 1) dt \\ &\leq \int_0^1 \frac{(1 - t)^{-1/2} - 1}{t} dt = 2 \log 2. \end{aligned}$$

The inequality uses $t^x \leq 1$; the integrand on the right is bounded at zero, since $(1 - t)^{-1/2} - 1 = t/2 + O(t^2)$, and is $O((1 - t)^{-1/2})$ at one. Thus it is integrable at both endpoints. The substitution $u = \sqrt{1 - t}$ reduces its integral to $\int_0^1 2/(1 + u) du$. For the numerical bound, the positive logarithmic series gives

$$\log 2 = 2 \sum_{j=0}^{\infty} \frac{3^{-2j-1}}{2j+1} < \frac{2}{3} + \frac{2}{3} \sum_{j=1}^{\infty} 3^{-2j-1} = \frac{25}{36} < \frac{7}{10}.$$

Consequently $B(p/2, 1/2) < 2/p + 7/5$.

For fixed $p \in (0, 1)$ the right side of the necessary pinching inequality increases with $\delta \in (0, 1)$. Indeed δ^{1-p} increases, as does $4/(1 - \delta)^2 + \delta(2 - \delta)$. At $\delta_* = 2/5$ the latter bracket is $100/9 + 16/25 = 2644/225 < 16$. Thus throughout $0 < \delta \leq 2/5$ the right side is strictly bounded above by

$$\begin{aligned} Q(p) &= 2^{p-3}(1 - p)(2 + p) \left(\frac{2}{p} + \frac{7}{5} \right) \left(\frac{2}{5} \right)^{1-p} 16^{p/2} \\ &= \frac{1}{20}(1 - p)(2 + p) \left(\frac{2}{p} + \frac{7}{5} \right) 20^p. \end{aligned}$$

At the endpoint

$$Q(1/3) = \frac{259}{90} \left(\frac{4}{25} \right)^{1/3} < \frac{259}{90} \frac{11}{20} = \frac{2849}{1800} < \frac{8}{5}.$$

The cube comparison is $(11/20)^3 - 4/25 = 51/8000 > 0$, and $8/5 - 2849/1800 = 31/1800 > 0$.

We next prove that Q decreases on the whole interval $[1/3, 1)$. Its logarithmic derivative is

$$\frac{Q'(p)}{Q(p)} = \log 20 - \frac{1}{1-p} + \frac{1}{2+p} - \frac{10}{p(10+7p)}.$$

After multiplication by the positive denominator $D = (1-p)(p+2)p(10+7p)$, the assertion

$$\frac{1}{1-p} - \frac{1}{2+p} + \frac{10}{p(10+7p)} > 3$$

is equivalent to positivity of

$$N(p) = 21p^4 + 65p^3 + 5p^2 - 60p + 20.$$

A direct polynomial identity gives a stronger positivity certificate:

$$\begin{aligned} N(p) = 84 \left(p - \frac{1}{2} + \frac{17}{672} \right)^2 + \frac{3407}{5376} \\ + \left(p - \frac{1}{2} \right)^2 \left(21p^2 + 86p + \frac{7}{4} \right). \end{aligned} \quad (1)$$

Expansion verifies the equality coefficient by coefficient. For every $p \geq 0$ the two squared terms multiplied by their displayed factors are nonnegative, whereas $3407/5376 > 0$. Thus $N(p) > 0$ for all $p \geq 0$, in particular on the interval $[1/3, 1)$ needed here. This argument is purely algebraic and requires no estimate of N'' . Finally $e^3 > \sum_{j=0}^8 3^j/j! = 89641/4480 > 20$, so $Q'/Q < \log 20 - 3 < 0$ as claimed.

The necessary pinching inequality is now impossible: its right side is less than $Q(1/3) < 8/5$, whereas its left side is at least $8/5$. Therefore $\delta = 0$, and Proposition 5.1 gives a half-space.

□

6. ESTIMATES UNIFORM AT ORDER ZERO

The parameter $p = 0$ below is used only to define comparison kernels and to evaluate bounds. No zero-order perimeter or zero-order stability assumption is used.

Lemma 6.1 (A rational bound for the order-zero row tail). *Define the order-zero comparison kernels, for $-1 \leq z < 1$, by*

$$A_0(z) = \int_0^\infty \frac{\rho^{1/2}}{(1 + \rho^2 - 2\rho z)^{3/2}} d\rho, \quad B_0(z) = \frac{1}{1-z}, \quad W_0 = B_0 - A_0.$$

For $0 < r \leq 2$, put

$$w_0(r) = W_0(1 - r^2/2) = \int_0^\infty F_1(u)(u^2 + r^2)^{-3/2} du, \quad F_1(u) = 2 - \frac{4}{\sqrt{u^2 + 4}},$$

and set

$$E_0 = \int_{\sqrt{2}}^2 [-w'_0(r)][-2 \sin(2\theta(r))] dr, \quad \theta(r) = 2 \arcsin(r/2).$$

Then

$$0 < E_0 \leq \frac{797893372445475270877}{14514159317550104576000} < \frac{11}{200}.$$

For $0 < p < 1$, use $b_p = 2/[(1+p)(2+p)]$ and the row constants K_p, H_p of Lemma 4.2. With

$$D_p = \frac{(1-p)^2(1+p)(2+p)(3+p)}{6},$$

one has $0 < D_p \leq 1$ and

$$0 \leq \frac{K_p - H_p}{b_p} \leq D_p E_0 < \frac{11}{200}.$$

Proof. We first verify the order-zero identities directly. For $-1 \leq z < 1$ write $D_z(\rho) = 1 + \rho^2 - 2\rho z$ and

$$\tilde{B}_0(z) = \int_0^\infty \rho D_z(\rho)^{-3/2} d\rho, \quad \tilde{A}_0(z) = \int_0^\infty \rho^{1/2} D_z(\rho)^{-3/2} d\rho.$$

Both integrals converge: at zero their integrands are respectively $O(\rho)$ and $O(\rho^{1/2})$, and at infinity they are $O(\rho^{-2})$ and $O(\rho^{-5/2})$. There is no interior singularity because $z < 1$. Inversion $\rho \mapsto 1/\rho$ gives $\tilde{B}_0 = \int_0^\infty D_z(\rho)^{-3/2} d\rho$. The identity

$$\frac{d}{d\rho} \frac{\rho - 1}{\sqrt{D_z(\rho)}} = (1 - z)(\rho + 1)D_z(\rho)^{-3/2}$$

therefore gives

$$\tilde{B}_0(z) = \frac{1}{2} \int_0^\infty (\rho + 1)D_z(\rho)^{-3/2} d\rho = \frac{1}{1 - z}.$$

Here the primitive $(\rho - 1)/\sqrt{D_z(\rho)}$ has endpoint limits -1 and 1 , so the calculation includes $z = -1$. Splitting both radial integrals at one and inverting their outer pieces also gives

$$\tilde{B}_0(z) - \tilde{A}_0(z) = \int_0^1 \frac{(\sqrt{\rho} - 1)^2}{D_z(\rho)^{3/2}} d\rho.$$

Set $z = 1 - r^2/2$ and $\rho = e^{-2t}$ in this folded integral. Since $D_z(e^{-2t}) = e^{-2t}(4 \sinh^2 t + r^2)$, it becomes

$$4 \int_0^\infty (\cosh t - 1)(4 \sinh^2 t + r^2)^{-3/2} dt.$$

Finally $u = 2 \sinh t$, $du = 2 \cosh t dt$, transforms its numerator and differential into

$$\frac{2(\cosh t - 1)}{\cosh t} du = \left(2 - \frac{4}{\sqrt{u^2 + 4}}\right) du = F_1(u) du.$$

Thus the positive F_1 integral is exactly $\tilde{B}_0(1 - r^2/2) - \tilde{A}_0(1 - r^2/2)$. In the notation of the statement, these identities say precisely $\tilde{B}_0 = B_0$, $\tilde{A}_0 = A_0$, and that both displayed definitions of w_0 agree. For positive p the same F_{1-p} representation is the conclusion of Lemma 2.1. Here and below differentiation of the positive integral has a uniform integrable majorant on the interval used for the row tail. Indeed, for $0 \leq p < 1$, $r \in [\sqrt{2}, 2]$, and $k = 1 - p$, the bound $0 \leq F_k(u) \leq F_1(u) \leq \min\{u^2/4, 2\}$ gives

$$\left| \frac{\partial}{\partial r} [F_k(u)(u^2 + r^2)^{-(3+p)/2}] \right| = (3+p)r F_k(u)(u^2 + r^2)^{-(5+p)/2} \leq \begin{cases} 2u^2, & 0 < u \leq 1, \\ 16u^{-5}, & u \geq 1. \end{cases}$$

Both pieces are integrable in u . The undifferentiated integrand is similarly bounded by $u^2/4$ on $(0, 1]$ and $2u^{-3}$ on $[1, \infty)$. On an arbitrary compact interval $r \in [a, b] \subset (0, \infty)$ the same argument gives $C_{a,b}u^2$ for $u \leq 1$ and $8bu^{-5}$ for $u \geq 1$. Dominated differentiation is therefore valid, including at $p = 0$, and gives

$$-w'_p(r) = (3+p)r \int_0^\infty F_{1-p}(u)(u^2 + r^2)^{-(5+p)/2} du.$$

In particular $-w'_0(r) > 0$ for every $r > 0$, since $F_1(u) > 0$ for $u > 0$. The remaining factor defining E_0 is nonnegative on $[\sqrt{2}, 2]$ and positive in its interior. Its product with $-w'_0$ is continuous and integrable there, and consequently $E_0 > 0$.

Put $\tau = (1+z)/2 = 1 - r^2/4$. On the interval relevant to E_0 , one has $0 \leq \tau \leq 1/2$. Write

$$a_n = \frac{(3/2)_n^2}{(2)_n n!}, \quad c_n = \frac{1}{2} - \frac{\pi}{8} a_n \quad (n \geq 0),$$

where $(a)_n$ is the rising factorial, with $(a)_0 = 1$. The binomial series and beta integration give

$$A_0(z) = \frac{\pi}{8} \sum_{n=0}^{\infty} a_n \tau^n, \quad B_0(z) = \frac{1}{2(1-\tau)}, \quad w_0(r) = \sum_{n=0}^{\infty} c_n \tau^n.$$

Indeed the coefficient of τ^n in A_0 is

$$\frac{4^n (3/2)_n}{n!} \int_0^{\infty} \frac{\rho^{n+1/2}}{(1+\rho)^{3+2n}} d\rho = \frac{4^n (3/2)_n \Gamma(n+3/2)^2}{n! \Gamma(2n+3)} = \frac{\pi}{8} a_n,$$

where the last equality uses

$$\Gamma(2n+3) = \frac{2^{2n+2}}{\sqrt{\pi}} \Gamma(n+3/2) \Gamma(n+2), \quad \Gamma(n+3/2) = \Gamma(3/2) (3/2)_n.$$

In the folded integral, the same expansion gives explicitly

$$c_n = \frac{4^n (3/2)_n}{n!} \int_0^1 \frac{(\sqrt{\rho}-1)^2 \rho^n}{(1+\rho)^{3+2n}} d\rho \geq 0.$$

For $0 \leq \tau < 1$, the binomial parameter $4\tau\rho/(1+\rho)^2$ is at most τ , so Tonelli's theorem justifies the nonnegative expansions. The resulting power series agree with $W_0 = B_0 - A_0$ on this interval, hence have the displayed coefficients c_n . Moreover

$$a_0 = 1, \quad \frac{a_n}{a_{n-1}} = \frac{(2n+1)^2}{4n(n+1)} > 1 \quad (n \geq 1),$$

so the sequence c_n decreases. In particular $0 \leq c_n \leq 1/2$. On $0 \leq \tau \leq 1/2$, the terms of the series are bounded by 2^{-n-1} and the terms of its derivative by $(n/2)2^{1-n}$ for $n \geq 1$. Both numerical series converge. The Weierstrass test therefore gives uniform convergence of the series and its derivative, and permits termwise differentiation.

Under the change from r to τ ,

$$-2 \sin(2\theta(r)) = 8(1-2\tau) \sqrt{\tau(1-\tau)}.$$

Since $d\tau/dr = -r/2$, termwise differentiation gives $[-w'_0(r)] dr = -\sum_{n \geq 1} n c_n \tau^{n-1} d\tau$. The endpoints $r = \sqrt{2}, 2$ correspond to $\tau = 1/2, 0$, respectively. Consequently

$$E_0 = 8 \int_0^{1/2} (1-2\tau) \sqrt{\tau(1-\tau)} \sum_{n=1}^{\infty} n c_n \tau^{n-1} d\tau.$$

All summands are nonnegative. The geometric factor $-2 \sin(2\theta)$ is at most 2, so the contribution from $n \geq 9$ is at most

$$2 \sum_{n=9}^{\infty} c_n 2^{-n} \leq 2c_8 \sum_{n=9}^{\infty} 2^{-n} = \frac{c_8}{128}.$$

For the first eight terms use

$$\sqrt{1-\tau} \leq 1 - \frac{\tau}{2} - \frac{\tau^2}{8} - \frac{\tau^3}{16} = \sum_{j=0}^3 d_j \tau^j, \quad (d_0, d_1, d_2, d_3) = (1, -1/2, -1/8, -1/16).$$

This is Taylor's upper bound, since the fourth derivative of $\sqrt{1-\tau}$ is negative on $[0, 1/2]$. The polynomial on the right decreases on this interval and its value at $1/2$ is $91/128 > 0$. For $s = n + j$,

$$\int_0^{1/2} (1-2\tau) \tau^{s-1/2} d\tau = \frac{4 \cdot 2^{-s-1/2}}{(2s+1)(2s+3)}.$$

Multiplying by $8n$ and summing against d_j therefore gives the moment bound

$$\begin{aligned} 8n \int_0^{1/2} (1-2\tau)\tau^{n-1/2}\sqrt{1-\tau} d\tau \\ \leq 16\sqrt{2}n \sum_{j=0}^3 \frac{d_j}{2^{n+j}(2n+2j+1)(2n+2j+3)}. \end{aligned}$$

Here the rational bounds $\sqrt{2} < 99/70$ and $\pi > 333/106$ suffice. The first follows by squaring. For an explicit certificate of the second, we use Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

To verify it, put $\alpha = \arctan(1/5)$ and $\beta = \arctan(1/239)$. The tangent addition formula gives

$$\tan(2\alpha) = \frac{5}{12}, \quad \tan(4\alpha) = \frac{120}{119}, \quad \tan(4\alpha - \beta) = \frac{120/119 - 1/239}{1 + 120/(119 \cdot 239)} = 1.$$

Moreover $0 < 4\alpha - \beta < 4/5 < \pi/2$, so the correct branch is $4\alpha - \beta = \pi/4$. The alternating arctangent series gives

$$\begin{aligned} \pi &> 16 \left(\frac{1}{5} - \frac{1}{375} + \frac{1}{15625} - \frac{1}{546875} \right) - \frac{4}{239} \\ &= \frac{1231847548}{392109375} > \frac{333}{106}. \end{aligned}$$

The last difference is $3418213/41563593750 > 0$. Define rational upper bounds

$$\bar{c}_n = \frac{1}{2} - \frac{333}{848}a_n \geq c_n.$$

For each $n \geq 1$, the inner rational sum in the moment bound is positive: by the preceding monomial computation it is a positive multiple of the integral of $(1-2\tau)\tau^{n-1/2} \sum_{j=0}^3 d_j \tau^j$. Thus the negative coefficients d_1, d_2, d_3 cause no reversal when we replace $\sqrt{2}$ by $99/70$ and c_n by $\bar{c}_n$. Combining the finite moments with the geometric remainder therefore gives the fully rational bound

$$\begin{aligned} E_0 &\leq \frac{1584}{70} \sum_{n=1}^8 n \bar{c}_n \sum_{j=0}^3 \frac{d_j}{2^{n+j}(2n+2j+1)(2n+2j+3)} + \frac{\bar{c}_8}{128} \\ &= \frac{797893372445475270877}{14514159317550104576000} < \frac{11}{200}. \end{aligned}$$

The exact difference in the last comparison is

$$\frac{385390019780480803}{14514159317550104576000} > 0.$$

As an alternative integer check of the finite sum, its eight summands and its remainder are, respectively, bounded above in units of 10^{-7} by

$$(388342, 109172, 33578, 11248, 4013, 1501, 582, 232; 1071).$$

These integers sum to $549739 < 550000$, the latter being $(11/200)10^7$. All these finite comparisons follow directly from the displayed recurrence for a_n .

Finally use the exact positive-order representation in Lemma 2.1. Its numerator satisfies $F_{1-p} \leq (1-p)^2 F_1$. Differentiation therefore gives, for $r \geq \sqrt{2}$,

$$\frac{-w'_p(r)}{b_p} \leq \frac{(1-p)^2}{b_p} \frac{3+p}{3} [-w'_0(r)] = D_p[-w'_0(r)].$$

The additional factor $(u^2 + r^2)^{-p/2}$ in the differentiated integral is at most one. The formula for $K_p - H_p$ in Lemma 4.2, whose geometric factor is nonnegative on this interval, yields $(K_p - H_p)/b_p \leq D_p E_0$. Furthermore $D_0 = 1$, and

$$\frac{D'_p}{D_p} = -\frac{2}{1-p} + \frac{1}{1+p} + \frac{1}{2+p} + \frac{1}{3+p}$$

has derivative

$$-\frac{2}{(1-p)^2} - \frac{1}{(1+p)^2} - \frac{1}{(2+p)^2} - \frac{1}{(3+p)^2} < 0.$$

Its value at zero is $-1/6$, so $D'_p/D_p < 0$ for $0 < p < 1$. Since $D_0 = 1$ and every factor in D_p is positive in this range, $0 < D_p \leq 1$ for $0 < p < 1$, proving the stated uniform tail bound. $\square$

Lemma 6.2 (A sharper cap-density bound). *Let Ω be a nonempty proper smooth open set of S^2 with compact embedded boundary Γ , outward spherical conormal ν , and arclength measure μ . All boundary components are included. For $0 < p < 1$, use B_p, b_p from Lemma 2.1 and put*

$$P = \iint_{\Gamma^2} B_p(x \cdot y)(1 - \nu_x \cdot \nu_y) d\mu_x d\mu_y, \quad C^2 = \frac{155\pi^2}{54} + \frac{2P}{b_p}.$$

Then P is finite and, for every $x \in S^2$ and $r > 0$,

$$\mu(\Gamma \cap B_r(x)) \leq Cr.$$

Also, with $L = \mu(\Gamma)$ and $V = \int_{\Gamma} \nu d\mu$,

$$|V| \leq 2\pi, \quad |V| \leq \frac{L^2}{2\pi}.$$

Proof. Smoothness and compactness imply that Γ has finitely many components, each of finite length. We first verify that P is finite from the radial definition in Lemma 2.1. Write $d = |x - y|$ and $m = (3 + p)/2$. In the defining radial integral,

$$1 + \rho^2 - 2\rho x \cdot y = (\rho - 1)^2 + \rho d^2.$$

For $0 < \rho < 1/2$ this denominator is at least $1/4$, and for $\rho > 2$ it is at least $\rho^2/4$. The corresponding integrals of $\rho[(\rho - 1)^2 + \rho d^2]^{-m}$ are therefore bounded by $4^m \int_0^{1/2} \rho d\rho$ and $4^m \int_2^\infty \rho^{-2-p} d\rho$, respectively, uniformly in x, y . On $1/2 \leq \rho \leq 2$, the same integral is at most

$$2 \int_{\mathbb{R}} (t^2 + d^2/2)^{-m} dt = C'_p d^{-2-p},$$

by $t = dv$. The integral defining C'_p is finite because $2m > 1$. Since $0 < d \leq 2$, the two bounded tail contributions can be absorbed by enlarging this constant to a finite C_p . Thus $B_p(x \cdot y) \leq C_p d^{-2-p}$. On each sufficiently small link chart the smooth unit normal obeys $1 - \nu_x \cdot \nu_y = \frac{1}{2}|\nu_x - \nu_y|^2 \leq Cd^2$, and chord distance is comparable to arclength separation. The diagonal integrand defining P is therefore bounded by Cd^{-p} , integrable in the arclength difference since $p < 1$. A finite chart cover gives uniform constants. Its complement in Γ^2 is compact and separated from the diagonal, including pairs on distinct components. Hence the full double integral P is finite.

Fix $x \in S^2$ and first suppose $0 < r \leq \sqrt{2}$. Let $\theta = 2 \arcsin(r/2) \in (0, \pi/2]$. Write

$$m_r = \mu(\Gamma \cap B_r(x)), \quad v_r = \int_{\Gamma \cap B_r(x)} \nu d\mu.$$

First take r^2 to be a regular value of the smooth function $y \mapsto |x - y|^2$ on Γ . The cap circle then meets the finitely many boundary components transversely at finitely many points. Hence $\Omega \cap B_r(x)$ has a boundary consisting of finitely many smooth arcs with corners, and the spherical divergence theorem applies. Sard's theorem makes these regular radii dense in $(0, 2)$. Apply the divergence theorem with the outward orientation on every boundary component,

including the boundary of any hole. For a unit vector $e \perp x$, choose polar coordinates (t, φ) around x with $e \cdot y = \sin t \cos \varphi$. The tangential projection of e has spherical divergence $-2e \cdot y$, and its scalar product with the outward cap conormal is $\cos \theta \cos \varphi$. Consequently

$$e \cdot v_r = -2 \int_{\Omega \cap B_r(x)} e \cdot y d\omega_y - \int_{\Omega \cap \partial B_r(x)} e \cdot \nu_{\text{cap}} d\mu.$$

The positive and negative parts of $\cos \varphi$ each have integral 2 on $[0, 2\pi)$. Thus maximizing the absolute value of each signed integral over arbitrary subsets gives

$$|e \cdot v_r| \leq 4 \int_0^\theta \sin^2 t dt + 2 \sin \theta \cos \theta = 2\theta.$$

Taking the supremum over unit $e \perp x$, and using that $\theta / \sin(\theta/2)$ increases on $(0, \pi/2]$, yields

$$\frac{|\text{proj}_{x^\perp} v_r|}{r} \leq \frac{\pi}{\sqrt{2}}.$$

Indeed, the derivative of that ratio has the sign of $\sin(\theta/2) - (\theta/2) \cos(\theta/2)$, which is positive for $\theta > 0$, and its value at $\theta = \pi/2$ is $\pi/\sqrt{2}$. For the radial component the same divergence identity reads

$$x \cdot v_r = -2 \int_{\Omega \cap B_r(x)} \cos t d\omega + \sin^2 \theta |S|,$$

where $S \subset [0, 2\pi)$ records the portion of the cap circle belonging to Ω , and $|S|$ is its angular Lebesgue measure. Here $x \cdot \nu_{\text{cap}} = -\sin \theta$ and the cap-circle arclength element is $\sin \theta d\varphi$, explaining the second term. Since $\cos t \geq 0$ on this cap, the first term belongs to $[-2\pi \sin^2 \theta, 0]$, because $\int_{B_r(x)} \cos t d\omega = \pi \sin^2 \theta$. The second term belongs to $[0, 2\pi \sin^2 \theta]$. Their sum therefore belongs to $[-2\pi \sin^2 \theta, 2\pi \sin^2 \theta]$. Hence

$$\frac{|x \cdot v_r|}{r} \leq 2\pi r(1 - r^2/4) \leq \frac{8\pi}{3\sqrt{3}},$$

the last maximum being attained at $r = 2/\sqrt{3}$: the derivative of $r(1 - r^2/4)$ is $1 - 3r^2/4$. Orthogonality of the radial and tangential parts, together with $1/2 + 64/27 = 155/54$, gives

$$|v_r|^2 \leq \frac{155\pi^2}{54} r^2.$$

For two points of this cap, their Euclidean distance is at most $2r$. The lower kernel bound in Lemma 2.1 therefore gives

$$P \geq \frac{b_p}{2r^2} \iint_{(\Gamma \cap B_r(x))^2} (1 - \nu_y \cdot \nu_z) d\mu_y d\mu_z = \frac{b_p}{2r^2} (m_r^2 - |v_r|^2).$$

This proves $m_r \leq Cr$ for $r \leq \sqrt{2}$. For $0 < r \leq \sqrt{2}$, choose regular radii $r_j \uparrow r$. The open-cap indicators increase to that of $B_r(x)$, and finiteness of μ gives $m_{r_j} \rightarrow m_r$ and $v_{r_j} \rightarrow v_r$ by monotone and dominated convergence. The bounds on the right are continuous in r . Thus the small-cap estimate holds for every open cap in this range, including a radius whose cap circle contains a boundary arc or component. For a closed cap with $r < \sqrt{2}$, use regular radii decreasing to r and continuity from above of the finite measure. The whole-phase bound below also covers closed caps with $r \geq \sqrt{2}$.

On the whole phase the divergence theorem applied to each constant direction's tangential projection gives $V = -2 \int_\Omega y d\omega_y$. For each unit direction e , the positive and negative parts of $e \cdot y$ on S^2 each have integral π . Thus $|e \cdot V| = 2 |\int_\Omega e \cdot y d\omega_y| \leq 2\pi$, and taking the supremum over e gives $|V| \leq 2\pi$. Since $B_p \geq b_p/2$, $P \geq (b_p/2)(L^2 - |V|^2)$, so

$$L^2 \leq 4\pi^2 + \frac{2P}{b_p}.$$

For $r \geq \sqrt{2}$ it follows that

$$\frac{m_r^2}{r^2} \leq \frac{L^2}{2} \leq 2\pi^2 + \frac{P}{b_p} \leq \frac{155\pi^2}{54} + \frac{2P}{b_p}.$$

This completes the local length estimate.

Finally parametrize a boundary component by arclength $\gamma : [0, \ell] \rightarrow S^2$, periodically. Up to orientation, its conormal integral is $\int_0^\ell \gamma \times \gamma' dt$. Put $\bar{\gamma} = \ell^{-1} \int_0^\ell \gamma dt$. Since $\int_0^\ell \gamma' dt = 0$, subtracting this average does not change the cross-product integral. The vector-valued periodic Wirtinger inequality, obtained by applying its scalar form to the three coordinates, gives

$$\int_0^\ell |\gamma - \bar{\gamma}|^2 dt \leq \left(\frac{\ell}{2\pi}\right)^2 \int_0^\ell |\gamma'|^2 dt.$$

The pointwise cross-product bound and Cauchy–Schwarz now yield

$$\left| \int_0^\ell \gamma \times \gamma' dt \right| \leq \frac{\ell}{2\pi} \int_0^\ell |\gamma'|^2 dt = \frac{\ell^2}{2\pi}.$$

Sum over the finitely many components and use $\sum \ell_i^2 \leq (\sum \ell_i)^2 = L^2$.

□

Lemma 6.3 (Comparison of truncated kernels). *For the kernels in Lemma 2.1, let*

$$w_p(r) = W_p(1-r^2/2), \quad T_p(a) = \int_0^a w_p(r) dr, \quad c_p = \frac{(1-p)^2}{b_p} \quad (0 < p < 1, \ 0 < a \leq 2).$$

Define the auxiliary order-zero kernel and its integral solely by

$$w_0(r) = \int_0^\infty F_1(u)(u^2 + r^2)^{-3/2} du, \quad T_0(a) = \int_0^a w_0(r) dr,$$

where $F_1(u) = 2 - 4/\sqrt{u^2 + 4}$. These integrals are finite for $a > 0$. Then

$$\frac{T_p(a)}{b_p} \leq c_p a^{-p} \left[T_0(a) + \frac{\pi a p}{16(1-p)} \right].$$

Furthermore

$$\frac{T_0(59/500)}{59/500} < \frac{14}{15}.$$

The quotient $T_p(a)/a$ is decreasing in a , also for $p = 0$.

Proof. Use the exact positive representation and the bounds $F_{1-p} \leq (1-p)^2 F_1 \leq (1-p)^2 u^2/4$ from Lemma 2.1. In the strip $u > 0$, $0 < r < a$, put $\rho = \sqrt{u^2 + r^2}$. For $\rho \geq a$ one has $\rho^{-p} \leq a^{-p}$. On the quarter disk $\rho < a$, bound only the positive difference $\rho^{-p} - a^{-p}$ by using $F_1 \leq u^2/4$. Polar coordinates have angular integral $\frac{1}{4} \int_0^{\pi/2} \cos^2 \theta d\theta = \pi/16$. It follows by Tonelli's theorem that

$$\begin{aligned} \frac{T_p(a)}{(1-p)^2} &\leq a^{-p} T_0(a) + \frac{\pi}{16} \int_0^a (\rho^{-p} - a^{-p}) d\rho \\ &= a^{-p} \left[T_0(a) + \frac{\pi a p}{16(1-p)} \right]. \end{aligned}$$

The radial integral is finite exactly in the range $p < 1$. For clarity, Tonelli's theorem writes $T_0(a)$ as the positive double integral over this strip. On its portion $\rho < 1$, the bound $F_1 \leq u^2/4$ and the polar Jacobian bound its integral by $\pi/16$. On $u < 1$, $\rho \geq 1$, its integrand is at most $u^2/4$, so that portion is at most $a/12$. On $u \geq 1$, $F_1 \leq 2$ bounds the integral by $2a \int_1^\infty u^{-3} du = a$. Thus $T_0(a) \leq \pi/16 + 13a/12 < \infty$. The preceding comparison and $p < 1$ now also give $T_p(a) < \infty$. For $\varepsilon \leq r \leq 2$, the defining integrands for w_p , including $p = 0$, are bounded by $C_\varepsilon u^2$ on $0 < u < 1$ and by $2u^{-3}$ on $u \geq 1$, using $F_{1-p} \leq F_1 \leq \min\{u^2/4, 2\}$. Dominated convergence proves that w_p is continuous on $(0, 2]$. Hence $T'_p(a) = w_p(a)$ for

$0 < a < 2$, with continuous extension to $a = 2$. For $0 < r < s$, the positive integral defining w_p has a smaller denominator at r , so $w_p(r) \geq w_p(s)$, also at $p = 0$. Consequently $T_p(a) \geq aw_p(a)$, and $(T_p(a)/a)' = (aw_p(a) - T_p(a))/a^2 \leq 0$.

Here is an elementary certificate for the last numerical constant. Integrating first in r gives

$$\frac{T_0(a)}{a} = \int_0^\infty \frac{F_1(u)}{u^2 \sqrt{u^2 + a^2}} du.$$

The rational upper bound $F_1(u) \leq 2u^2/(u^2 + 8)$ gives, with $q = \sqrt{1 - a^2/8}$ and $0 < a \leq 2$,

$$2 \int_0^\infty \frac{du}{(u^2 + 8)\sqrt{u^2 + a^2}} = \frac{\operatorname{arctanh} q}{4q}.$$

Indeed $u = a \sinh t$ followed by $v = \tanh t$ transforms the integral into $\frac{1}{4} \int_0^1 (1 - q^2 v^2)^{-1} dv$. Since $0 < a \leq 2$ gives $0 < q < 1$, evaluating the primitive $\operatorname{arctanh}(qv)/(4q)$ at $v = 0, 1$ gives the displayed value. Retain part of the exact difference between the rational upper bound and F_1 . For any $u_0 > 0$, let $z_0 = \sqrt{u_0^2 + 4}$. To see the subtraction explicitly, put $z = \sqrt{u^2 + 4}$. Then

$$\frac{2u^2}{u^2 + 8} - F_1(u) = \frac{4(z - 2)^2}{z(z^2 + 4)} = \frac{4u^4}{(u^2 + 8)\sqrt{u^2 + 4}(\sqrt{u^2 + 4} + 2)^2}.$$

Dividing by $u^2 \sqrt{u^2 + a^2}$ gives the difference of the two integrands. Restricting that nonnegative difference to $u \geq u_0$ gives

$$\frac{T_0(a)}{a} \leq \frac{\operatorname{arctanh} q}{4q} - \frac{u_0}{\sqrt{u_0^2 + a^2}} D(u_0),$$

where

$$D(u_0) = \int_{z_0}^\infty \frac{4}{(z^2 + 4)(z + 2)^2} dz = \frac{1}{8} \log \frac{z_0^2 + 4}{(z_0 + 2)^2} + \frac{1}{2(z_0 + 2)}.$$

Indeed the exact difference before substitution is

$$\int_0^\infty \frac{4u^2 du}{(u^2 + 8)\sqrt{u^2 + 4}(\sqrt{u^2 + 4} + 2)^2 \sqrt{u^2 + a^2}}.$$

Restrict to $u \geq u_0$, bound $u/\sqrt{u^2 + a^2}$ below by its value at u_0 , and set $z = \sqrt{u^2 + 4}$. To verify its sign and endpoint, put $h(z) = \frac{1}{8} \log[(z^2 + 4)/(z + 2)^2] + 1/[2(z + 2)]$. Direct differentiation gives $-h'(z) = 4/[(z^2 + 4)(z + 2)^2]$, and $h(z) \rightarrow 0$ as $z \rightarrow \infty$. Thus $D(u_0) = h(z_0)$, as displayed.

Now take $a = 59/500$ and $u_0 = 2/5$. Squaring verifies

$$q > \frac{9991}{10000}, \quad \frac{u_0}{\sqrt{u_0^2 + a^2}} > \frac{959}{1000}.$$

The positive square margins are

$$q^2 - \left(\frac{9991}{10000}\right)^2 = \frac{5869}{100000000}, \quad \frac{u_0^2}{u_0^2 + a^2} - \left(\frac{959}{1000}\right)^2 = \frac{11350439}{43481000000}.$$

Here $z_0 = 2\sqrt{26}/5 < 51/25$, since $(51/25)^2 - z_0^2 = 1/625 > 0$. Also $(z_0^2 + 4)/(z_0 + 2)^2 \geq 1/2$, since its equivalent inequality is $(z_0 - 2)^2 \geq 0$. Hence

$$D(2/5) > -\frac{\log 2}{8} + \frac{25}{202} > \frac{25}{202} - \frac{1733}{20000} > \frac{371}{10000}.$$

The last rational difference is $1/80800 > 0$. For the logarithm bound used here, the positive series for $\log 2 = 2 \sum_{j \geq 0} [(2j + 1)3^{2j+1}]^{-1}$ has tail

$$2 \sum_{j=4}^\infty \frac{1}{(2j + 1)3^{2j+1}} < \frac{2}{9} \sum_{j=4}^\infty 3^{-(2j+1)} = \frac{1}{4 \cdot 3^9}.$$

Therefore

$$\log 2 < 2 \sum_{j=0}^3 \frac{1}{(2j+1)3^{2j+1}} + \frac{1}{4 \cdot 3^9} = \frac{1910051}{2755620} < \frac{1733}{2500}.$$

The last difference is $9049/172226250 > 0$. Moreover

$$\operatorname{arctanh} q = \log \frac{1+q}{\sqrt{1-q^2}} < \log \frac{2000\sqrt{2}}{59} < \frac{387}{100}.$$

For the first inequality, $1+q < 2$ and $\sqrt{1-q^2} = a/\sqrt{8}$ bound the logarithm's argument by $2000\sqrt{2}/59$. For the last inequality, $2000\sqrt{2}/59 < 2397/50$ follows from the positive square margin

$$\left(\frac{2397}{50}\right)^2 - \left(\frac{2000\sqrt{2}}{59}\right)^2 = \frac{464929}{8702500} > 0.$$

The finite positive-series check is

$$\sum_{j=0}^{16} \frac{(387/100)^j}{j!} > \frac{2397}{50}.$$

For a short rational certificate, its consecutive groups $j = 0, \dots, 4, 5, \dots, 8, 9, \dots, 12$, and $13, \dots, 16$ are respectively greater than $31364/1000$, $15727/1000$, $840/1000$, and $9/1000$. These lower bounds sum to $2397/50$; each group comparison is a finite comparison of positive rationals. Therefore $e^{387/100} > 2397/50$ and the logarithmic bound follows. Combining these rational bounds gives

$$\frac{T_0(59/500)}{59/500} < \frac{9675}{9991} - \frac{959}{1000} \frac{371}{10000} < \frac{14}{15}.$$

For example the difference between the last right side and the preceding rational expression is $162063697/299730000000 > 0$. □

Lemma 6.4 (Uniform row coefficients). *Use η_p, K_p from Lemma 4.2, H_p, b_p from Lemma 3.2, and c_p, T_0, w_0 from Lemma 6.3. Put $\eta_0 = \pi - 4 \log 2$. For every $0 < p < 1$,*

$$\eta_p \leq c_p \left[\eta_0 + \frac{p}{24(2-p)} \right] \leq \eta_0 < \frac{3691}{10000}, \quad \frac{H_p}{b_p} < \frac{359}{250}(1-p).$$

Consequently, for every nontrivial stationary stable cone with the smooth compact link hypotheses of Proposition 2.3,

$$\frac{P}{b_p L} < \frac{(359/250)(1-p) + 11/200}{6309/10000}, \quad L = |\Gamma|.$$

Proof. Use c_p, w_0 and the truncated kernels from Lemma 6.3. The positive representation in Lemma 2.1 gives

$$\eta_p \leq c_p \int_0^2 \int_0^\infty r F_1(u) (u^2 + r^2)^{-(3+p)/2} du dr.$$

Compare this double integral with its order-zero value. Outside the unit quarter disk, the extra factor $(u^2 + r^2)^{-p/2}$ is at most one. On the disk, bound its positive difference from one using $F_1 \leq u^2/4$. Polar coordinates now have angular factor $\frac{1}{4} \int_0^{\pi/2} \cos^2 \theta \sin \theta d\theta = 1/12$. Therefore

$$\eta_p \leq c_p \left[\int_0^2 r w_0(r) dr + \frac{1}{12} \int_0^1 (\rho^{1-p} - \rho) d\rho \right] = c_p \left[\eta_0 + \frac{p}{24(2-p)} \right].$$

For the order-zero integral, Tonelli's theorem and integration in r give

$$\int_0^2 r w_0(r) dr = \int_0^\infty F_1(u) \left(\frac{1}{u} - \frac{1}{\sqrt{u^2 + 4}} \right) du.$$

Set $u = 2 \sinh t$ and then $v = e^{-t}$. The integral becomes

$$\begin{aligned} & 2 \int_0^\infty \left(1 - \frac{1}{\cosh t}\right) \left(\frac{\cosh t}{\sinh t} - 1\right) dt \\ &= \int_0^1 \frac{4v(1-v)}{(1+v^2)(1+v)} dv = \int_0^1 \left(\frac{4}{1+v^2} - \frac{4}{1+v}\right) dv = \pi - 4 \log 2. \end{aligned}$$

The substitutions may first be made on compact intervals and then extended by monotone convergence. The integrand in v is nonnegative, is $O(v)$ at zero and $O(1-v)$ at one. This proves finiteness as well as the value of the order-zero integral. The polynomial identity

$$c_p - \left(1 - \frac{p}{2}\right) = \frac{p^2}{2}(p^2 + p - 3) \leq 0$$

holds because $0 < p < 1$. Also $p/[24(2-p)] \leq p/24$. Since $\eta_0 > 1/12$, these bounds give

$$\begin{aligned} c_p \left[\eta_0 + \frac{p}{24(2-p)} \right] &\leq \left(1 - \frac{p}{2}\right) \left(\eta_0 + \frac{p}{24}\right) \\ &= \eta_0 + p \left(\frac{1}{24} - \frac{\eta_0}{2}\right) - \frac{p^2}{48} \leq \eta_0. \end{aligned}$$

For example $\pi > 3$ and $\log 2 < 7/10$ imply $\eta_0 > 1/5 > 1/12$. The sharper upper bound follows from

$$\pi < \frac{355}{113}, \quad \log 2 > 2 \sum_{j=0}^4 \frac{1}{(2j+1)3^{2j+1}},$$

by rational substitution: the resulting upper bound and its positive margin are

$$\frac{355}{113} - 8 \sum_{j=0}^4 \frac{1}{(2j+1)3^{2j+1}} = \frac{258533563}{700616385}, \quad \frac{3691}{10000} - \frac{258533563}{700616385} = \frac{127889407}{1401232770000} > 0.$$

The stated bound on π , as well as $157/50 < \pi < 22/7$ used below, follows from Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and alternating series: use terms through degree nine for the upper bound on $\arctan(1/5)$ and through degree three for the lower bound on $\arctan(1/239)$; terms through degree seven and the first term, respectively, give the needed lower bound on π .

Write $h(p) = H_p/b_p$. Duplication in the exact gamma formula of Lemma 3.2 gives

$$h(p) = \pi(2+p)2^{-1-p} \frac{\Gamma(a)}{\Gamma(a+1/2)} \frac{\Gamma(d+1/2)}{\Gamma(d)}, \quad a = \frac{3+p}{4} = 1-d, \quad d = \frac{1-p}{4}.$$

Here $3/4 < a < 1$ and $0 < d < 1/4$, so every gamma argument is positive and logarithmic differentiation is legitimate. The formula extends continuously to $p = 0$, where $a = 3/4$ and $d = 1/4$. Let $\psi = \Gamma'/\Gamma$ and $\Delta(t) = \psi(t+1/2) - \psi(t)$ for $t > 0$. The trigamma series

$$\psi'(t) = \sum_{n=0}^{\infty} \frac{1}{(n+t)^2}$$

converges locally uniformly, and shows that ψ' is strictly decreasing. Hence $\Delta'(t) = \psi'(t+1/2) - \psi'(t) < 0$. Logarithmically differentiating the formula for $h(p)$, and using $\psi(d+1) = \psi(d) + 1/d$ with $4d = 1-p$, gives

$$\frac{d}{dp} \log \frac{h(p)}{1-p} = \frac{1}{2+p} - \log 2 + \frac{1}{4} [\psi(a) - \psi(a+1/2) + \psi(d+1) - \psi(d+1/2)].$$

Since $a < 1$, monotonicity of Δ gives

$$\psi(a) - \psi(a+1/2) = -\Delta(a) \leq -\Delta(1) = 2 \log 2 - 2.$$

Since $d > 0$, it also gives

$$\psi(d+1) - \psi(d+1/2) = \Delta(d+1/2) \leq \Delta(1/2) = 2 \log 2.$$

Here the endpoint values follow from $\psi(1) - \psi(1/2) = 2 \log 2$ and the digamma recurrence. Thus the displayed logarithmic derivative is at most $1/(2+p) - 1/2 < 0$. In particular

$$h(p) \leq h(0)(1-p), \quad h(0) = \frac{4\pi\Gamma(3/4)^2}{\Gamma(1/4)^2}.$$

These are limits of explicit integrals or gamma expressions; no surface equation of order zero is being invoked.

To bound $h(0)$ rationally, let

$$I = \int_0^{\pi/2} (1 + \sin^2 t)^{-1/2} dt.$$

For the beta evaluation, first substitute $x = \sin t$. Since $dt = dx/\sqrt{1-x^2}$, one obtains

$$I = \int_0^1 \frac{dx}{\sqrt{1-x^2}\sqrt{1+x^2}} = \int_0^1 (1-x^4)^{-1/2} dx.$$

Next set $y = x^4$, so $dx = \frac{1}{4}y^{-3/4} dy$. Hence

$$I = \frac{1}{4} \int_0^1 y^{-3/4} (1-y)^{-1/2} dy = \frac{1}{4} B(1/4, 1/2) = \frac{\sqrt{\pi} \Gamma(1/4)}{4\Gamma(3/4)}.$$

The substitutions can first be made on compact subintervals and then extended by monotone convergence. The endpoint powers $-3/4$ and $-1/2$ are both greater than -1 , so the beta integral is finite. Substituting this identity into the already computed expression for $h(0)$ gives

$$\frac{\pi^2}{4I^2} = \frac{4\pi\Gamma(3/4)^2}{\Gamma(1/4)^2} = h(0).$$

Expand

$$(1 + \sin^2 t)^{-1/2} = \sqrt{\frac{2}{3}} \left(1 - \frac{1}{3} \cos 2t\right)^{-1/2}.$$

The binomial series

$$\left(1 - \frac{1}{3} \cos 2t\right)^{-1/2} = \sum_{n=0}^{\infty} \frac{(1/2)_n}{n!} \left(\frac{\cos 2t}{3}\right)^n$$

converges uniformly on $[0, \pi/2]$: its terms are bounded by 3^{-n} because $(1/2)_n/n! \leq 1$. Termwise integration is therefore valid. Odd powers integrate to zero under $t \mapsto \pi/2 - t$, and all even contributions are positive. In particular

$$\int_0^{\pi/2} \cos^2(2t) dt = \frac{\pi}{4}, \quad \int_0^{\pi/2} \cos^4(2t) dt = \frac{3\pi}{16}.$$

Using the binomial coefficients $3/8$ and $35/128$ for powers two and four, and retaining these together with the constant term, gives

$$I \geq \frac{\pi}{\sqrt{6}} \left(1 + \frac{1}{48} + \frac{35}{27648}\right),$$

and hence the rational estimate

$$h(0) \leq \frac{3}{2(1 + 1/48 + 35/27648)^2} = \frac{1146617856}{798571081} < \frac{359}{250}.$$

The difference in the last comparison is $32554079/199642770250 > 0$. This proves the claimed bound for H_p/b_p .

For the stable cone in the statement, the link is nonempty, so $0 < L < \infty$. Its spherical phase $\Omega = E \cap S^2$ is a nonempty proper smooth open set with boundary Γ and outward conormal ν . Thus Lemma 6.2 gives $P < \infty$ throughout $0 < p < 1$. Finally the constant angular test in Proposition 2.3 and the row inequality of Lemma 4.2 imply $(1 - \eta_p)P \leq K_p L$.

By Lemma 6.1, $(K_p - H_p)/b_p < 11/200$ for every $0 < p < 1$. Since $1 - \eta_p > 6309/10000 > 0$, division is legitimate and gives

$$\frac{P}{b_p L} \leq \frac{K_p/b_p}{1 - \eta_p} < \frac{(359/250)(1 - p) + 11/200}{6309/10000},$$

as claimed. Only the comparison integrals and the continuous gamma formula have been evaluated at $p = 0$; stability was used solely at the given positive order. $\square$

7. EXACT NORMAL-FLUX PINCHING AT SMALL ORDERS

Lemma 7.1 (Pinching with the exact truncated kernel). *Let E be a nontrivial stationary stable p -perimeter cone in $\mathbb{R}^3$, $0 < p < 1$, with locally one-sided smooth boundary away from zero and compact smooth link Γ . With $L = |\Gamma|$, $V = \int_{\Gamma} \nu d\mu$, set*

$$\delta = 1 - \frac{|V|}{L}, \quad \alpha = \frac{155}{216}.$$

For $0 < \delta < 1$ define

$$D(\delta) = \sqrt{\frac{\alpha}{(1 - \delta)^2} + \delta(2 - \delta)}, \quad a(\delta) = \frac{\delta}{2D(\delta)}.$$

If $\delta = 0$, then E is a halfspace. If $0 < \delta < 1$, then, for T_p from Lemma 6.3,

$$1 \leq \Phi_p(\delta) := \frac{2\delta}{2 - \delta} \frac{T_p(a(\delta))}{b_p a(\delta)}.$$

For each fixed p , Φ_p is increasing on $(0, 1)$. The test below requires $V \neq 0$ and makes no assertion at $\delta = 1$; both final flatness arguments first exclude that value by their row bounds.

Proof. Use w_p, T_p and their decreasing-mean property from Lemma 6.3. Use P, q_B, q_A from Proposition 2.3. Nontriviality and the smooth compact link give $0 < L < \infty$. The phase $\Omega = E \cap S^2$ is a nonempty proper smooth open set with boundary Γ , so Lemma 6.2 states that $P < \infty$. The formula for F_{1-p} in Lemma 2.1 shows that it is strictly positive for $u > 0$, since $0 < 1 - p < 1$. Hence $w_p(r) > 0$ for $0 < r \leq 2$, and the finite integrals of Lemma 6.3 satisfy $T_p(a) > 0$ for $0 < a \leq 2$.

The positive representation in Lemma 2.1 also justifies local absolute continuity of w_p . For $\varepsilon \leq r \leq 2$, its integrand and its r derivative are bounded by constant multiples of

$$\frac{u^2}{(u^2 + \varepsilon^2)^{(3+p)/2}}, \quad \frac{u^2}{(u^2 + \varepsilon^2)^{(5+p)/2}},$$

respectively. Both are integrable on $(0, \infty)$ since $p > 0$. Dominated differentiation therefore gives $w_p \in C^1((0, 2])$ and

$$w'_p(r) = -(3 + p)r \int_0^\infty \frac{F_{1-p}(u)}{(u^2 + r^2)^{(5+p)/2}} du \leq 0.$$

The bound $F_{1-p}(u) \leq (1 - p)^2 u^2 / 4$ and the substitution $u = rv$ give $|w'_p(r)| \leq C_p r^{-1-p}$. The same substitution in the undifferentiated integral gives $w_p(r) \leq C_p r^{-p}$: the resulting integral is $\int_0^\infty v^2(1 + v^2)^{-(3+p)/2} dv < \infty$ because its tail is $O(v^{-1-p})$. Thus $r|w'_p(r)|$ is integrable at zero because $p < 1$, and $T'_p(a) = w_p(a)$ for $a > 0$. These bounds justify the derivatives and improper integrations by parts below. If $\delta = 0$, then $V \neq 0$, and with $e = V/|V|$ one has

$$\int_{\Gamma} |\nu - e|^2 d\mu = 2(L - |V|) = 0.$$

Continuity gives $\nu = e$ on every component. Since the cone's radial direction is tangent to its boundary, $x \cdot \nu_x = 0$, so the link lies in the orthogonal great circle $G = S^2 \cap e^\perp$. A nonempty component of the compact embedded one-dimensional link is open in G by the inverse function theorem, and is closed by compactness. Connectedness of G makes that

component all of G , hence $\Gamma = G$. Each complementary hemisphere has no phase boundary, so its membership in the open phase is constant. Local one-sidedness forces exactly one hemisphere to belong to the phase. Thus E is a halfspace.

Suppose $0 < \delta < 1$, choose $e = V/|V|$, and put $g = 1 - \nu \cdot e$. Then $0 \leq g \leq 2$ and

$$M := \int_{\Gamma} g d\mu = L\delta.$$

With $h = \nu \cdot e$, the translation Jacobi identity in Proposition 2.3 gives $q_B(h, 1) = q_B[h] = 0$, while $q_B[1] = -P$. Polarization yields $q_B[g] = -P$. Its stated angular stability identity therefore gives $0 \leq q_A[g] = -P + \iint W_p g_x g_y$, or

$$P \leq \iint_{\Gamma^2} W_p(x \cdot y) g_x g_y d\mu_x d\mu_y.$$

The measure $\lambda = g\mu$ has mass M and, by Lemma 6.2, satisfies $\lambda(B_r(x)) \leq \min\{2Cr, M\}$ with $C^2 = 155\pi^2/54 + 2P/b_p$. The cap-density bound at radius two gives $L \leq 2C$: the only point of the sphere omitted from $B_2(x)$ is $-x$, which has zero arclength measure. Since $M = L\delta > 0$, the cutoff below belongs to $(0, \delta] \subset (0, 1)$; in particular

$$a := \frac{M}{2C} \leq 1.$$

For a fixed $x \in \Gamma$, put $m_x(r) = \lambda(B_r(x))$. The measure λ has no atoms because $0 \leq g \leq 2$ and arclength is nonatomic. In particular $m_x(2) = M$ up to the harmless choice of including the antipodal point. At the lower endpoint, the positive kernel representation and $F_{1-p} \leq (1-p)^2 u^2/4$ give $w_p(r) = O_p(r^{-p})$, whereas $m_x(r) \leq 2Cr$. Thus $w_p(r)m_x(r) \rightarrow 0$ as $r \downarrow 0$. For $d = |x - y| > 0$, local absolute continuity gives $w_p(d) = w_p(2) + \int_d^2 [-w'_p(r)] dr$. Integrating this nonnegative layer identity and applying Tonelli gives the first equality below. It is also the Stieltjes integration-by-parts formula, and holds when m_x has jumps at positive radii: the value of m_x at a jump does not affect the Lebesgue integral in r . Thus

$$\begin{aligned} \int_{\Gamma} W_p(x \cdot y) d\lambda(y) &= Mw_p(2) + \int_0^2 [-w'_p(r)] m_x(r) dr \\ &\leq Mw_p(2) + 2C \int_0^a r [-w'_p(r)] dr + M \int_a^2 [-w'_p(r)] dr \\ &= Mw_p(2) + 2C \left[-aw_p(a) + \int_0^a w_p(r) dr \right] + M[w_p(a) - w_p(2)] \\ &= 2C T_p(a), \end{aligned}$$

where the last equality uses $M = 2Ca$. The lower endpoint $rw_p(r) \rightarrow 0$ in the second integration by parts follows again from $p < 1$. This bound is uniform in x . Integrating it against $d\lambda(x)$, whose total mass is M , proves

$$P \leq 2CM T_p\left(\frac{M}{2C}\right).$$

Set $R = 2P/(b_p L^2)$. The lower bound $B_p \geq b_p/2$ from Lemma 2.1 and the flux bound in Lemma 6.2 give

$$R \geq \delta(2 - \delta), \quad L \geq 2\pi(1 - \delta), \quad \frac{C}{L} \leq \sqrt{\frac{\alpha}{(1 - \delta)^2}} + R.$$

For the length bound, specifically use $|V| \leq L^2/(2\pi)$ from Lemma 6.2: because $|V| = L(1 - \delta)$ and $L > 0$, this gives $L \geq 2\pi(1 - \delta)$. Consequently

$$\left(\frac{C}{L}\right)^2 = \frac{155\pi^2}{54L^2} + R \leq \frac{155}{216(1 - \delta)^2} + R,$$

which explains the constant α . For fixed $\delta > 0$, consider $zT_p(\delta/(2z))$ on $z > \delta/4$, where its argument $a = \delta/(2z)$ belongs to $(0, 2)$. Its derivative is $T_p(a) - aw_p(a) \geq 0$ by the decreasing-mean property. The actual value $z = C/L \geq 1/2$ and its upper bound $\sqrt{\alpha/(1-\delta)^2 + R}$ both lie in this domain. Consequently the preceding inequality for P implies

$$R \leq \frac{4\delta}{b_p} \sqrt{\frac{\alpha}{(1-\delta)^2} + R} T_p\left(\frac{\delta}{2\sqrt{\alpha/(1-\delta)^2 + R}}\right).$$

One can replace R here by its lower bound. Indeed, put $A = \alpha/(1-\delta)^2$ and $G(R) = \sqrt{A+R}T_p(\delta/(2\sqrt{A+R}))$ for $R > 0$. All its cutoffs satisfy

$$0 < \frac{\delta}{2\sqrt{A+R}} \leq \frac{\delta(1-\delta)}{2\sqrt{\alpha}} \leq \frac{1}{8\sqrt{\alpha}} < \frac{1}{4} < 2,$$

since $\alpha > 1/4$. Thus $G(R) > 0$, and all derivatives and divisions below are within the domain of T_p . Then

$$0 \leq \frac{G'(R)}{G(R)} = \frac{1}{2(A+R)} \left(1 - \frac{aw_p(a)}{T_p(a)}\right) \leq \frac{1}{2(A+R)} < \frac{1}{R}.$$

Thus $R/G(R)$ increases. Put $R_0 = \delta(2-\delta) > 0$. The preceding bound on R and $R \geq R_0$ imply

$$\frac{R_0}{G(R_0)} \leq \frac{R}{G(R)} \leq \frac{4\delta}{b_p}.$$

Since $G(R_0) = D(\delta)T_p(a(\delta))$, division by R_0 gives

$$1 \leq \frac{4D(\delta)T_p(a(\delta))}{b_p(2-\delta)} = \frac{2\delta}{2-\delta} \frac{T_p(a(\delta))}{b_p a(\delta)}.$$

Finally $D(\delta)$ increases, since

$$\frac{d}{d\delta} D(\delta)^2 = \frac{2\alpha}{(1-\delta)^3} + 2(1-\delta) > 0.$$

On its cutoff domain, $zT_p(\delta/(2z))$ is nondecreasing in z , as already shown, and its derivative in δ at fixed z is $w_p(\delta/(2z))/2 > 0$. For $0 < \delta_1 < \delta_2 < 1$, first increase z from $D(\delta_1)$ to $D(\delta_2)$ with $\delta = \delta_1$, then increase δ to δ_2 at that fixed z . On the first segment the cutoff is at most $a(\delta_1) < 2$; on the second it is at most $a(\delta_2) < 2$. Thus both comparisons are legitimate. Since $2-\delta$ decreases, the equivalent formula

$$\Phi_p(\delta) = \frac{4D(\delta)T_p(\delta/(2D(\delta)))}{b_p(2-\delta)}$$

therefore proves its monotonicity. □

Theorem 7.2 (Flatness for $0 < p \leq 1/3$). *Let $0 < p \leq 1/3$. Every nontrivial stationary p -perimeter cone in $\mathbb{R}^3$ with locally one-sided smooth boundary away from zero, compact smooth spherical link, and stability under compact variations away from zero is a halfspace. No convexity, symmetry, or link connectivity is assumed.*

Proof. Let $\Omega = E \cap S^2$, $\Gamma = \partial\Omega$, $A = |\Omega|$, $L = |\Gamma|$, $V = \int_{\Gamma} \nu d\mu$, and $v = |V|/L$. The nontrivial locally one-sided cone has a nonempty proper smooth spherical phase, with all boundary components included. Thus $0 < A < 4\pi$ and $0 < L < \infty$, and its outward normal is the spherical conormal of this phase. The triangle inequality gives $0 \leq v \leq 1$. Use P, b_p as in Proposition 2.3. Lemma 6.2 ensures $P < \infty$ for the positive order under consideration. If $v = 1$, Lemma 7.1 already gives the conclusion. Suppose $v < 1$. The kernel lower bound in Lemma 2.1, the area estimate Lemma 4.1, and the spherical isoperimetric inequality for the whole smooth phase give

$$\frac{P}{b_p} \geq \frac{(1-v^2)L^2}{2}, \quad \frac{P}{b_p} \geq (2\pi - A)^2 \geq 4\pi^2 - L^2.$$

For the spherical isoperimetric step, Ω has finite perimeter L . Choose its equal-area cap with polar angle $\theta \in (0, \pi)$: $A = 2\pi(1 - \cos \theta)$ and $L \geq 2\pi \sin \theta$. Hence $L^2 \geq A(4\pi - A)$, equivalently $(2\pi - A)^2 \geq 4\pi^2 - L^2$. This inequality allows disconnected phases, holes, and area larger than a hemisphere. For fixed $v < 1$, minimize the maximum of these bounds divided by $L > 0$. The increasing expression $(1 - v^2)L/2$ and the decreasing expression $4\pi^2/L - L$ cross at $L^2 = 8\pi^2/(3 - v^2)$. The crossing is unique: the difference of the increasing and decreasing expressions goes from negative to positive as L runs from zero to infinity. Below the crossing the decreasing expression is the larger and is at least its crossing value; above it the increasing expression is the larger and is at least that value. Thus

$$\frac{P}{b_p L} \geq F(v) := \frac{\pi\sqrt{2}(1 - v^2)}{\sqrt{3 - v^2}}.$$

This function is strictly decreasing on $[0, 1)$, since

$$F'(v) = \frac{\pi\sqrt{2}v(v^2 - 5)}{(3 - v^2)^{3/2}} < 0 \quad (0 < v < 1).$$

It is also strictly concave:

$$F''(v) = -\frac{\pi\sqrt{2}(v^2 + 15)}{(3 - v^2)^{5/2}} < 0.$$

Let

$$v_*(p) = \frac{38}{125} + p, \quad d(p) = 1 - v_*(p) = \frac{87}{125} - p, \quad c_*(p) = \frac{(359/250)(1 - p) + 11/200}{6309/10000}.$$

The coefficient estimate Lemma 6.4 gives $F(v) < c_*(p)$. On the other hand,

$$F(v_*(p)) > c_*(p) \quad (0 \leq p \leq 1/3).$$

To check this last inequality, put $J(p) = F(38/125 + p) - c_*(p)$. Since c_* is affine, $J''(p) = F''(38/125 + p) < 0$ on $[0, 1/3]$. A concave function lies above its endpoint chord, so

$$J(p) \geq (1 - 3p)J(0) + 3pJ(1/3).$$

It therefore suffices to prove both endpoint values positive. The scalar endpoint $p = 0$ invokes no zero-order stability. The lower bound $\pi > 157/50$ reduces these two checks, respectively, to the rational inequalities

$$\frac{2(157/50)^2(1 - (38/125)^2)^2}{3 - (38/125)^2} > \left(\frac{(359/250) + 11/200}{6309/10000} \right)^2$$

and

$$\frac{2(157/50)^2(1 - (239/375)^2)^2}{3 - (239/375)^2} > \left(\frac{(359/250)(2/3) + 11/200}{6309/10000} \right)^2.$$

The differences of left and right sides in these two squared comparisons are, respectively,

$$\frac{4868190251937401}{3924288075761718750} > 0, \quad \frac{67604925960406735676}{638019092666337890625} > 0.$$

All unsquared quantities are positive, so the squared comparisons give the required strict endpoint inequalities. As F decreases, this proves $v > v_*(p) > 0$. In particular

$$0 \leq \delta := 1 - v < d(p) < 1.$$

It remains to exclude $\delta > 0$.

Use $\alpha = 155/216$ and $a(d)$ from Lemma 7.1. For $0 \leq p \leq 1/3$, the number $d = d(p)$ lies in $[136/375, 87/125]$, and

$$a(d) \geq \frac{59}{500}.$$

Here is an exact endpoint check for this bound. Write $H(d) = d(1-d)^3 + \alpha(1-2d)$. Direct differentiation gives

$$a'(d) = \frac{H(d)}{2(1-d)^3 D(d)^3}.$$

The denominator is positive, and

$$H'(d) = (1-d)^2(1-4d) - 2\alpha < 0,$$

because $d \geq 136/375 > 1/4$. Thus H has at most one zero, and any sign change is from positive to negative. Consequently a increases and then decreases, or is monotone; in all cases its minimum on the closed interval is attained at an endpoint. At both endpoints, squaring reduces the claimed lower bound to

$$\frac{d^2}{4[\alpha/(1-d)^2 + d(2-d)]} > \frac{3481}{250000}, \quad d \in \left\{ \frac{136}{375}, \frac{87}{125} \right\},$$

The differences of left and right sides, in the displayed order of endpoints, are

$$\frac{242842143943}{37921011624250000} > 0, \quad \frac{425942995981}{10566226874750000} > 0.$$

Thus the cutoff lower bound holds throughout the interval.

If $\delta > 0$, monotonicity in Lemma 7.1 gives $1 \leq \Phi_p(\delta) \leq \Phi_p(d)$. The definition gives $D(d) \geq \sqrt{\alpha}/(1-d)$, so

$$0 < a(d) \leq \frac{d(1-d)}{2\sqrt{\alpha}} \leq \frac{1}{8\sqrt{\alpha}} < \frac{1}{4} < 2,$$

where $\alpha > 1/4$ and $0 < d < 1$. Thus both cutoff hypotheses hold. Apply Lemma 6.3 at $a = a(d)$. After division by $b_p a > 0$, that comparison gives

$$\frac{T_p(a)}{b_p a} \leq c_p a^{-p} \left[\frac{T_0(a)}{a} + \frac{\pi p}{16(1-p)} \right].$$

Here $a = a(d) \geq 59/500$, so $a^{-p} \leq (500/59)^p$. The decreasing quotient $T_0(a)/a$ and its strict rational certificate at $59/500$ give $T_0(a)/a < 14/15$. Multiplying by the positive factor $2d/(2-d)$ therefore yields

$$1 < \frac{2d}{2-d} c_p \left(\frac{500}{59} \right)^p \left[\frac{14}{15} + \frac{\pi p}{16(1-p)} \right].$$

It is enough for the contradiction to bound this last expression by a number smaller than one. Using $\pi < 22/7$ and $c_p = (1-p)^2(1+p)(2+p)/2$, it is strictly smaller than

$$\mathcal{B}(p) = \frac{d(p)}{2-d(p)} (1-p)(1+p)(2+p) \left(\frac{14}{15} - \frac{619p}{840} \right) \left(\frac{500}{59} \right)^p.$$

The bracket results from the exact identity

$$\frac{14}{15}(1-p) + \frac{11p}{56} = \frac{14}{15} - \frac{619p}{840}.$$

All factors are positive on $[0, 1/3]$: in particular the bracket is decreasing and has positive value $1733/2520$ at $p = 1/3$. Its logarithmic derivative is

$$\begin{aligned} (\log \mathcal{B})'(p) = & -\frac{1}{d} - \frac{1}{2-d} - \frac{1}{1-p} + \frac{1}{1+p} + \frac{1}{2+p} \\ & - \frac{619}{784-619p} + \log \frac{500}{59}. \end{aligned}$$

This derivative is strictly decreasing: the derivative of its first pair is $-1/d^2 + 1/(2-d)^2 < 0$, and the derivatives of each of its remaining nonconstant rational terms are negative. At $p = 0$ it is strictly below $-1/5$. Indeed

$$\frac{125}{87} > \frac{7}{5}, \quad \frac{125}{163} > \frac{3}{4}, \quad \frac{619}{784} > \frac{3}{4}, \quad \log(500/59) < \log 9 < \frac{11}{5},$$

which give the claimed strict bound by substitution. Substitution gives $(\log \mathcal{B})'(0) < -7/5 - 3/4 + 1/2 - 3/4 + 11/5 = -1/5$. For the last logarithm inequality, the first six positive terms of the exponential series give

$$e^{11/10} > \sum_{j=0}^5 \frac{(11/10)^j}{j!} = \frac{36015101}{12000000} = 3 + \frac{15101}{12000000} > 3.$$

Thus $\log 9 = 2 \log 3 < 11/5$. It follows that $\mathcal{B}$ decreases on $[0, 1/3]$. Finally

$$\mathcal{B}(0) = \frac{812}{815} < 1.$$

This is incompatible with the necessary pinching inequality. Thus $\delta = 0$, and Lemma 7.1 makes the cone a halfspace. $\square$

Proof of Theorem 1.3. If $0 < p \leq 1/3$, apply Theorem 7.2. If $1/3 \leq p < 1$, apply Theorem 5.3. Both results have exactly the geometric hypotheses stated here, and their intervals cover $(0, 1)$, including their common endpoint. $\square$

8. FROM CONES TO THE FRACTIONAL ALLEN–CAHN EQUATION

The reference locators in this section use the arXiv versions specified in the bibliography. In the geometric and phase-transition references whose operator has symbol $|\xi|^p$, the parameter is our $p = 2s$. Cabré–Sire uses the fractional exponent s and the symbol $|\xi|^{2s}$.

Existence and uniqueness in Definition 1.1. Cabré–Sire [2, Theorem 2.4] gives existence and uniqueness up to translation for the increasing layer under the double-well conditions $G' = -f$, $G'(\pm 1) = 0$, $G > G(\pm 1)$ in $(-1, 1)$ and $f'(\pm 1) < 0$. Here $f(t) = t - t^3$ and $G(t) = (1 - t^2)^2/4$; all these conditions hold, with $G(\pm 1) = 0$ and $f'(\pm 1) = -2$. Choose $\max\{0, 1 - 2s\} < \gamma < 1$ and $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ near $[-1, 1]$. Set $\tilde{f}(t) = \chi(t)(t - t^3)$. Its derivatives are bounded, and $|\tilde{f}'(x) - \tilde{f}'(y)| \leq C \min\{|x - y|, 1\} \leq C|x - y|^\gamma$ for a finite C ; hence its global $C^{1,\gamma}(\mathbb{R})$ norm is finite. The potential $\tilde{G}$, defined by $\tilde{G}' = -\tilde{f}$ and $\tilde{G}(0) = 1/4$, agrees with G on $[-1, 1]$, so every displayed well condition remains true. The theorem gives a profile with derivative strictly positive and limits ± 1 , hence with values strictly between the wells. It therefore solves precisely the cubic equation. The Fourier calculation in Section 3 of the cited paper fixes its operator symbol as $|\xi|^{2s}$, the normalization here. Continuity, strict increase, and the endpoint limits give a unique zero; translating it to zero fixes the remaining translation freedom. Thus this construction exists and is unique, and $q'_s > 0$. $\square$

Lemma 8.1 (Local finiteness and the C^2 cone interface). *Fix $p \in (0, 1)$. Let $E \subset \mathbb{R}^3$ be a nontrivial cone whose boundary is locally one-sided and C^2 away from zero. Its spherical link is compact, embedded, and C^2 , and $\text{Per}_p(E; B_R) < \infty$ for every $R > 0$. Suppose that, for every smooth vector field X supported in $B_1 \setminus \{0\}$, with flow Φ_t ,*

$$\liminf_{t \rightarrow 0} \frac{\text{Per}_p(\Phi_t(E); B_1) - \text{Per}_p(E; B_1)}{t^2} \geq 0.$$

Then E has smooth boundary away from zero and is stationary and stable under every compact smooth variation away from zero. Conversely, a smooth cone with these stationarity and stability properties satisfies the displayed condition. In particular, Theorem 1.3 supplies the nontrivial dimension-three cone hypothesis in Cabré–Cinti–Serra [4, Definitions 1.6 and 2.8].

Proof. Lemma 2.2 proves that the link is compact and C^2 , that the localized perimeters are finite, and that perimeter along every stated flow is twice continuously differentiable. Thus the displayed lower-limit condition involves finite quantities.

Here is a justification of the first-variation formula at this regularity. Replace the kernel by $K_\varepsilon(Z) = (|Z|^2 + \varepsilon^2)^{-(3+p)/2}$ and denote its localized perimeter by $\text{Per}_{p,\varepsilon}$. Its first variation under a field X supported away from zero is, by the divergence theorem,

$$\left. \frac{d}{dt} \right|_{t=0} \text{Per}_{p,\varepsilon}(\Phi_t(E); B_1) = \int_{\partial E} H_{p,\varepsilon,E}(Z) X(Z) \cdot \nu(Z) d\sigma_Z,$$

where $H_{p,\varepsilon,E}(Z) = \int (\chi_{E^c} - \chi_E)(Y) K_\varepsilon(Z - Y) dY$. One may first truncate the divergence integrals near zero and infinity; the field is zero near both truncation boundaries, and the remaining kernel tails converge. The differentiated pullback kernels are bounded, uniformly in ε , by the majorant of Lemma 2.2. Their first variations therefore converge as $\varepsilon \downarrow 0$. On the compact surface patch meeting the field, subtract the tangent halfspace, whose regularized curvature is zero by reflection symmetry. The C^2 graph differs from it by height $O(\rho^2)$, giving a uniform local bound $C \int_0^r \rho^{-p} d\rho$. This tends to zero with r , while off that neighborhood dominated convergence applies, including the integrable tail. Thus $H_{p,\varepsilon,E}$ converges uniformly on the patch to the physical principal-value curvature $H_{p,E}$. The same split into a small graph neighborhood and its complement proves continuity of $H_{p,E}$. Passing to the limit gives

$$\left. \frac{d}{dt} \right|_{t=0} \text{Per}_p(\Phi_t(E); B_1) = \int_{\partial E} H_{p,E}(Z) X(Z) \cdot \nu(Z) d\sigma_Z.$$

If the first variation of a fixed flow were $A \neq 0$, its expansion would be $tA + o(t)$, and one of the two signs of t would contradict the assumed lower limit. Hence the first variation vanishes. The first-variation formula and continuity show that the pointwise nonlocal curvature is zero. More explicitly, if it had one strict sign near a boundary point, choose a nonnegative ambient bump supported in that patch and multiply it by the normal at its center. Its scalar product with the local normal stays positive on a smaller patch, giving a nonzero first variation, a contradiction. This uses only smooth ambient tests, not a smooth extension of the merely C^1 normal field. Dilation carries an arbitrary compact support away from zero into $B_1 \setminus \{0\}$ and multiplies the perimeter by the positive factor R^{3-p} . Thus the same conclusion holds at every nonzero boundary point.

We give the precise regularity interface, rather than applying a minimizer-only regularity statement to a merely stationary cone. At a fixed nonzero point, rotate and shrink a graph cylinder until its C^1 slope is as small as required below. Write the graph as $x_3 = u(x')$, with $u(0) = 0$ and $Du(0) = 0$ after translation. On a sufficiently small disk B'_{2r} we have $|Du| \leq \varepsilon$ and $|u| \leq 2r\varepsilon$. Multiplication by a smooth cutoff equal to one on B'_r , supported in B'_{2r} , and with gradient bounded by C/r , gives a bounded C^2 extension $\tilde{u}$ satisfying $\|D\tilde{u}\|_\infty \leq C\varepsilon$. Thus the extension, as well as the original patch, can be chosen with arbitrarily small slope. Up to reversing the phase, let $F = \{x_3 < \tilde{u}(x')\}$. The sets E and F agree in a neighborhood of the smaller graph patch. The difference of their curvatures is therefore a smooth function of the ambient point X :

$$\Psi(X) = 2 \int_{\mathbb{R}^3} \frac{\chi_F(Y) - \chi_E(Y)}{|X - Y|^{3+p}} dY.$$

For X in an even smaller neighborhood the numerator vanishes within a fixed positive distance r_0 of X . Differentiation of order j is dominated by $C_{j,p}|X - Y|^{-3-p-j}$ on the remaining region; its integral is bounded by $C_{j,p}r_0^{-p-j}$. Thus Ψ is C^∞ in the ambient variables. This does not assert that $\Psi(x', u(x'))$ is smooth before u has been bootstrapped.

Use the sign convention $H_{p,E}(X) = \text{PV} \int (\chi_{E^c} - \chi_E)(Y) |X - Y|^{-3-p} dY$. Then $H_{p,E} - H_{p,F} = \Psi$, so stationarity gives $H_{p,F} = -\Psi$. First exclude a horizontal disk of radius ε and integrate in the vertical variable. Subtracting the tangent halfspace shows that this cutoff and the physical-ball cutoff have the same limit: their remaining local absolute contribution

is bounded by $C \int_0^{C\varepsilon} r^{-p} dr \rightarrow 0$. Thus the curvature of F is twice the principal value of

$$\int_{\mathbb{R}^2} G_p \left(\frac{\tilde{u}(x') - \tilde{u}(y')}{|x' - y'|} \right) \frac{dy'}{|x' - y'|^{2+p}}, \quad G_p(t) = \int_0^t (1 + \tau^2)^{-(3+p)/2} d\tau.$$

Consequently the displayed integral equals $-\Psi(x', u(x'))/2$. Choose an additional smooth ambient cutoff supported in the neighborhood where Ψ is smooth and equal to one near the smaller graph patch. Its product with Ψ , extended by zero, is smooth in (x', z) on all the variables required for a local Schauder theorem. The equation on the graph is unchanged. Its linearized order is $1 + p$. It also holds in the viscosity sense used by that theorem: if a smooth function touches u from above at x' , replace u by that function in a sufficiently small disk and keep u outside. At the contact point, monotonicity of G_p makes the graph integral no larger; the reverse inequality holds for a contact from below. The right-hand side has the same value at contact. This is precisely the corresponding viscosity inequality, and the original C^2 principal value converges. The localization and bootstrap in Barrios–Figalli–Valdinoci [1, Theorem 6 and Section 3, equations (49), (51)–(55)] apply to this equation: their graph-kernel construction requires only the local graph and this zero-curvature equation, not minimization. Small slope supplies their kernel-perturbation hypothesis. Starting with C^2 regularity, choose $p/2 < \beta < p$; the graph is $C^{1,\beta}$, and their antisymmetric remainder has exponent $2\beta - p > 0$. In using their Schauder theorem, retain a strict loss in every Hölder exponent: from a remainder in $C^{2\beta-p}$ use $0 < \gamma < 2\beta - p$, rather than claiming an endpoint estimate. One can choose $\max\{0, \beta - p\} < \gamma < 2\beta - p$ to obtain a strict gain over $C^{1,\beta}$. Choose γ away from the exceptional value for which $1 + p + \gamma$ is an integer; an open interval of choices is available. The analogous noninteger choice is made at each later Schauder step. The localized iteration and its differentiated versions then give every finite regularity order. At each finite order use a finite covering of an arbitrary compact subpatch; the neighborhoods and bounds may depend on that order. Thus the graph is C^∞ locally, without claiming a uniform estimate over infinitely many derivatives. This is also the stationary-cone bootstrap used in Tsiamis [9, Lemma 2.4]. All parameters here are fixed; no estimate uniform as $p \downarrow 0$ is needed for this regularity step.

Twice differentiability was proved before the bootstrap in Lemma 2.2. Taylor’s formula and vanishing first variation identify the displayed lower limit with one half of the second variation. Lemma A.4 also identifies the local-in-time variation class with the globally time-parametrized class in the draft specification, without changing the family near time zero. The same lemma on perimeter differentiation removes any distinction between autonomous flows and time-dependent smooth deformations at a stationary cone. Dilation handles arbitrary compact supports away from zero. The converse follows from the same Taylor expansion for a smooth stationary stable cone. $\square$

Lemma 8.2 (Strict monotonicity of a nonconstant profile). *Fix $0 < p < 1$ and $n \geq 1$. Suppose that $v(x) = h(e \cdot x)$, $e \in S^{n-1}$, is a bounded entire solution of $(-\Delta)^{p/2}v = v - v^3$, and that h is nondecreasing and nonconstant. Then h satisfies the one-dimensional equation with the same Fourier normalization and $h'(t) > 0$ for every $t \in \mathbb{R}$.*

Proof. A smooth cutoff of the cubic outside the bounded range of v gives a global $C^{1,\gamma}$ nonlinearity with $1 - p < \gamma < 1$. Cabré–Sire [2, Lemma 2.1] gives $v \in C^{2,\beta}(\mathbb{R}^n)$ with finite global norm for some $\beta > 0$. Restriction to the line te gives the same bounds for h . In the Fourier normalization write

$$c_{n,p} = \frac{p2^{p-1}\Gamma((n+p)/2)}{\pi^{n/2}\Gamma(1-p/2)}.$$

For $1 \leq m < n$ and $z \in \mathbb{R}^m \setminus \{0\}$, Gaussian integration or the beta integral gives

$$\begin{aligned} \int_{\mathbb{R}^{n-m}} (|z|^2 + |y|^2)^{-(n+p)/2} dy &= \pi^{(n-m)/2} \frac{\Gamma((m+p)/2)}{\Gamma((n+p)/2)} |z|^{-m-p}, \\ c_{n,p} \int_{\mathbb{R}^{n-m}} (|z|^2 + |y|^2)^{-(n+p)/2} dy &= c_{m,p} |z|^{-m-p}. \end{aligned} \quad (2)$$

For completeness, insert $A^{-a} = \Gamma(a)^{-1} \int_0^\infty t^{a-1} e^{-tA} dt$ with $a = (n+p)/2$, integrate the nonnegative Gaussian in y , and then integrate in t . All gamma arguments are positive. The bound $|h(t+\tau) - h(t)| \leq \min\{\|h'\|_\infty |\tau|, 2\|h\|_\infty\}$ is integrable against $|\tau|^{-1-p}$ because $0 < p < 1$. Fubini and the identity with $m = 1$ therefore yield the asserted one-dimensional equation. For $n = 1$ this step is immediate.

Put $w = h' \geq 0$. A difference quotient of $h(t) - h(t+\tau)$ is bounded in absolute value by

$$\min\{\|h''\|_\infty |\tau|, 2\|h'\|_\infty\}.$$

The same integrability justifies differentiation of the equation, giving $(-\partial_t^2)^{p/2} w = (1-3h^2)w$. Nonconstancy of h implies $w \not\equiv 0$. If $w(t_0) = 0$, its absolutely convergent integral satisfies

$$(-\partial_t^2)^{p/2} w(t_0) = -c_{1,p} \int_{\mathbb{R}} \frac{w(t)}{|t-t_0|^{1+p}} dt < 0.$$

Strict negativity follows because the continuous nonnegative function w is positive on a nonempty interval. The equation instead gives zero at t_0 , a contradiction. Hence $h' > 0$ everywhere. $\square$

Proposition 8.3 (The monotone reduction). *Fix $p \in (0, 1)$. Assume every nontrivial stationary p -perimeter cone in $\mathbb{R}^3$, with locally one-sided smooth boundary away from its vertex, compact smooth spherical link, and stability under compactly supported variations away from that vertex, is a halfspace. Then every bounded entire classical solution*

$$(-\Delta)^{p/2} U = U - U^3 \quad \text{in } \mathbb{R}^4, \quad U_{x_4} > 0,$$

is a one-dimensional layer: $U(X) = q(e \cdot X)$ for some $e \in S^3$ with $e_4 > 0$, where $q' > 0$, $(-\partial_t^2)^{p/2} q = q - q^3$, and $q(\pm\infty) = \pm 1$. No directional-limit hypothesis is prescribed.

Proof. All external perimeter parameters in this proof are p .

We make explicit the transport between the operator normalizations in the cited results. Let $c_{n,p}$ be the kernel constant for the Fourier symbol $|\xi|^p$, and suppose a cited result uses

$$L_a v(x) = a_{n,p} \text{PV} \int_{\mathbb{R}^n} \frac{v(x) - v(y)}{|x-y|^{n+p}} dy, \quad a_{n,p} > 0.$$

Set

$$\lambda = \left(\frac{a_{n,p}}{c_{n,p}} \right)^{-1/p}, \quad v(x) = u(\lambda x).$$

Then $a_{n,p} \lambda^p = c_{n,p}$ and $L_a v(x) = ((-\Delta)^{p/2} u)(\lambda x)$. For $W(t) = (1-t^2)^2/4$ and a positive coefficient b , write

$$\mathcal{E}_b[z; D] = \frac{b}{4} \iint_{\mathbb{R}^{2n} \setminus (D^c \times D^c)} \frac{|z(x) - z(y)|^2}{|x-y|^{n+p}} dx dy + \int_D W(z) dx.$$

For the bounded solutions used below, local finiteness of this energy follows from their global Lipschitz regularity. Indeed $|z(x+h) - z(x)|^2 \leq C \min\{|h|^2, 1\}$, and hence its kinetic integral over the interaction region of a bounded D is at most

$$2C|D| |S^{n-1}| \left(\int_0^1 r^{1-p} dr + \int_1^\infty r^{-1-p} dr \right) < \infty.$$

The potential is bounded on the solution's range. Competitors of infinite energy cannot improve such a solution; the scaling identity below is also valid for nonnegative extended-valued energies. With $X = \lambda x$ and $Y = \lambda y$, the interaction integral acquires the factor λ^{p-n} ,

its integration region becomes the corresponding region for λD , and the potential integral acquires λ^{-n} . Since $a_{n,p}\lambda^p = c_{n,p}$, this gives the exact identity

$$\mathcal{E}_{a_{n,p}}[v; D] = \lambda^{-n} \mathcal{E}_{c_{n,p}}[u; \lambda D].$$

Competitors and compact perturbations are carried bijectively by this dilation; consequently minimization and stability are preserved. The dilation also preserves monotonicity and one-dimensionality. It only relabels blow-down radii and preserves asymptotic slab flatness. We apply every cited result in its own normalization and then return to the Fourier normalization. In particular, Cabré–Cinti–Serra [4, Definition 1.12], uses $a_{n,p} = 2 - p$.

Choose $M_0 > \|U\|_\infty$, $1 - p < \gamma < 1$, and $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ near $[-M_0, M_0]$. Use $\tilde{f}(t) = \chi(t)(t - t^3)$; its bounded derivatives give a finite global $C^{1,\gamma}$ norm. With $s = p/2$, Cabré–Sire [2, Lemma 2.1] applies to this $C^{1,\gamma}(\mathbb{R})$ nonlinearity and gives $U \in C^{2,\beta}(\mathbb{R}^4)$ for some $\beta > 0$. Its global norm is unchanged by translation; no estimate uniform in p is needed.

If $M = \sup U > 1$, choose X_j with $U(X_j) \rightarrow M$. Arzelà–Ascoli and a diagonal extraction give $U(X_j + \cdot) \rightarrow V$ in C_{loc}^2 , where $V(0) = M$ and $V \leq M$. To pass the equation, first differences are bounded by $C|Z|$ for $|Z| < 1$, using the common global gradient bound, and by $2M_0$ for $|Z| \geq 1$. Against $|Z|^{-4-p}$ these are integrable because $0 < p < 1$. Dominated convergence therefore gives $(-\Delta)^{p/2}V = V - V^3$. At its maximum the kernel integral is nonnegative, contradicting $M - M^3 < 0$. The infimum is analogous, so $|U| \leq 1$. If U attains 1 or -1 , its equation is a zero integral with a one-signed integrand. Positivity of the kernel and continuity force the corresponding constant well, contrary to $U_4 > 0$. Hence $-1 < U < 1$.

Set $w = U_4 > 0$. The preceding $C^{2,\beta}$ regularity gives a global bound and a global Lipschitz bound for w . Difference quotients in the x_4 direction are dominated near the diagonal by $C|X - Y|$ and in the tail by a constant, so differentiation under the absolutely convergent kernel gives $(-\Delta)^{p/2}w = (1 - 3U^2)w$. For $\phi \in C_c^\infty(\mathbb{R}^4)$, put $\eta = \phi^2/w$. Positivity and continuity of w on a neighborhood of the compact support make η a compactly supported C^1 test. Multiplying the linearized equation by η and symmetrizing gives

$$\int_{\mathbb{R}^4} (1 - 3U^2)\phi^2 = \frac{c_{4,p}}{2} \iint_{\mathbb{R}^8} \frac{(w_X - w_Y)(\eta_X - \eta_Y)}{|X - Y|^{4+p}} dX dY.$$

The algebraic identity

$$(\phi_X - \phi_Y)^2 - (w_X - w_Y) \left(\frac{\phi_X^2}{w_X} - \frac{\phi_Y^2}{w_Y} \right) = w_X w_Y \left(\frac{\phi_X}{w_X} - \frac{\phi_Y}{w_Y} \right)^2$$

then yields the exact ground-state formula

$$\|(-\Delta)^{p/4}\phi\|_{L^2}^2 - \int_{\mathbb{R}^4} (1 - 3U^2)\phi^2 = \frac{c_{4,p}}{2} \iint_{\mathbb{R}^8} \frac{w_X w_Y}{|X - Y|^{4+p}} \left(\frac{\phi_X}{w_X} - \frac{\phi_Y}{w_Y} \right)^2 dX dY \geq 0.$$

Choose a compact neighborhood K of $\text{supp } \phi$ with positive distance between $\text{supp } \phi$ and K^c . On K , w has a positive minimum, and the differences of w , η , and ϕ/w are $O(|X - Y|)$. Thus the paired integrands above are $O(|X - Y|^{2-4-p})$ near the diagonal, integrable since $p < 2$. If one point, say Y , lies outside K , a nonzero test factor requires $X \in \text{supp } \phi$. The squared ground-state numerator then equals $\phi_X^2 w_Y / w_X$, bounded by the global bound on w and its positive minimum on the support. The symmetrized linearized numerator is bounded in the same region. The distance is bounded away from zero and the kernel tail is integrable because $p > 0$; if both tests vanish both numerators vanish. Diagonal truncation and dominated convergence therefore justify the testing, symmetrization, and displayed identity. Finally $\|(-\Delta)^{p/4}\phi\|_2^2 = (c_{4,p}/2) \iint (\phi_X - \phi_Y)^2 |X - Y|^{-4-p}$, which fixes its normalization and proves stability.

This inequality also holds for compactly supported Lipschitz tests, as required by Cabré–Cinti–Serra [4, Definition 1.12]. Such a test belongs to $H^{p/2}$: its squared L^2 first differences

are bounded by $C \min\{|Z|^2, 1\}$, whose integral against $|Z|^{-n-p}$ is finite for $0 < p < 2$. Smooth mollifications have support in a fixed compact set and converge in $H^{p/2}$ and L^2 . The kinetic form therefore converges, while the potential integral converges because $1 - 3U^2$ is bounded. This proves the claimed extension of the test class. The same approximation applies to all translates and endpoint limits below, in either dimension three or four, because their potentials are bounded as well.

Bounded monotonicity gives pointwise limits $U^\pm(x') = \lim_{t \rightarrow \pm\infty} U(x', t)$. Every sequence of vertical translates tending to either end has a C_{loc}^2 convergent subsequence by the common global $C^{2,\beta}$ norm. Its limit is the same pointwise monotone limit, independent of x_4 . Uniqueness of that limit gives convergence of the full family in C_{loc}^2 . The first-difference and tail domination used above passes the four-dimensional equation to these limits.

They also solve the three-dimensional equation with the Fourier normalization. The normalized constants are

$$c_{n,p} = \frac{p2^{p-1}\Gamma((n+p)/2)}{\pi^{n/2}\Gamma(1-p/2)}.$$

For $z \neq 0$, the beta integral gives

$$\int_{\mathbb{R}} (|z|^2 + t^2)^{-(4+p)/2} dt = \frac{\sqrt{\pi}\Gamma((3+p)/2)}{\Gamma((4+p)/2)} |z|^{-3-p},$$

and multiplication by $c_{4,p}$ gives exactly $c_{3,p}|z|^{-3-p}$. Bounded locally Lipschitz profiles have absolutely integrable first differences near the diagonal for $p < 1$ and an integrable tail; Fubini therefore gives the reduced equation directly.

For a fixed compactly supported test, stability of the translated solutions passes to either limit by local uniform convergence of $1 - 3U^2$. Thus each limiting profile, viewed in four dimensions, is stable. To obtain three-dimensional stability, fix $\phi \in C_c^\infty(\mathbb{R}^3)$ and a nonzero $\chi \in C_c^\infty(\mathbb{R})$, and test with $\phi(x')\chi(x_4/T)$. The potential term divided by $T\|\chi\|_2^2$ is exactly $\int_{\mathbb{R}^3} (1 - 3(U^\pm)^2)\phi^2$. With the unitary Fourier transform, the normalized kinetic term is

$$\frac{1}{\|\chi\|_2^2} \iint (|\xi'|^2 + \tau^2/T^2)^{p/2} |\widehat{\phi}(\xi')|^2 |\widehat{\chi}(\tau)|^2 d\xi' d\tau.$$

Its multiplier converges to $|\xi'|^p$ and is bounded for $T \geq 1$ by $|\xi'|^p + |\tau|^p$, since $0 < p/2 < 1$. Schwartz decay supplies integrable domination, and Plancherel gives the limit $\|(-\Delta)^{p/4}\phi\|_2^2$. The resulting inequality is precisely three-dimensional stability.

First classify the endpoints in the Fourier normalization in dimension three. If an endpoint attains either well, the strong maximum principle makes it that constant well; otherwise its values lie strictly in $(-1, 1)$.

Lemma 8.1 verifies the entire C^2 cone interface in Cabré–Cinti–Serra [4, Definitions 1.6 and 2.8]: their cones have finite local perimeter and become smooth, stationary, and stable in precisely the class used here. In dimension three their Definition 1.6 requires only the cone classification for $m = 3$, not an additional classification of punctured stable cones in dimension two. Thus the cone hypothesis supplies exactly the assumption of their Theorem 1.8.

Apply Cabré–Cinti–Serra [4, Theorem 1.8] to each such endpoint, using the dimension-three dilation described above. Returning to the Fourier normalization, each nonconstant endpoint is a one-dimensional nondecreasing profile. Lemma 8.2 proves its normalized one-dimensional equation and strictly positive derivative before it is used as an endpoint layer below.

We justify both the equation and stability at the profile endpoints. Let $h(x) = q(e \cdot x)$, with $e \in S^2$, be the bounded stable solution in $\mathbb{R}^3$ under consideration, in the Fourier normalization, where q is increasing and nonconstant. Bounded monotonicity gives limits $\ell_- < \ell_+$: choose $t_1 < t_2$ with $q(t_1) < q(t_2)$ and use $\ell_- \leq q(t_1) < q(t_2) \leq \ell_+$. For $h_T(x) = h(x + Te) = q(e \cdot x + T)$, monotonicity bounds h_T on $|x| \leq R$ between $q(T - R)$ and $q(T + R)$. Consequently $h_T \rightarrow \ell_\pm$ locally uniformly as $T \rightarrow \pm\infty$, and these translates have a common global bound.

For every $\psi \in C_c^\infty(\mathbb{R}^3)$, the weak equation is justified by Lemma A.2: the translates are bounded and locally C^2 , the paired integral is absolutely convergent, and the cutoff limit can be taken before or after testing. Thus

$$\int_{\mathbb{R}^3} h_T(-\Delta)^{p/2}\psi = \int_{\mathbb{R}^3} (h_T - h_T^3)\psi.$$

The function $(-\Delta)^{p/2}\psi$ is integrable and has integral zero. Indeed,

$$\int_{\mathbb{R}^3} |\psi(x) - \psi(x+z)| dx \leq \min\{|z|\|\nabla\psi\|_{L^1}, 2\|\psi\|_{L^1}\}.$$

After integration against $|z|^{-3-p}$ this bound is finite since $0 < p < 1$. Fubini and translation invariance then give the zero integral. Dominated convergence on the left, and local uniform convergence on the right, imply $0 = (\ell_\pm - \ell_\pm^3) \int \psi$. Choosing $\int \psi = 1$ proves that both limits are roots of $r - r^3$.

Translation invariance preserves stability. Thus, for each fixed $\varphi \in C_c^\infty(\mathbb{R}^3)$,

$$\|(-\Delta)^{p/4}\varphi\|_{L^2}^2 \geq \int_{\mathbb{R}^3} (1 - 3h_T^2)\varphi^2 \longrightarrow (1 - 3\ell_\pm^2)\|\varphi\|_{L^2}^2.$$

Hence each limiting constant is stable. The root zero is unstable: for nonzero φ and $\varphi_R(x) = \varphi(x/R)$, its stability quadratic form is exactly

$$R^{3-p}\|(-\Delta)^{p/4}\varphi\|_{L^2}^2 - R^3\|\varphi\|_{L^2}^2 < 0$$

for all sufficiently large R . The distinct ordered limits must therefore be $\ell_- = -1$ and $\ell_+ = 1$.

Apply this argument to each nonconstant endpoint $h = U^+$ or $h = U^-$. Thus each such endpoint is a layer joining -1 to 1 . A constant endpoint is a root of $r - r^3$, and the same dilated stability test excludes zero. Thus constant endpoints are wells.

Now set $\lambda_{4,p} = (a_{4,p}/c_{4,p})^{-1/p}$ and use the dimension-four dilation $v(X) = U(\lambda_{4,p}X)$ to apply their Proposition 7.9. Its endpoint functions are $v^\pm(x') = U^\pm(\lambda_{4,p}x')$. Each remains an increasing one-dimensional function or a constant well, and each, viewed as a function independent of x_4 , solves the same four-dimensional equation as v because it is a vertical translation limit of v . These are exactly the endpoint hypotheses of Proposition 7.9; no equality between the external dimension-three and dimension-four kernel constants is asserted or needed. That proposition makes v an entire minimizer in its external normalization, and the displayed energy identity makes U an entire minimizer in the Fourier normalization. No directional endpoint values have been prescribed.

There is a point where $U = 0$. Otherwise continuity and connectedness of $\mathbb{R}^4$ give either $U > 0$ everywhere or $U < 0$ everywhere. In the first case both endpoints are nonnegative. A nonconstant layer takes negative values, so the endpoint classification forces $U^- = U^+ = 1$. The inequalities $U^- \leq U \leq U^+$ then make $U \equiv 1$, contrary to $U_4 > 0$. The second case similarly forces $U \equiv -1$. Translate a zero to the origin.

For every sequence $R_j \rightarrow \infty$, pass first to an increasing subsequence. In the dimension-four Cabré–Cinti–Serra normalization already fixed, set $S_j = R_j/\lambda_{4,p}$. Then $S_j \rightarrow \infty$ and

$$v(S_j \cdot) = U(R_j \cdot).$$

Their stable blow-down compactness and conicality results [4, Theorem 2.10 and Proposition 6.1] give a subsequence for which these functions tend in L_{loc}^1 to a nontrivial conical phase $2\chi_C - 1$. Proposition 6.1 states local convergence on all balls, conicality, and nontriviality on this single subsequence. Theorem 2.10 adds its density-boundary characterization and level-set convergence. All subsequent extractions are subsequences of this one. Theorem 2.10 includes nontriviality; its density estimates also exclude the empty and full limits at the fixed zero.

For minimizing compactness we keep the coefficient in the cited functional explicit. In Dipierro–Serra–Valdinoci [7, equations (1.9)–(1.10)], write $a_{4,p}^D > 0$ for the coefficient of the

double integral. Their quartic potential is $(1 - z^2)^2$, so that functional is exactly $4\mathcal{E}_{a_{4,p}^D}$ in the notation above. Indeed its first variation against a compact smooth ϕ is

$$4 \int_{\mathbb{R}^4} \left[a_{4,p}^D \text{PV} \int_{\mathbb{R}^4} \frac{z(X) - z(Y)}{|X - Y|^{4+p}} dY - (z - z^3)(X) \right] \phi(X) dX.$$

This follows by differentiating the squared difference, symmetrizing, and differentiating the potential; the paired integrals are finite by the local energy bounds already proved. Thus the same coefficient occurs in the equation and in $\mathcal{E}_{a_{4,p}^D}$. Put

$$\mu_{4,p}^D = (c_{4,p}/a_{4,p}^D)^{1/p}, \quad z_D(X) = U(\mu_{4,p}^D X), \quad \varepsilon_j = \mu_{4,p}^D / R_j.$$

The energy scaling identity makes z_D a minimizer of precisely this functional, and $z_D(\cdot/\varepsilon_j) = U(R_j \cdot)$. Their minimizing phase compactness [7, Lemma 8.1], along these same scales and after extracting further, gives a perimeter-minimizing phase limit. Its almost-everywhere convergence and the bound $|z_D| < 1$ imply L_{loc}^1 convergence by dominated convergence. Uniqueness of the already fixed L_{loc}^1 limit therefore identifies this phase with C up to null sets. Thus this same cone C minimizes perimeter in every ball. To pass to a bounded open set D , choose a ball $B \supset D$ and a competitor $F = C$ on D^c . Set $\mathcal{R}_D = \mathbb{R}^8 \setminus (D^c \times D^c)$ and define $\mathcal{R}_B$ similarly. On $\mathcal{R}_B \setminus \mathcal{R}_D$ both points lie outside D , so the corresponding interaction integral is the same finite nonnegative number A for F and C . Consequently

$$\text{Per}_p(F; B) = \text{Per}_p(F; D) + A, \quad \text{Per}_p(C; B) = \text{Per}_p(C; D) + A.$$

If $\text{Per}_p(F; D) = \infty$, the desired comparison is immediate. Otherwise both ball perimeters are finite and ball minimality gives $\text{Per}_p(C; D) \leq \text{Per}_p(F; D)$ by subtracting the finite A . Thus no cancellation of infinite terms is involved.

For each fixed $t > 0$, let $u_j(x) = U(R_j x)$. Monotonicity gives $u_j(x + te_4) \geq u_j(x)$. Translation is continuous as a map from L_{loc}^1 to L_{loc}^1 at this fixed t : convergence on a compact set and its translate gives convergence of both terms. Taking the negative part of their difference, a 1-Lipschitz operation, shows that

$$\chi_C(x + te_4) \geq \chi_C(x) \quad \text{for almost every } x.$$

No pointwise subsequence depending on t is needed. Choose once and for all the density-one representative

$$C^{(1)} = \left\{ x : \lim_{r \downarrow 0} \frac{|C \cap B_r(x)|}{|B_r|} = 1 \right\}.$$

For this fixed t , integrating the almost-everywhere inequality over an arbitrary ball gives

$$|C \cap B_r(x + te_4)| \geq |C \cap B_r(x)| \quad (x \in \mathbb{R}^4, r > 0).$$

Thus $x \in C^{(1)}$ implies $x + te_4 \in C^{(1)}$. The representative is independent of t , and the argument applies to every $t > 0$, so $C^{(1)} + te_4 \subset C^{(1)}$ for all such t . Likewise, dilation invariance of C up to null sets gives, for every $\lambda > 0$,

$$\frac{|C \cap B_r(\lambda x)|}{|B_r|} = \frac{|C \cap B_{r/\lambda}(x)|}{|B_{r/\lambda}|},$$

so $C^{(1)}$ is an exact cone. The boundary characterization in Cabré–Cinti–Serra [4, Theorem 2.10, equation (2.9)] identifies this representative with the interior of their phase representative: a density-one point cannot lie on that boundary or in the interior of the complementary phase, and every interior phase point has density one. Thus it is open. Every boundary point has both phases of positive density. The Lebesgue differentiation theorem then places density-one points of each phase in every neighborhood of such a point, so passing to the interior does not change the boundary. We henceforth write C for $C^{(1)}$; null-set changes preserve all perimeter comparisons and L^1 limits.

Cozzi–Farina–Lombardini [6, Proposition 3.1], applies to this minimizing cone and this nonzero translating vector. It says that C is a halfspace or a cylinder $D \times \mathbb{R}$ in the e_4 direction.

In the latter case the cylinder theorem of Caffarelli–Roquejoffre–Savin [5, Theorem 10.1], makes $D \subset \mathbb{R}^3$ a minimizing cone. By Savin–Valdinoci [8, Corollary 2], its singular set has $\mathcal{H}^d$ measure zero for every $d > 0$, since the ambient dimension is three. In particular it has Hausdorff dimension at most zero; outside this set the boundary is locally a $C^{1,\alpha}$ graph. Dilation preserves regularity and singularity, so a nonzero singular point of the cone would generate a singular ray. Such a ray has Hausdorff dimension one, contradicting the dimension bound. Thus D is locally a $C^{1,\alpha}$ graph away from zero. At any such point, after translating and rotating the tangent plane, differentiability gives $\partial D \cap B_r \subset \{|x_3| \leq \varepsilon r\}$ for sufficiently small r . In Barrios–Figalli–Valdinoci [1, Corollary 2], take the fixed lower parameter $p_0 = p/2$ and then ε below its flatness threshold. That result makes the graph smooth in $B_{r/2}$. Consequently D has smooth, locally one-sided boundary away from zero. Its nontriviality follows from that of $C = D \times \mathbb{R}$, and its compact smooth link follows by intersecting this closed boundary with the sphere transversely to the radial tangent. Minimality implies stationarity and stability for the required compact variations, so the cone hypothesis forces D to be a halfspace. In either case C is a halfspace through the origin.

Apply Cabré–Cinti–Serra [4, Theorem 2.10] also to $-v$, along the same S_j and extracting a further subsequence if needed. Since $-v(S_j \cdot) = -U(R_j \cdot)$, its phase limit is $1 - 2\chi_C$ by uniqueness of the already fixed L_{loc}^1 limit. The theorem’s equation (2.10) uses closed superlevels $\{v(S_j \cdot) \geq c\}$, so it applies at $c = -4/5$ with precisely the threshold convention used below, for both v and $-v$. Thus the positive- and negative-phase level-set convergences hold along the same scales and give the threshold inclusions below on compact sets separated from the limiting interface. To see this directly, suppose that the limiting positive phase is $\{e \cdot x > 0\}$. A point of $\overline{B}_1$ with $e \cdot x \leq -\varepsilon/2$ and $U(R_j x) \geq -4/5$ would belong to the superlevel set $\{U(R_j \cdot) \geq -4/5\}$ at distance at least $\varepsilon/2$ from the limiting positive halfspace. This contradicts its Hausdorff convergence on B_2 . Apply the same argument to $-U$ to rule out $U(R_j x) \leq 4/5$ when $e \cdot x \geq \varepsilon/2$. Thus, uniformly on these two compact regions, $U(R_j x) < -4/5$ and $U(R_j x) > 4/5$, respectively. For $R > 0$, let $F(R)$ be the infimum of the numbers $a \in [0, 1]$ for which some $e \in S^3$ gives the inclusions

$$\{e \cdot x \leq -aR\} \subset \{U \leq -4/5\} \subset \{U \leq 4/5\} \subset \{e \cdot x \leq aR\} \quad \text{in } B_R.$$

The admissible set is nonempty, since $a = 1$ works in the open ball. If $F(R)$ did not tend to zero, there would be $\varepsilon \in (0, 1]$ and scales $R_j \rightarrow \infty$ with $F(R_j) \geq \varepsilon$. Extract a halfspace sublimit and orient its normal e toward its positive phase. The threshold inclusions just proved on $\overline{B}_1$ give the displayed inclusions with $a = \varepsilon/2$ at all sufficiently large j , a contradiction. Hence $F(R) \rightarrow 0$. Choose $R_0 > 2$ so that $\sup_{\rho \geq R_0} F(\rho) < 1/2$, and set

$$a(R) = \begin{cases} 1, & 1 < R < R_0, \\ \sup_{\rho \geq R} F(\rho) + R^{-1}, & R \geq R_0. \end{cases}$$

This function is nonincreasing, belongs to $(0, 1]$, and tends to zero. For $R < R_0$, the slab inclusions are vacuous at width R in the open ball. For $R \geq R_0$, one has $a(R) > F(R)$; the definition of the infimum supplies an admissible width smaller than $a(R)$ and hence an admissible direction at width $a(R)$ itself. Thus the slab condition holds at every required scale. We check the hypotheses of Dipierro–Serra–Valdinoci [7, Theorem 1.2] explicitly. For this separate application set $\mu_{4,p} = c_{4,p}^{1/p}$ and $z(X) = U(\mu_{4,p} X)$. This is the unit-kernel normalization of their operator (1.2), independently of the coefficient just retained for minimizing compactness. The same scaling identity gives $Lz = z - z^3$. Their convex body is the unit ball, which is even and satisfies their uniform interior-ball condition (H1). The cubic satisfies (H2) with $\kappa = 1/5$ and $c_\kappa = 9/10$, because $f(\pm 1) = 0$ and

$$f'(t) = 1 - 3t^2 \leq -\frac{23}{25} < -\frac{9}{10} \quad \text{when} \quad |t| \geq \frac{4}{5}.$$

For (H3), the layer in Definition 1.1, with $s = p/2$, and Cabré–Sire [2, Theorem 2.4] give $\phi_0(t) = q_{p/2}(c_{1,p}^{1/p}t)$ in their unnormalized one-dimensional operator: the same dilation identity shows that it solves $\mathcal{L}\phi_0 = f(\phi_0)$, and it has positive derivative, zero at zero, and limits ± 1 . Finally, the displayed slab inclusions for U give those for z at radius R with width function $a(\mu_{4,p}R)$, for all sufficiently large R . This function remains nonincreasing, belongs to $(0, 1]$, and tends to zero. If a width function on all $R > 1$ is desired, set it equal to one below a sufficiently large transition radius and equal to $a(\mu_{4,p}R)$ above that radius; it is still nonincreasing, and the width-one inclusions are vacuous in the open ball. Their direction may depend on R , exactly as here. Neither a unique blow-down normal nor a quantitative rate of $a(R) \rightarrow 0$ has been assumed. Thus their Theorem 1.2 applies, equivalently in the cubic normalization restated by Cabré–Cinti–Serra [4, Theorem 7.1], and makes z , hence U , one-dimensional. Write $U(X) = q(e \cdot X)$ with $|e| = 1$. The identity $U_4 = e_4 q'$ rules out $e_4 = 0$; reversing e and the profile if necessary gives $e_4 > 0$ and then $q' > 0$ everywhere. Rotation invariance and integration in $e^\perp$ give the exact one-dimensional equation. Indeed the beta integral and the normalized kernel constants satisfy, for $\tau \neq 0$,

$$c_{4,p} \int_{\mathbb{R}^3} (\tau^2 + |z|^2)^{-(4+p)/2} dz = c_{1,p} |\tau|^{-1-p}.$$

At $X = te$, write $Y = (t + \tau)e + z$. The bounded profile is locally Lipschitz, so its first differences are absolutely integrable against $|\tau|^{-1-p}$ near zero because $p < 1$, and boundedness controls the tail. Fubini therefore gives

$$(-\Delta)^{p/2} U(te) = (-\partial_t^2)^{p/2} q(t) = q(t) - q(t)^3.$$

Bounded strict monotonicity gives distinct ordered limits $\ell_- < \ell_+$: for $t_1 < t_2$, $\ell_- \leq q(t_1) < q(t_2) \leq \ell_+$. Since $e_4 > 0$, writing $X = (x', x_4)$ gives $U(x', x_4 + T) = q(e \cdot X + T e_4)$, which tends locally uniformly to $\ell_\pm$ as $T \rightarrow \pm\infty$. Thus the previously constructed $U^\pm$ are these constant endpoints. Their equations make them roots of $r - r^3$, and they inherit four-dimensional stability as already shown. For a nonzero $\psi \in C_c^\infty(\mathbb{R}^4)$ and $\psi_R(X) = \psi(X/R)$, the stability form of the constant zero is

$$R^{4-p} \|(-\Delta)^{p/4} \psi\|_2^2 - R^4 \|\psi\|_2^2 < 0$$

for sufficiently large R . Hence zero is excluded in this dimension as well. Both endpoints are wells, and their strict ordering gives $\ell_- = -1$ and $\ell_+ = 1$, as required for a layer. $\square$

Proof of Corollary 1.4. Put $p = 2s$ and choose $M_0 > \|u\|_\infty$. Choose $1 - p < \gamma < 1$ and $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ near $[-M_0, M_0]$. Use $\tilde{f}(t) = \chi(t)(t - t^3)$; its bounded derivatives give a finite global $C^{1,\gamma}$ norm. Cabré–Sire [2, Lemma 2.1] gives $u \in C^{2,\beta}(\mathbb{R}^3)$ for some $\beta > 0$. The $C^{2,\beta}$ norm is unchanged by translation. If $M = \sup u > 1$, take x_j with $u(x_j) \rightarrow M$. Arzelà–Ascoli and a diagonal extraction give $u(x_j + \cdot) \rightarrow v$ in C_{loc}^2 , with $v(0) = M$ and $v \leq M$. The uniform first-derivative bound controls the first differences by $C|h|$ for $|h| < 1$, whose product with $|h|^{-3-p}$ is integrable because $p < 1$. On $|h| \geq 1$, use the common global bound $2M_0$. Dominated convergence therefore passes the fractional equation to v . At its maximum, $(-\Delta)^{p/2} v(0) \geq 0$, contradicting $M - M^3 < 0$. The infimum is treated in the same way, so $|u| \leq 1$. If u attains either well, its equation has a zero kernel integral with a one-signed integrand; positivity of the kernel and continuity force u to be that constant well. Otherwise $|u| < 1$. A constant solution must be a root of $r - r^3$; the root zero is excluded by the dilated stability test computed below. Thus it remains to treat nonconstant u . We make explicit the transport between the operator normalizations in the cited results. Let $c_{n,p}$ be the kernel constant for the Fourier symbol $|\xi|^p$, and suppose a cited result uses

$$L_a v(x) = a_{n,p} \text{PV} \int_{\mathbb{R}^n} \frac{v(x) - v(y)}{|x - y|^{n+p}} dy, \quad a_{n,p} > 0.$$

Set

$$\lambda = \left(\frac{a_{n,p}}{c_{n,p}} \right)^{-1/p}, \quad v(x) = u(\lambda x).$$

Then $a_{n,p}\lambda^p = c_{n,p}$ and $L_a v(x) = ((-\Delta)^{p/2}u)(\lambda x)$. For $W(t) = (1 - t^2)^2/4$ and a positive coefficient b , write

$$\mathcal{E}_b[z; D] = \frac{b}{4} \iint_{\mathbb{R}^{2n} \setminus (D^c \times D^c)} \frac{|z(x) - z(y)|^2}{|x - y|^{n+p}} dx dy + \int_D W(z) dx.$$

Changing variables gives the exact identity

$$\mathcal{E}_{a_{n,p}}[v; D] = \lambda^{-n} \mathcal{E}_{c_{n,p}}[u; \lambda D].$$

Competitors and compact perturbations are carried bijectively by this dilation; consequently minimization and stability are preserved. The dilation also preserves monotonicity and one-dimensionality. It only relabels blow-down radii and preserves asymptotic slab flatness. We apply every cited result in its own normalization and then return to the Fourier normalization. In particular, Cabré–Cinti–Serra [4, Definition 1.12], uses $a_{n,p} = 2 - p$. Their test class consists of compactly supported Lipschitz functions. The stability inequality here extends to that class: such a function φ belongs to $H^{p/2}(\mathbb{R}^3)$, and its smooth mollifications converge in that space and in L^2 , with support in one fixed compact set. Since $1 - 3u^2$ is bounded, both terms of the quadratic form converge. The dilated solution is therefore stable in exactly the sense of their definition.

The cone classification Theorem 1.3 will supply the hypothesis of Cabré–Cinti–Serra [4, Definition 1.6 and Theorem 1.8] in dimension three. Lemma 8.1 verifies the entire C^2 cone interface in Cabré–Cinti–Serra [4, Definitions 1.6 and 2.8]: their cones have finite local perimeter and become smooth, stationary, and stable in precisely the class used here. In dimension three their Definition 1.6 requires only the cone classification for $m = 3$, not an additional classification of punctured stable cones in dimension two. Thus the cone hypothesis supplies exactly the assumption of their Theorem 1.8. A cone smooth away from zero has a compact smooth link: its radial tangent is transverse to the unit sphere. Hence the classification theorem applies to every nontrivial cone in this class. Apply Cabré–Cinti–Serra’s theorem to the dimension-three rescaling $v(x) = u(\lambda x)$. It is a stable solution of their equation by the operator and energy identities and the test-class approximation just proved. Their conclusion makes v , and hence u , one-dimensional and increasing in a unit direction. Returning to u gives an increasing profile solving the original Fourier-normalized equation. To justify the last assertion, write $u(X) = q(e \cdot X)$ with $|e| = 1$. The beta integral gives

$$c_{3,p} \int_{\mathbb{R}^2} (\tau^2 + |z|^2)^{-(3+p)/2} dz = c_{1,p} |\tau|^{-1-p}, \quad \tau \neq 0.$$

The profile is bounded and locally Lipschitz by the preceding regularity. Since $p < 1$, its first differences are absolutely integrable at zero; boundedness controls the tail. Fubini at $X = te$, with $Y = (t + \tau)e + z$, therefore gives $(-\partial_t^2)^{p/2}q = q - q^3$ in the stated Fourier normalization.

We justify both the equation and stability at the profile endpoints. Let $h(x) = q(e \cdot x)$, with $e \in S^2$, be the bounded stable solution in $\mathbb{R}^3$ under consideration, in the Fourier normalization, where q is increasing and nonconstant. Bounded monotonicity gives limits $\ell_- < \ell_+$: choose $t_1 < t_2$ with $q(t_1) < q(t_2)$ and use $\ell_- \leq q(t_1) < q(t_2) \leq \ell_+$. For $h_T(x) = h(x + Te) = q(e \cdot x + T)$, monotonicity bounds h_T on $|x| \leq R$ between $q(T - R)$ and $q(T + R)$. Consequently $h_T \rightarrow \ell_\pm$ locally uniformly as $T \rightarrow \pm\infty$, and these translates have a common global bound.

For every $\psi \in C_c^\infty(\mathbb{R}^3)$, the weak equation is justified by Lemma A.2: the translates are bounded and locally C^2 , the paired integral is absolutely convergent, and the cutoff limit can be taken before or after testing. Thus

$$\int_{\mathbb{R}^3} h_T (-\Delta)^{p/2} \psi = \int_{\mathbb{R}^3} (h_T - h_T^3) \psi.$$

The function $(-\Delta)^{p/2}\psi$ is integrable and has integral zero. Indeed,

$$\int_{\mathbb{R}^3} |\psi(x) - \psi(x+z)| dx \leq \min\{|z| \|\nabla \psi\|_{L^1}, 2\|\psi\|_{L^1}\}.$$

After integration against $|z|^{-3-p}$ this bound is finite since $0 < p < 1$. Fubini and translation invariance then give the zero integral. Dominated convergence on the left, and local uniform convergence on the right, imply $0 = (\ell_{\pm} - \ell_{\pm}^3) \int \psi$. Choosing $\int \psi = 1$ proves that both limits are roots of $r - r^3$.

Translation invariance preserves stability. Thus, for each fixed $\varphi \in C_c^\infty(\mathbb{R}^3)$,

$$\|(-\Delta)^{p/4}\varphi\|_{L^2}^2 \geq \int_{\mathbb{R}^3} (1 - 3h_T^2)\varphi^2 \longrightarrow (1 - 3\ell_{\pm}^2)\|\varphi\|_{L^2}^2.$$

Hence each limiting constant is stable. The root zero is unstable: for nonzero φ and $\varphi_R(x) = \varphi(x/R)$, its stability quadratic form is exactly

$$R^{3-p}\|(-\Delta)^{p/4}\varphi\|_{L^2}^2 - R^3\|\varphi\|_{L^2}^2 < 0$$

for all sufficiently large R . The distinct ordered limits must therefore be $\ell_- = -1$ and $\ell_+ = 1$.

Apply this argument with $h = u$. Thus its profile has precisely the required limits. For the uniqueness step, use the layer q_s of Definition 1.1 and Cabré–Sire [2, Theorem 2.4]. That theorem gives uniqueness up to translation for increasing layers when $f \in C^{1,\gamma}(\mathbb{R})$, $\gamma > \max\{0, 1 - 2s\}$, a potential G satisfying $G' = -f$ has equal strict wells at ± 1 , and $f'(\pm 1) < 0$. Here $f(t) = t - t^3$, $G(t) = (1 - t^2)^2/4$, $G(\pm 1) = G'(\pm 1) = 0$, $G > 0$ on $(-1, 1)$, and $f'(\pm 1) = -2$. Choose $\max\{0, 1 - 2s\} < \gamma < 1$ and $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ near $[-1, 1]$. Use $\tilde{f} = \chi f$; its bounded derivatives give a finite global $C^{1,\gamma}$ norm. Both profiles remain in $(-1, 1)$, so this extension changes neither equation. Lemma 8.2 proves $q' > 0$; the existence proof gives $q'_s > 0$. Thus both profiles are layers in the cited theorem's sense, and strict monotonicity is not being inferred merely from nondecreasing behavior. The theorem uses the same Fourier normalization. It identifies the profile with a translate of q_s . The only remaining stable constants are ± 1 . $\square$

Proof of Theorem 1.2. Put $p = 2s \in (0, 1)$. The classification Theorem 1.3 supplies the stable smooth-cone hypothesis of Proposition 8.3: for every cone smooth away from zero its radial tangent is transverse to the unit sphere, so its spherical link is compact and smooth. Consequently that reduction yields

$$U(x) = q(e \cdot x)$$

for a unit vector e with $e_4 > 0$ and an increasing one-dimensional solution q having limits -1 and $+1$. Its one-dimensional operator has exactly the Fourier symbol $|\xi|^{2s}$, since restriction to a unit direction preserves that normalization. For the uniqueness step, use the layer q_s of Definition 1.1 and Cabré–Sire [2, Theorem 2.4]. Choose $\max\{0, 1 - 2s\} < \gamma < 1$ and $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ near $[-1, 1]$. Use $\tilde{f}(t) = \chi(t)(t - t^3)$, whose bounded derivatives give a finite global $C^{1,\gamma}$ norm. A primitive of the negative of this extension, chosen with value $1/4$ at zero, agrees there with $G(t) = (1 - t^2)^2/4$. Thus $G(\pm 1) = G'(\pm 1) = 0$, $G > 0$ on $(-1, 1)$, and $f'(\pm 1) = -2 < 0$, exactly the theorem's hypotheses. The reduction already gives $q' > 0$ and limits ± 1 ; the existence assertion of the cited theorem gives $q'_s > 0$ for the normalized profile of Definition 1.1, whose limits are ± 1 . Both profiles lie in $(-1, 1)$, so the extension changes neither equation. The two equations use the same Fourier normalization. Uniqueness up to translation therefore gives $a \in \mathbb{R}$ with $q(t) = q_s(t - a)$. Consequently $U(x) = q_s(e \cdot x - a)$, with precisely the direction, translation, and normalization in the theorem. No directional-limit assumption was used. $\square$

Remark 8.4 (Scope of the geometric hypothesis). The cone theorem concerns stability under compact variations away from the vertex, without a minimizing assumption. After the monotone solution has been shown minimizing, its phase cones have the stronger minimizing

property. A cylindrical base in dimension three is regular away from the origin by two-dimensional cone classification and dimension reduction. Thus the argument does not require classification of arbitrary nonsmooth stable cones.

APPENDIX A. INTERFACES FOR A FULL FORMALIZATION

This appendix separates the conventional proof above from its proposed proof-assistant implementation. It supplies one additional algebraic proof and fixes the obligations that a full implementation must discharge. Neither the presence of Lean source nor the successful execution of an arithmetic checker is a certification of the geometric or PDE theorem. The accompanying Lean development is an *unexecuted formalization draft*.

A.1. An algebraic version of the spherical minimax step. The following formulation avoids square roots and the differentiation of a maximum. It is useful when transferring the common geometric lower bound in the two parameter ranges into a formal ordered-field argument.

Lemma A.1 (Radical-free minimax interface). *Let $L, y, \pi_0, t \in \mathbb{R}$ satisfy $L > 0$, $y \geq 0$, $t > 0$, and*

$$tL \leq 2y, \quad 4\pi_0^2 \leq yL + L^2.$$

Then

$$2\pi_0^2 t^2 \leq (t + 2)y^2.$$

In particular, if $0 \leq v < 1$ and

$$y \geq \frac{(1 - v^2)L}{2}, \quad y \geq \frac{4\pi^2}{L} - L,$$

then

$$2\pi^2(1 - v^2)^2 \leq (3 - v^2)y^2.$$

Proof. The three quantities

$$\begin{aligned} A &= (yL + L^2 - 4\pi_0^2)t^2, \\ B &= (2y - tL)yt, \\ C &= (2y - tL)(2y + tL) \end{aligned}$$

are nonnegative. For A this is the second hypothesis multiplied by a square. For B , both $2y - tL$ and yt are nonnegative. For C , the first factor is nonnegative and the second is positive since $tL > 0$ and $y \geq 0$. Direct expansion gives the exact identity

$$A + B + C = 2((t + 2)y^2 - 2\pi_0^2 t^2).$$

This proves the first conclusion without dividing by a possibly zero quantity. For the second, put $t = 1 - v^2 > 0$ and $\pi_0 = \pi$; multiplication of the second lower bound by $L > 0$ gives the required second hypothesis. $\square$

In the geometric applications, $y = P/(b_p L) \geq 0$ and $t = 1 - v^2 > 0$. Since $3 - v^2 > 0$, taking the nonnegative square root recovers

$$\frac{P}{b_p L} \geq \frac{\pi\sqrt{2}(1 - v^2)}{\sqrt{3 - v^2}}.$$

The case $v = 1$ remains separate and is not divided out. The associated Lean declarations are named `minimax_squared` and `spherical_flux_squared` in the scalar bridge module. Their source proofs have not been run through Lean here. The polynomial identity in the proof has been independently replayed with exact symbolic arithmetic.

A.2. Fourier normalization and legitimate weak testing.

Lemma A.2 (Normalization and testing across the full fractional range). *Let $n \geq 1$, $0 < p < 2$, and*

$$c_{n,p} = \frac{p2^{p-1}\Gamma((n+p)/2)}{\pi^{n/2}\Gamma(1-p/2)}.$$

For $\phi \in C_c^\infty(\mathbb{R}^n)$ set

$$\mathcal{L}_p\phi(x) = c_{n,p} \text{PV} \int_{\mathbb{R}^n} \frac{\phi(x) - \phi(y)}{|x-y|^{n+p}} dy.$$

This is a continuous L^1 function. With $\widehat{\phi}(\xi) = \int e^{-ix \cdot \xi} \phi(x) dx$ it satisfies

$$\widehat{\mathcal{L}_p\phi}(\xi) = |\xi|^p \widehat{\phi}(\xi), \quad \int_{\mathbb{R}^n} \mathcal{L}_p\phi = 0.$$

If $u \in L^\infty(\mathbb{R}^n) \cap C_{\text{loc}}^2(\mathbb{R}^n)$, then its physical principal value exists, is continuous, and

$$\begin{aligned} \int \phi \mathcal{L}_p u &= \frac{c_{n,p}}{2} \iint \frac{(u(x) - u(y))(\phi(x) - \phi(y))}{|x-y|^{n+p}} dx dy \\ &= \int u \mathcal{L}_p \phi. \end{aligned} \tag{3}$$

The double integral is absolutely convergent. In particular, no global bound on the derivatives of u is needed for this testing identity.

Proof. Write $\delta_h^2 \phi(x) = 2\phi(x) - \phi(x+h) - \phi(x-h)$. Taylor's formula along a line segment and translation invariance of Lebesgue measure give

$$\|\delta_h^2 \phi\|_1 \leq \min\{C_\phi |h|^2, 4\|\phi\|_1\}, \quad C_\phi = \sum_{i,j=1}^n \|\partial_{ij}\phi\|_1.$$

For example the Taylor remainder is the integral of $h^T D^2 \phi(x \pm th) h$ against $1-t$, for $0 \leq t \leq 1$; $|h_i h_j| \leq |h|^2$ proves the first bound. Hence

$$\int_{\mathbb{R}^n} \frac{\|\delta_h^2 \phi\|_1}{|h|^{n+p}} dh \leq |S^{n-1}| \left(\frac{C_\phi}{2-p} + \frac{4\|\phi\|_1}{p} \right) < \infty.$$

Reflection $h \mapsto -h$ at every positive cutoff gives

$$\mathcal{L}_p \phi(x) = \frac{c_{n,p}}{2} \int_{\mathbb{R}^n} \frac{\delta_h^2 \phi(x)}{|h|^{n+p}} dh.$$

The Taylor bound on compact sets proves pointwise convergence, uniformly on compact sets. The preceding L^1 estimate proves convergence in L^1 as well. The truncated functions are continuous, so the limit is continuous. Fubini is now legitimate both for its integral in x and for its Fourier transform. The integral is zero because every translated second difference has zero integral. Its Fourier transform is

$$\widehat{\mathcal{L}_p\phi}(\xi) = c_{n,p} \widehat{\phi}(\xi) J_{n,p}(\xi), \quad J_{n,p}(\xi) = \int_{\mathbb{R}^n} \frac{1 - \cos(h \cdot \xi)}{|h|^{n+p}} dh.$$

Put $a = p/2 \in (0, 1)$. The nonnegative Gaussian representation of the kernel and Tonelli's theorem give

$$\begin{aligned} J_{n,p}(\xi) &= \frac{\pi^{n/2}}{\Gamma((n+p)/2)} \int_0^\infty t^{a-1} (1 - e^{-|\xi|^2/(4t)}) dt \\ &= \frac{\pi^{n/2}\Gamma(1-a)}{a4^a\Gamma((n+p)/2)} |\xi|^p = c_{n,p}^{-1} |\xi|^p. \end{aligned}$$

For $\xi \neq 0$ the second equality follows by the substitution $r = |\xi|^2/(4t)$ and integration by parts in $\int_0^\infty (1 - e^{-r}) r^{-1-a} dr = \Gamma(1-a)/a$. At zero the integrand is $O(r^{-a})$, and at infinity it

is $O(r^{-1-a})$, so the integral and the discarded boundary terms have the required convergence. For $\xi = 0$ both sides vanish. This proves the normalization in every dimension, including $1 \leq p < 2$ where an unpaired first-difference estimate would not suffice.

For bounded locally C^2 u , use the same symmetric expression. On a fixed compact set of x and $|h| < 1$, its second difference is bounded by $C_K|h|^2$ using the Hessian on a larger compact set; the large- h part is bounded by $4\|u\|_\infty$. This proves existence, continuity, and locally uniform convergence of the cutoffs. To justify the paired double integral in (3), observe that a nonzero test difference has at least one point in $\text{supp } \phi$. For $|x - y| < 1$ both points lie in a fixed compact neighborhood, and both first differences are bounded by a constant times $|x - y|$. Their product gives the integrable radial power r^{1-p} . For $|x - y| \geq 1$, boundedness of u bounds the product by $2\|u\|_\infty(|\phi(x)| + |\phi(y)|)$, with integrable kernel tail. For each cutoff $\varepsilon > 0$, Fubini and symmetry give both equalities in (3) with that cutoff. On the first side use locally uniform convergence on $\text{supp } \phi$; in the middle use the absolute majorant just proved; and on the last side use the L^1 convergence of $\mathcal{L}_{p,\varepsilon}\phi$ and boundedness of u . $\square$

Remark A.3 (The regularity boundary of the testing lemma). The last assertion assumes local C^2 regularity, which has already been established before its uses in Section 8. It is not a proof that an arbitrary bounded continuous function with pointwise principal values is C^2 , or that its principal values may be integrated term by term. The broader pointwise solution class in the formal draft still needs its regularity/weak-formulation bridge before the external PDE results can be imported. That bridge remains an explicit formalization obligation; it is not an additional assumption on the manuscript's solutions.

A.3. Local and global time parametrizations of variations.

Lemma A.4 (Preserving the germ of a local deformation). *Let Ψ_t be a smooth family of diffeomorphisms of $\mathbb{R}^n$ for $|t| < \tau$, $\tau > 0$, with $\Psi_0 = \text{Id}$ and with $\Psi_t = \text{Id}$ outside a fixed compact set K . Suppose $\Psi_t(x)$ and $\Psi_t^{-1}(x)$ are jointly smooth on this time interval. There is a globally time-parametrized smooth family with smooth inverse, with the same support K , which agrees with Ψ_t for $|t| < \tau/4$. Consequently first and second variations at zero agree for the two families whenever those variations exist.*

Proof. Choose $\chi \in C_c^\infty((-\tau/2, \tau/2))$ with $0 \leq \chi \leq 1$ and $\chi = 1$ on $[-\tau/4, \tau/4]$. Put $\theta(t) = t\chi(t)$. Then $|\theta(t)| < \tau/2$ for every real t , and $\theta(t) = t$ near zero. Define

$$\tilde{\Psi}_t(x) = \Psi_{\theta(t)}(x), \quad \tilde{\Xi}_t(x) = \Psi_{\theta(t)}^{-1}(x).$$

The two maps are jointly smooth by composition, are inverse at every t , and fix every point outside K . They are the identity when $|t| \geq \tau/2$, and coincide with the original maps near zero. Thus any scalar perimeter function formed from them has exactly the same germ at zero, not merely the same first velocity. Its derivatives and lower-limit quotients at zero are therefore unchanged. Conversely, a globally parametrized family restricts to every sufficiently small time interval. A support condition such as $K \subset D \setminus \{0\}$ is preserved by the construction. Autonomous flows of compact smooth fields have a smooth inverse Ψ_{-t} , so they satisfy the joint inverse hypothesis. If joint inverse smoothness is not part of the local-family convention, it follows from the inverse function theorem applied to $(t, x) \mapsto (t, \Psi_t(x))$, whose derivative is block triangular with invertible spatial block; uniqueness of the inverse makes the local inverse maps agree on overlaps. $\square$

A.4. Scalar bridges are not geometric theorems. The new scalar module connects the previously checked integer polynomials to the actual rational expressions in the proof. For example, with

$$A(\delta) = \frac{155}{216} + \delta(2 - \delta)(1 - \delta)^2,$$

it gives a proof draft for

$$\left(\frac{59}{500}\right)^2 < \frac{\delta^2(1 - \delta)^2}{4A(\delta)}, \quad \frac{136}{375} \leq \delta \leq \frac{87}{125},$$

and separately verifies positivity of $A(\delta)$. The identification of this quotient with $a(\delta)^2$ still uses the definition of the cutoff and its positive denominator. The geometric cap-density estimate remains a different theorem.

Similarly, the logarithmic-slope module clears its denominator only after showing that every factor is positive on $[0, 1/3]$. Its final comparison with the expression containing $\log(500/59)$ explicitly takes the logarithm bound as an argument. That conditional statement does not prove the logarithm bound, the differentiated barrier identity, or the analytic estimates preceding the barrier. Those proofs must be supplied before the conditional lemma can contribute to an unconditional target.

The scalar closure module proves only implications from the displayed pinching, comparison, and barrier inequalities to $\delta = 0$. These hypotheses are visible parameters. Such implications cannot be presented as proofs of stable cone flatness. The full-target checks are designed to reject a proposed main theorem that still requires any of these extra hypotheses.

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